

THORLABS

Coupling Light Into Optical Fiber

Overview

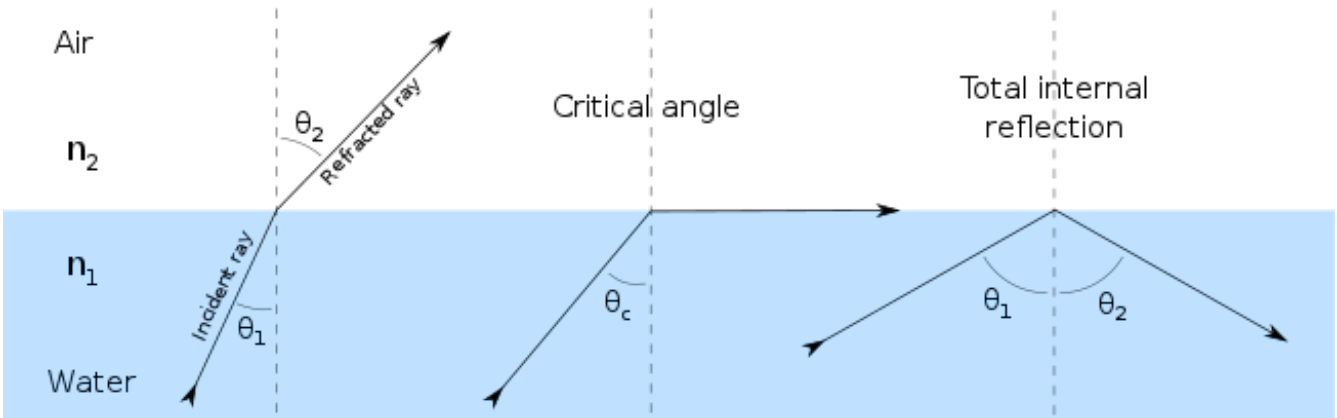
- Fiber Basics
- Beam Diameter Standards
- Multimode Fibers
- Single Mode Fibers
- Fiber Alignment & Degrees of Freedom
- Optical Aberration & Optic Selection
- Lab Work

Fiber Basics

Fiber Optic Concepts: Total Internal Reflection (TIR)

- When light is incident from a high index to a low index medium, there is a critical angle of incidence where light will no longer transmits and will 100% reflect (total internal reflection)

$$\theta_{critical} = sin^{-1}(\frac{n_2}{n_1}) \; for \; n_1 > n2$$

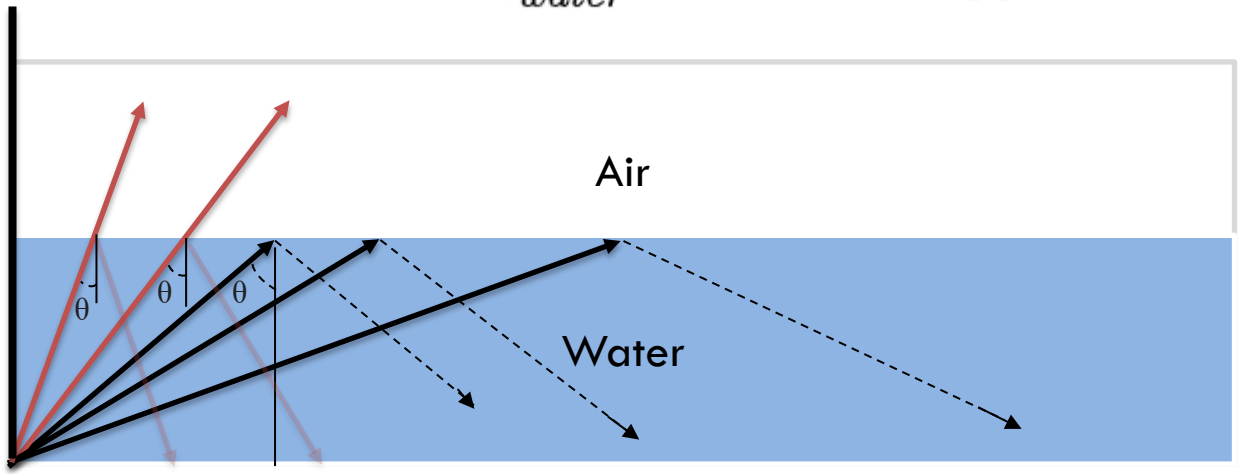


https://en.wikipedia.org/wiki/Angle_of_incidence_%28optics%29

Fiber Optic Concepts: Total Internal Reflection (TIR)

- Critical Angle for Water/Air Interface:

$$\theta_{Critical} = sin^{-1}(\frac{n_{air}}{n_{water}}) = \sin^{-1}(\frac{1}{1.33}) = 48.75°$$

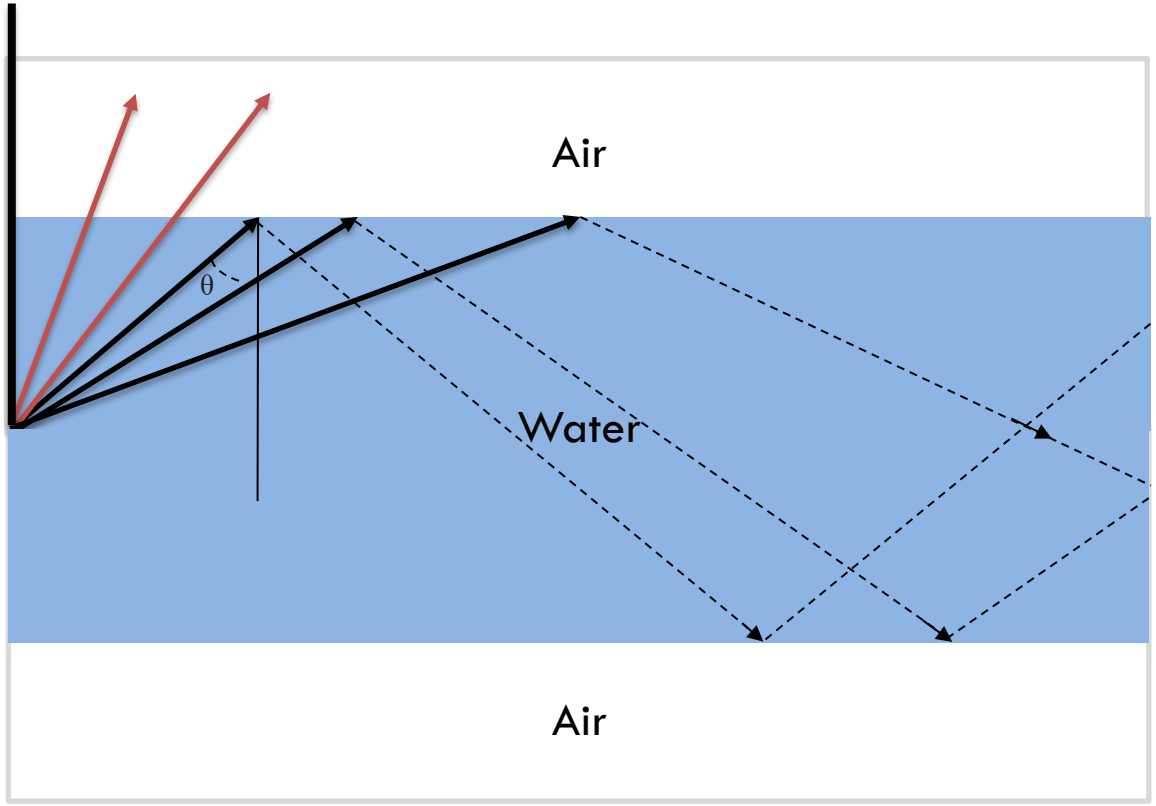


Below Critical Angle – Some light will transmit and refract

Above Critical Angle – Light totally internally reflects; NO transmission

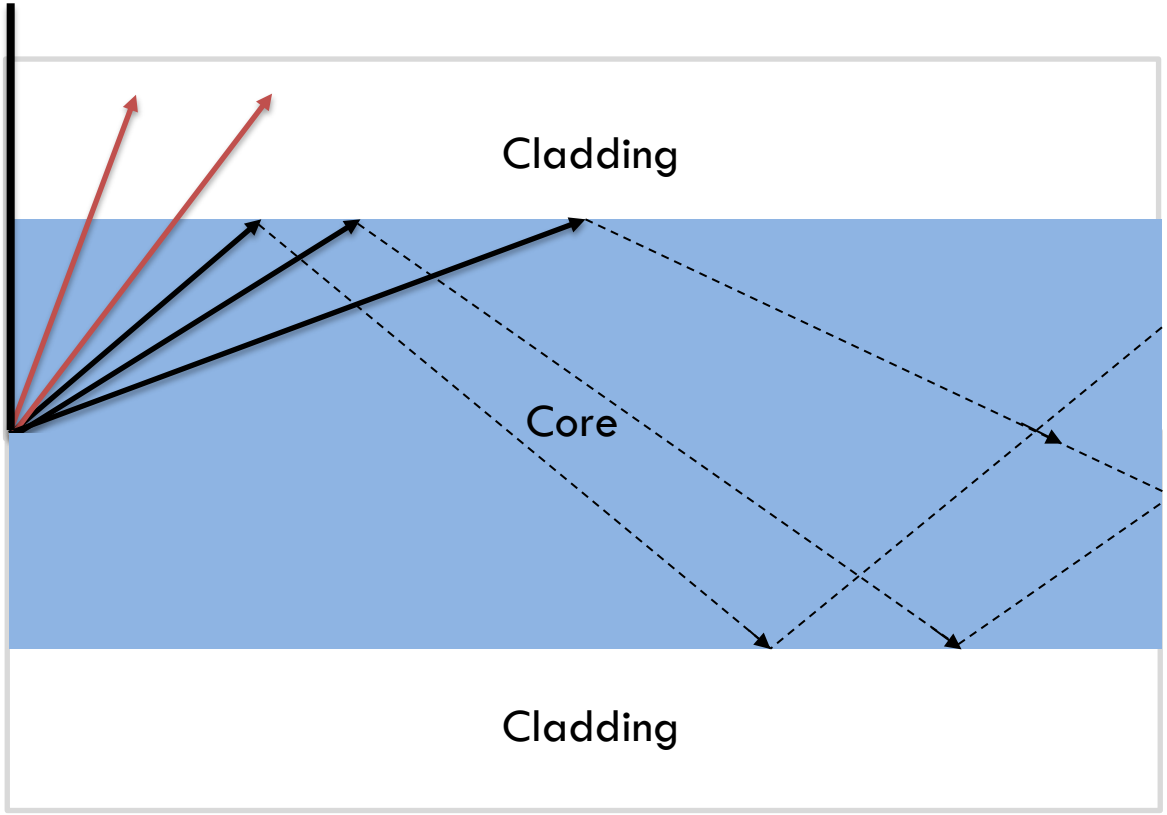
Fiber Optic Concepts: Total Internal Reflection (TIR)

- Surrounding the higher medium by a lower index medium keeps light bound by TIR



Fiber Optic Concepts: Total Internal Reflection (TIR)

- Step-Index optical fibers use a higher index glass or polymer core and lower index glass or polymer cladding to maintain TIR for guiding light propagation

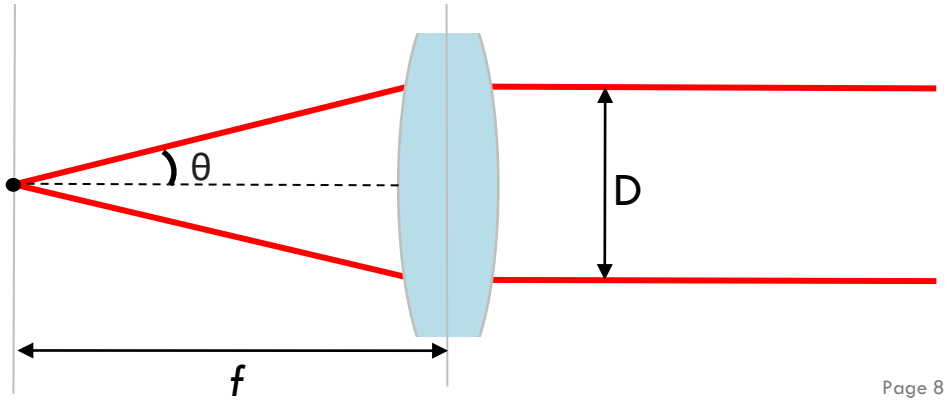


Optics Concepts: Numerical Aperture

- Numerical Aperture (NA) is a dimensionless parameter that defines the maximum angle (θ) of acceptance or emission for an optical system
- The NA of a light source is calculated with the divergence half-angle (θ) of the optical system and the refractive index of the propagating medium (n)
- The NA of an optical system is defined using the clear aperture beam diameter and the length focal length
- While a lens may have an NA specification, this serves as a maximum acceptable NA of the lens and not necessarily the NA of the full optical system

NA	Divergence Half Angle, Θ
0.10	5.7°
0.15	8.6°
0.20	11.5°
0.25	14.4°
0.30	17.4°
0.35	20.4°
0.40	23.5°
0.45	26.7°
0.50	30.0°

$$\mathrm{NA} = n \sin\theta = n \sin\left[\arctan\left(\frac{D}{2f}\right)\right] \approx n\frac{D}{2f}$$



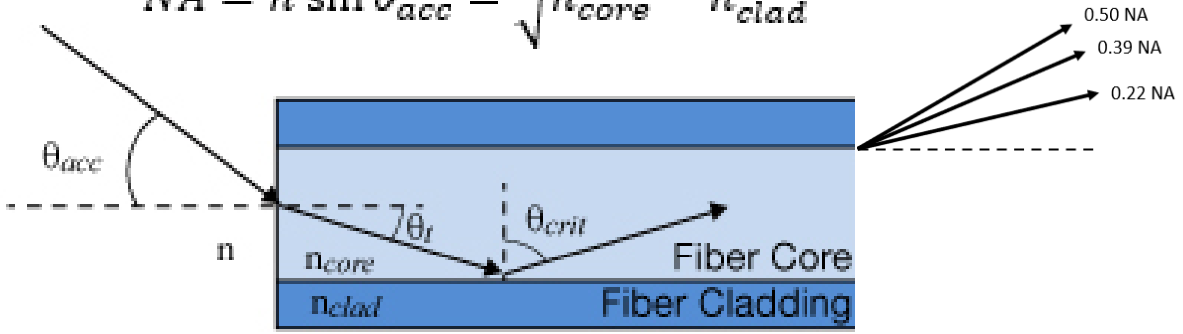
Fiber Optic Concepts: Multimode Numerical Aperture

- Numerical Aperture (NA) describes the maximum angle of incidence that light rays will maintain TIR for propagation in a multimode fiber/waveguide
- The acceptance angle (θ_{acc}) for maintaining TIR is calculated by the refractive indices of the incident medium, fiber core, and fiber cladding
- Light rays at angles exceeding the acceptance angle won't maintain propagation in the fiber

NA	Divergence Half Angle, Θ
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0.20	11.5°
0.25	14.4°
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0.45	26.7°
0.50	30.0°

$$\sin\theta_{crit} = \frac{n_{clad}}{n_{core}} = \cos\theta_t \qquad n\sin\theta_{acc} = n_{core}\sqrt{1-\cos^2\theta_t} = \sqrt{n_{core}^2 - n_{clad}^2}$$

$$NA = n\sin\theta_{acc} = \sqrt{n_{core}^2 - n_{clad}^2}$$

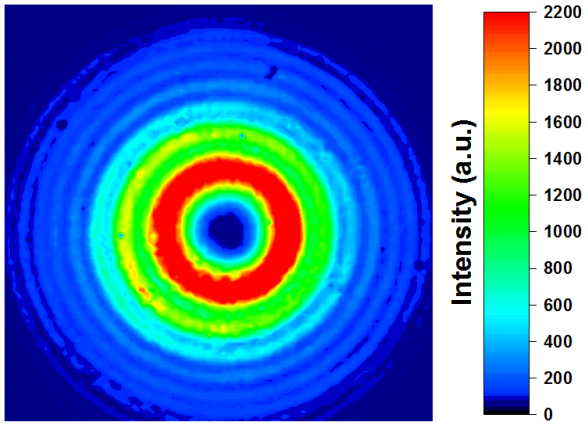


Beam Diameter Standards

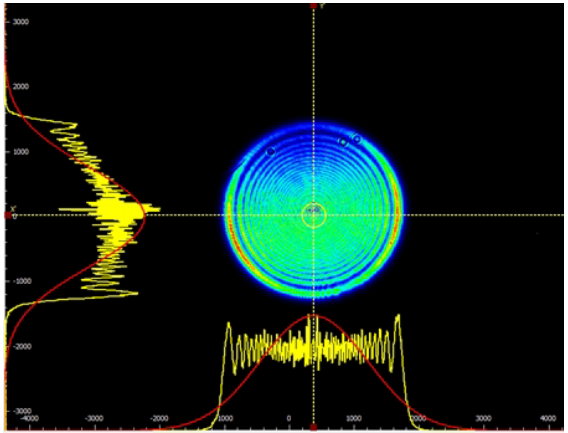
Beam Diameter

- A unified method to define beam diameter is critical, but is not always obvious due to beam intensity profiles
- The most common calculation methods are:
  - **Full Width at Half Max(FWHM) -** Beam width defined by the intensity falling 50% of the peak intensity in each axis
  - **$1/e^2$ -** Beam width defined by the intensity falling $1/e^2$ (~13.5%) of the peak intensity in each axis
  - **D4σ** (ISO 11146 standard) **-** Beam width defined by 4 times the standard deviation of the intensity distribution in each axis

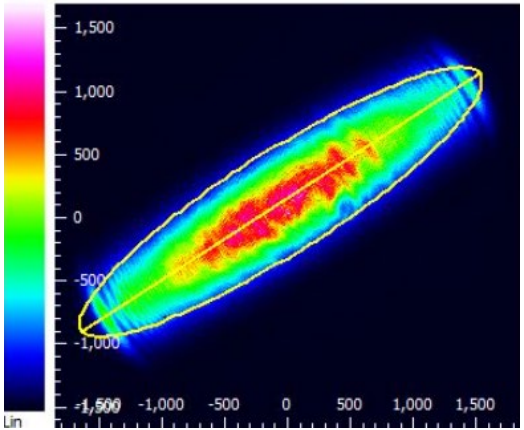
Donut Mode Beam



Irregular Round Beam



Elliptical Gaussian Beam

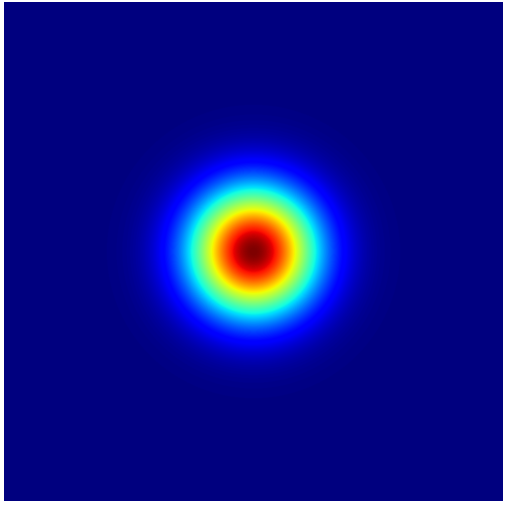


Beam Diameter

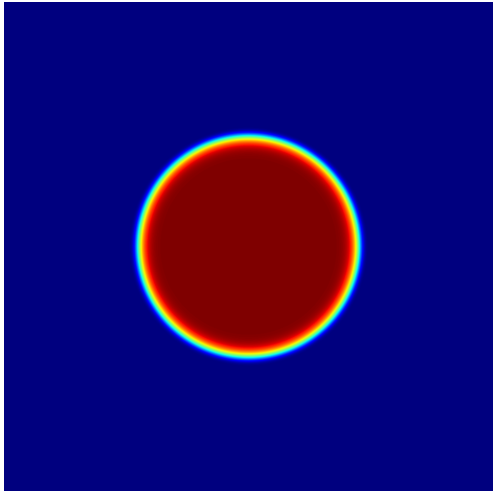
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How should the diameter of these beam profiles be calculated?

Gaussian Profile

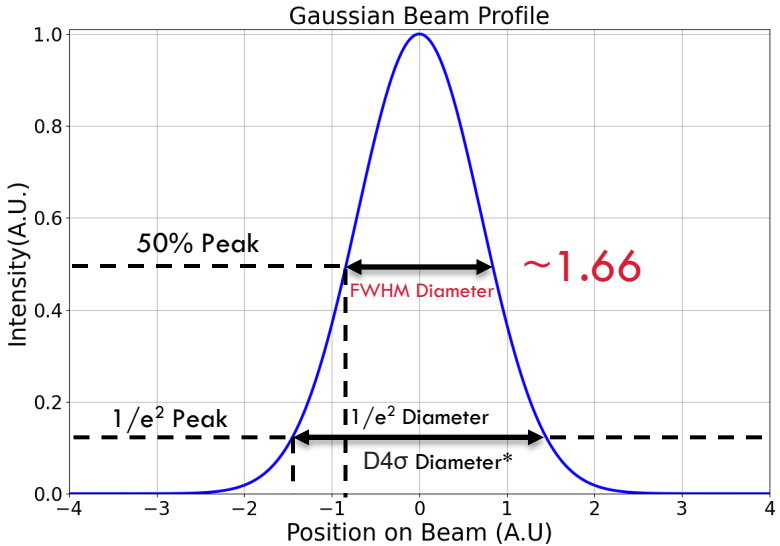


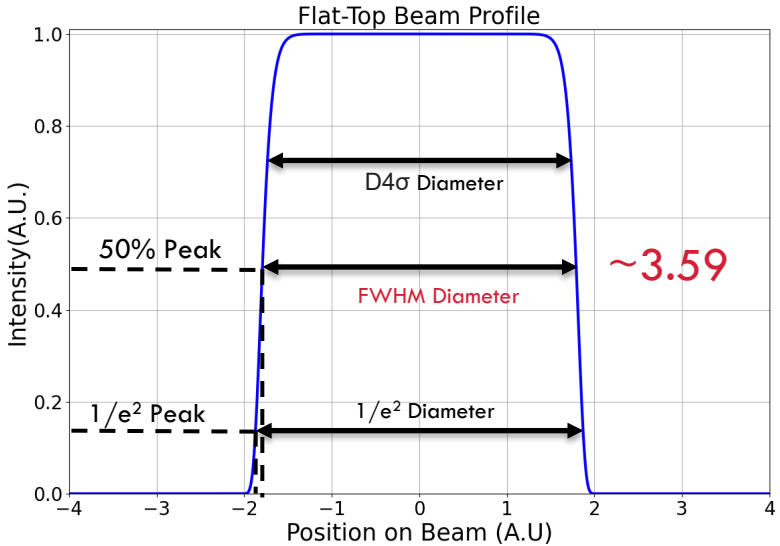
Flat-Top Profile



Beam Diameter

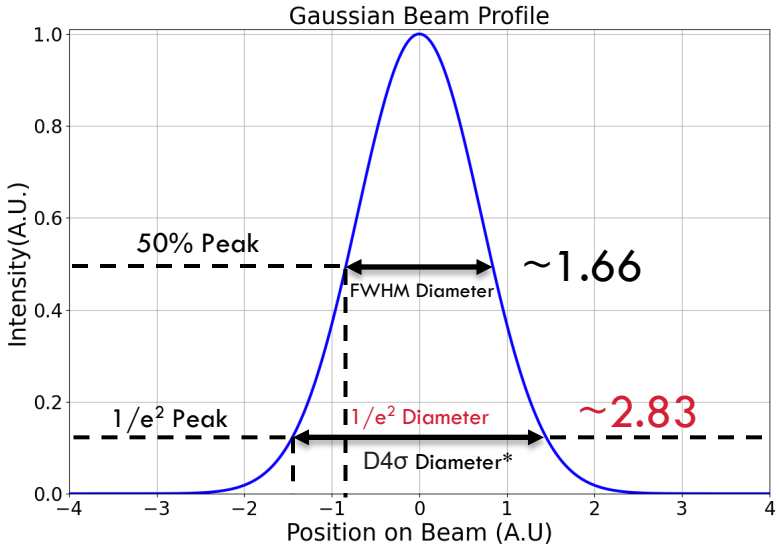
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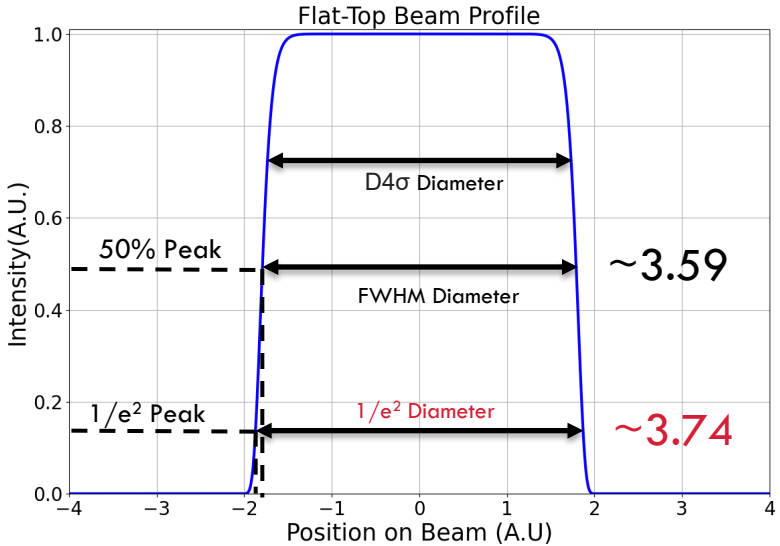




Beam Diameter

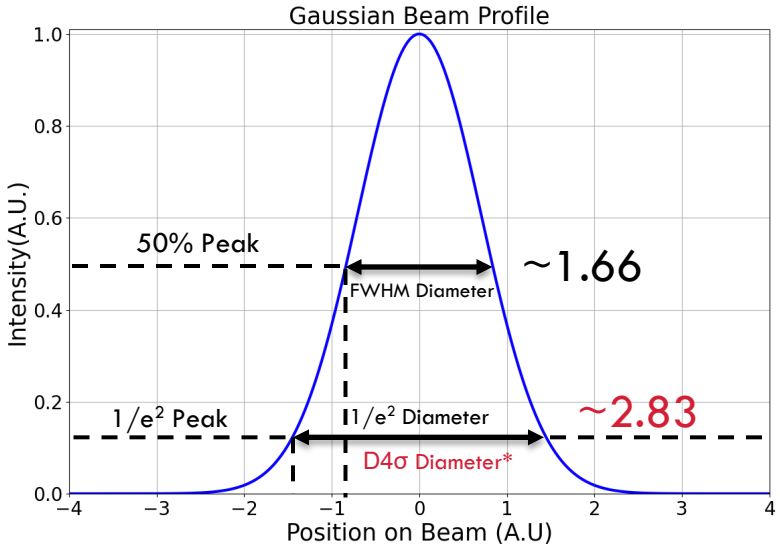
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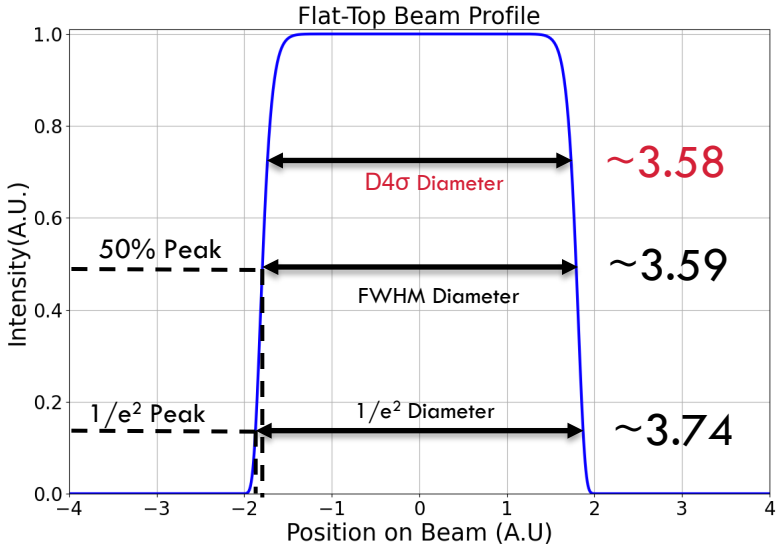




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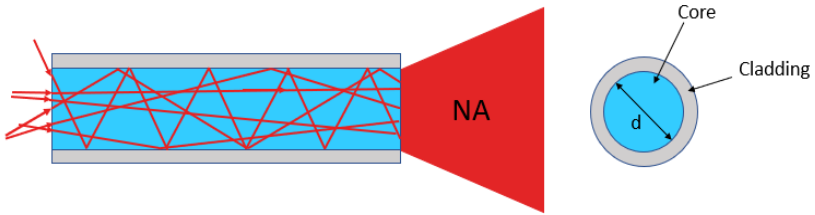


* For Gaussian beams, the D4σ diameter is equal to the $1/e^2$ diameter

Multimode Fibers

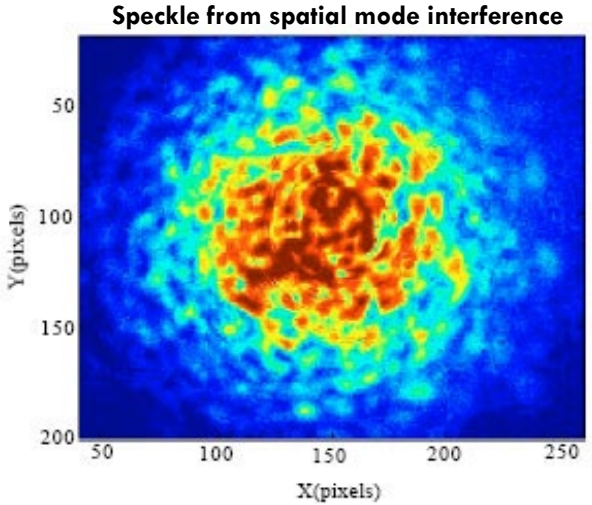
Fiber Optic Concepts: Fiber Optic Modes

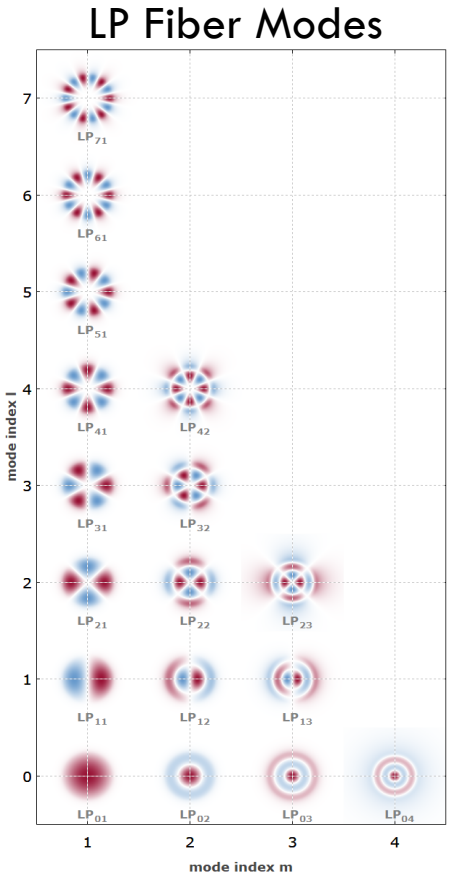
- For a given multimode fiber, there are a finite number of paths that can be taken while supporting propagation; referred to as 'modes', 'transverse modes', or 'spatial modes'
- Modes are stable patterns that waves form as they pass through a waveguide*
- In cylindrical waveguide step-index fibers, these modes are described by LP modes (right figure)
- The number of supported modes in a fiber is dependent on the fiber's NA, core diameter (d), and wavelength (λ)
- These modes overlap while propagating and the constructive and destructive interference between these modes is visible as a speckled structure when using a coherent source (below)



$$Number\ of\ Modes \approx \frac{\pi^2 d^2 (NA)^2}{2\lambda^2}$$

50 µm Core , 0.22 NA @ 633 nm → **~ 1490 Modes!**



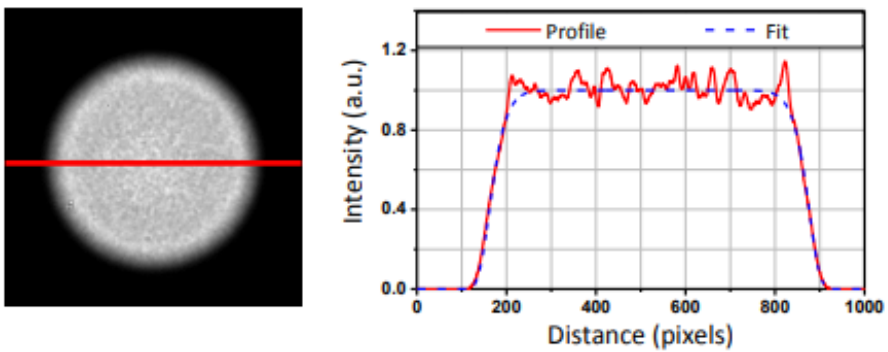


*Hecht, J. (2015). *Understanding fiber optics*. Createspace Independent Publishing Platform.

Fiber Optic Concepts: Mode Mixing and Launch Conditions

- When light is distributed equally among all modes of a multimode fiber, the resulting intensity distribution and output beam will approach being a flat top profile
- The launch conditions coupling into a fiber may not fill all the modes and may preferentially fill more of the lower order or higher modes
- The fiber modes the incident light launches into are dependent on two types of rays*:
  - Meridional rays pass through the central axis of the fiber after each TIR
  - Skew rays never pass through the central axis of the fiber and propagate in a helical path along the fiber in a series of paths tangent to an inner circle with radius 'r'

Beam Profile - Well Mode Mixed Flat Top

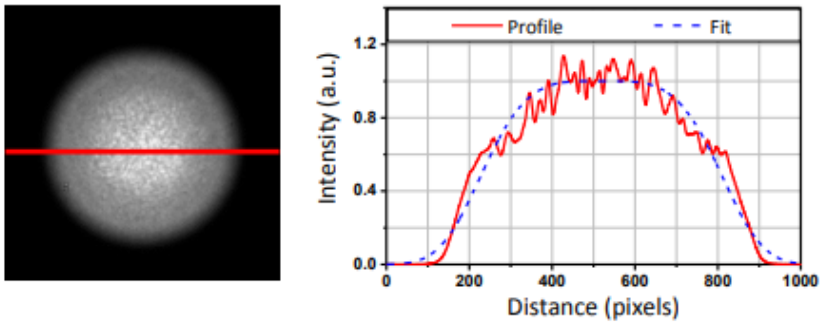


* Snyder, A. W., & Love, J.D. (1983). Optical Waveguide Theory (pp. 27-31). Chapman & Hall.

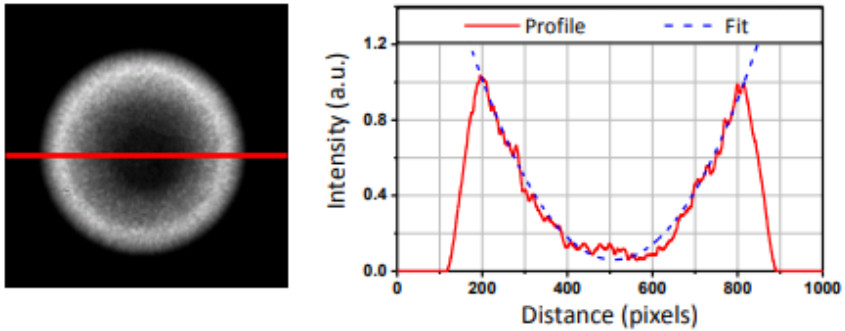
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Beam Profile - Low Order Modes



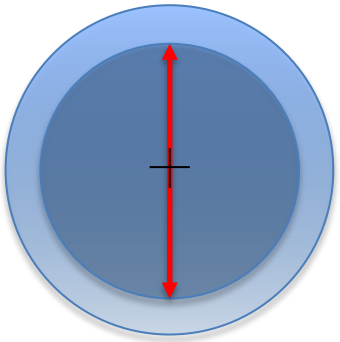
Beam Profile - High Order Modes

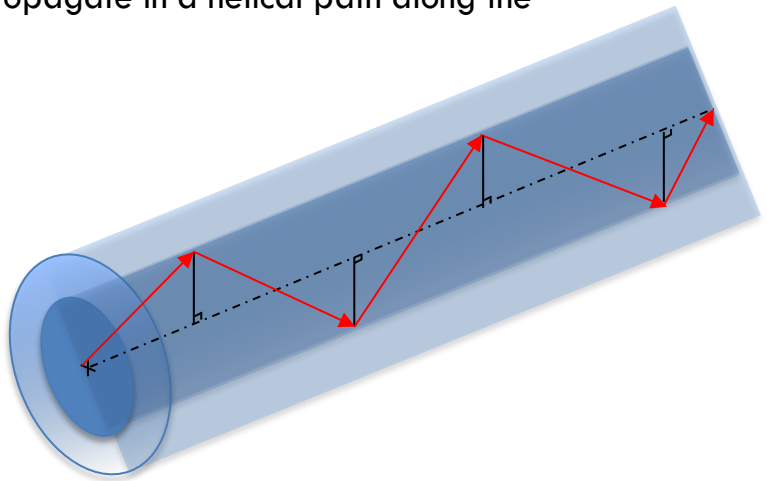


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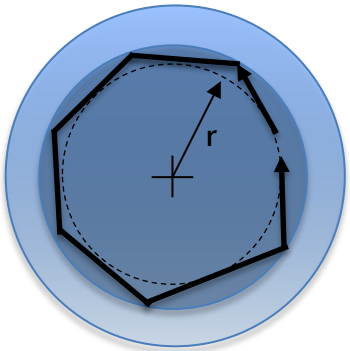


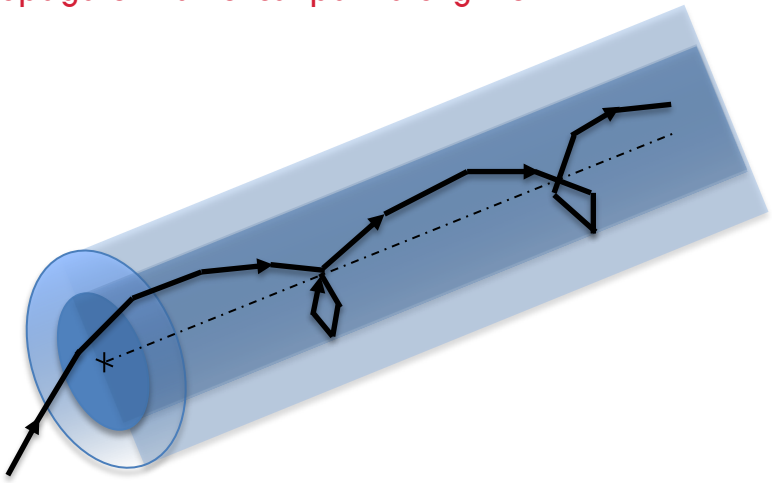


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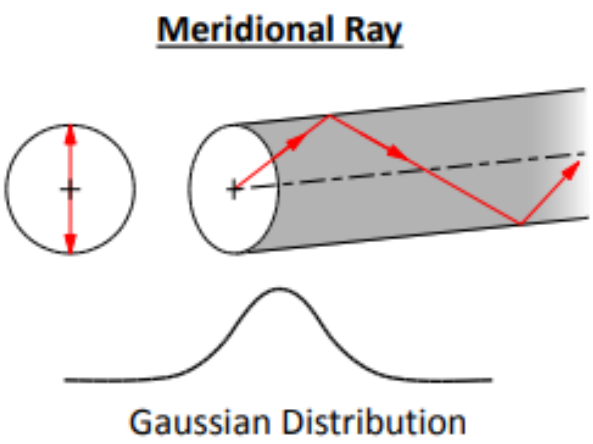


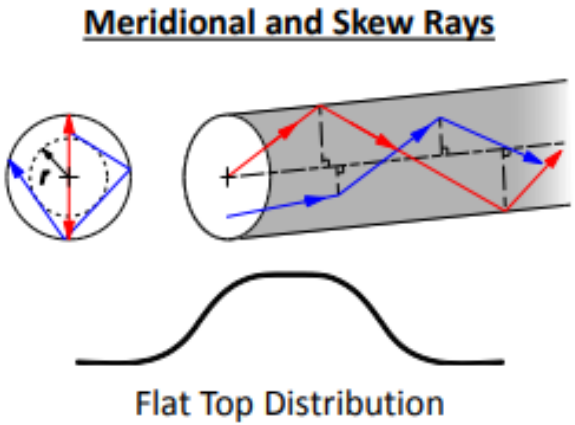


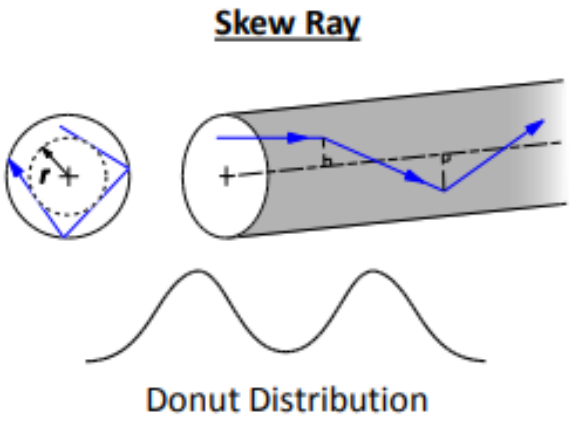
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Fiber Optic Concepts: Mode Mixing and Launch Conditions

- Meridional rays pass through the central axis of the core and tend to fill the lower order modes
- Skewed rays travel in helical paths near the edged of the core and tend to fill higher order modes
- By launching into the fiber off-axis, the relative proportion of meridional rays to skewed rays can be adjusted to influence the fiber's mode distribution as well as its resulting output profile

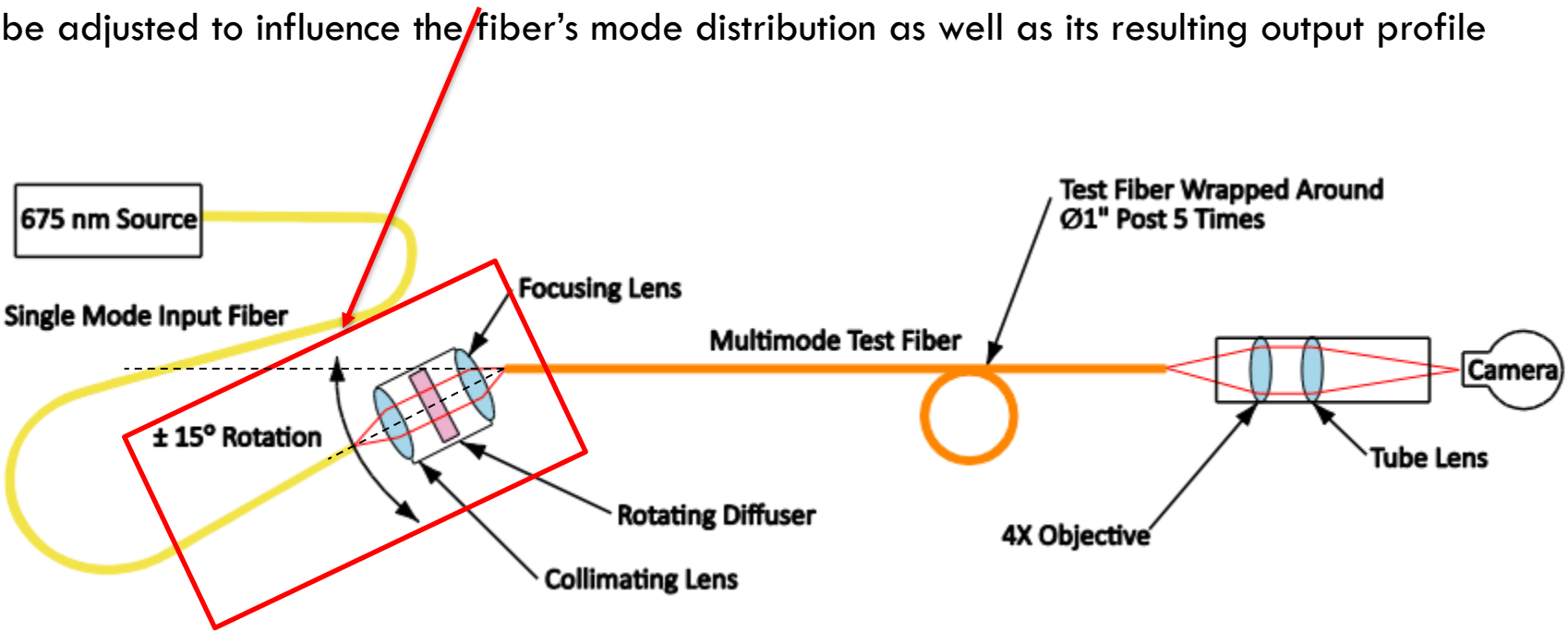






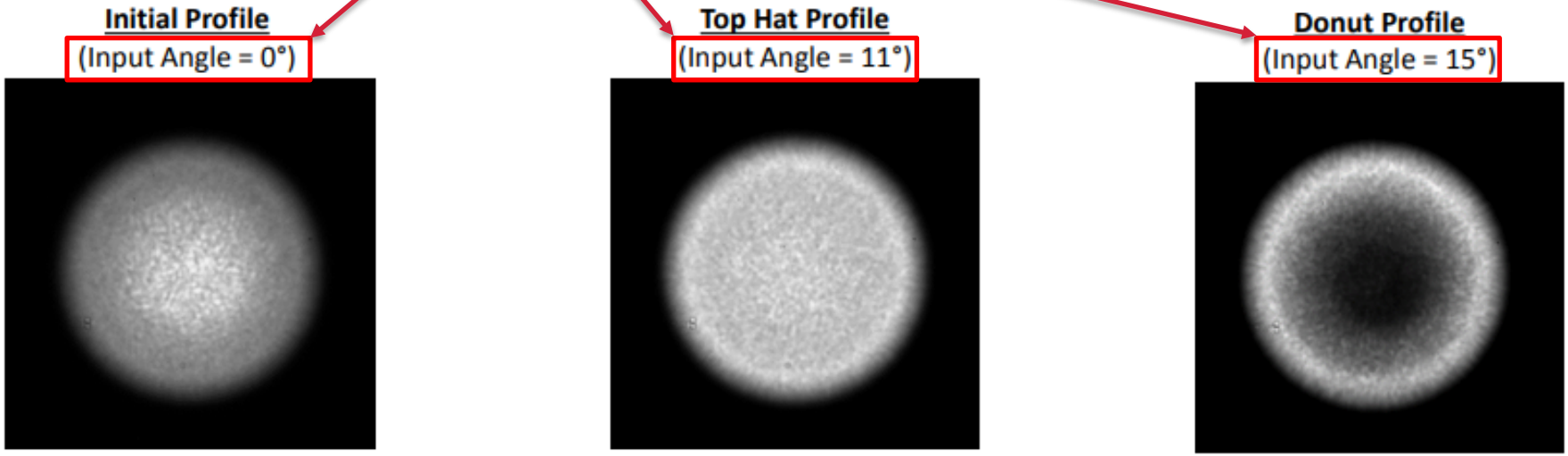
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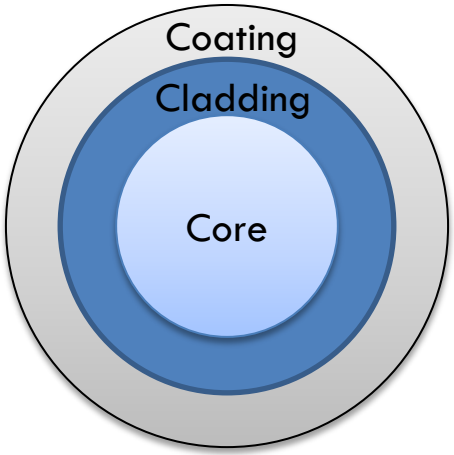
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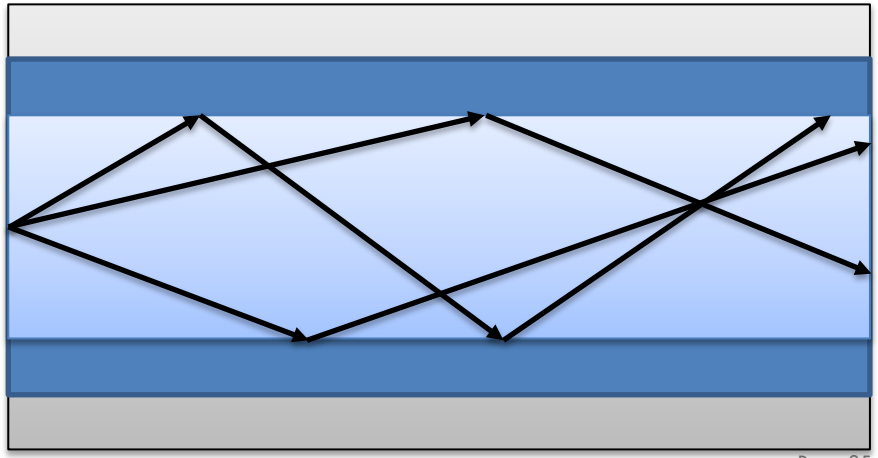
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Fiber Optic Concepts: Cladding Modes

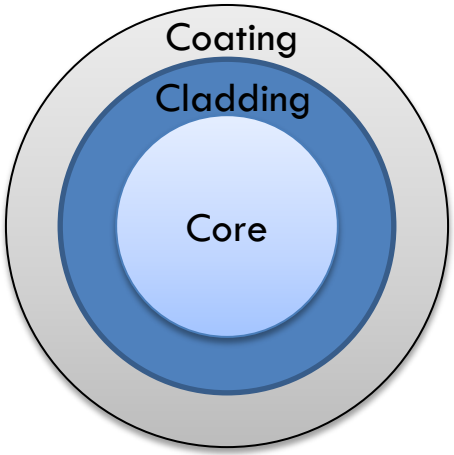
- Optical fibers contain a core for light propagation, cladding to maintain TIR, and a coating for protection
- Light can couple into the cladding, propagate as cladding modes, and worsen the output beam quality
- The two major causes of cladding modes are:
  - Exceeding the fiber NA
  - Over filling the core
- As best practice, it's recommended to stay within the fiber NA and fill up to 80% of the core to minimize coupling into cladding modes and minimize risk of damage due to high power density

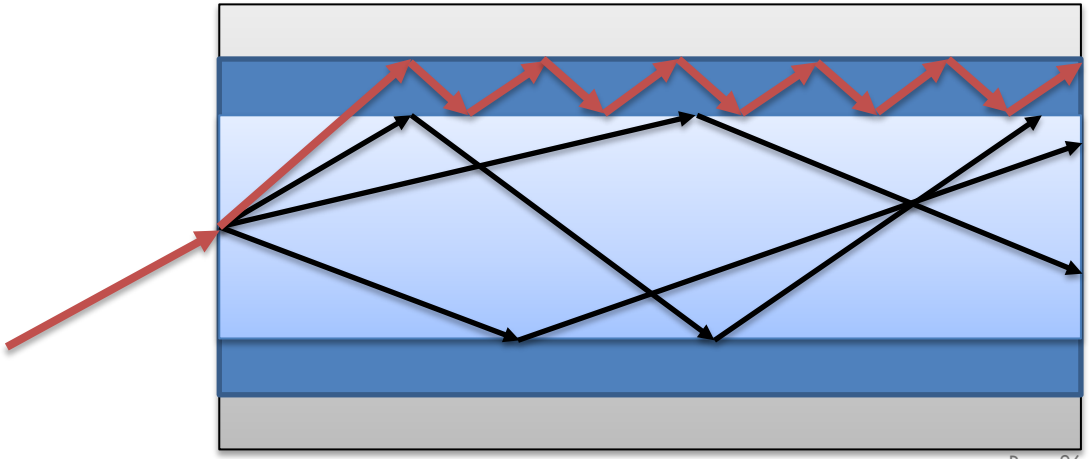




Fiber Optic Concepts: Cladding Modes

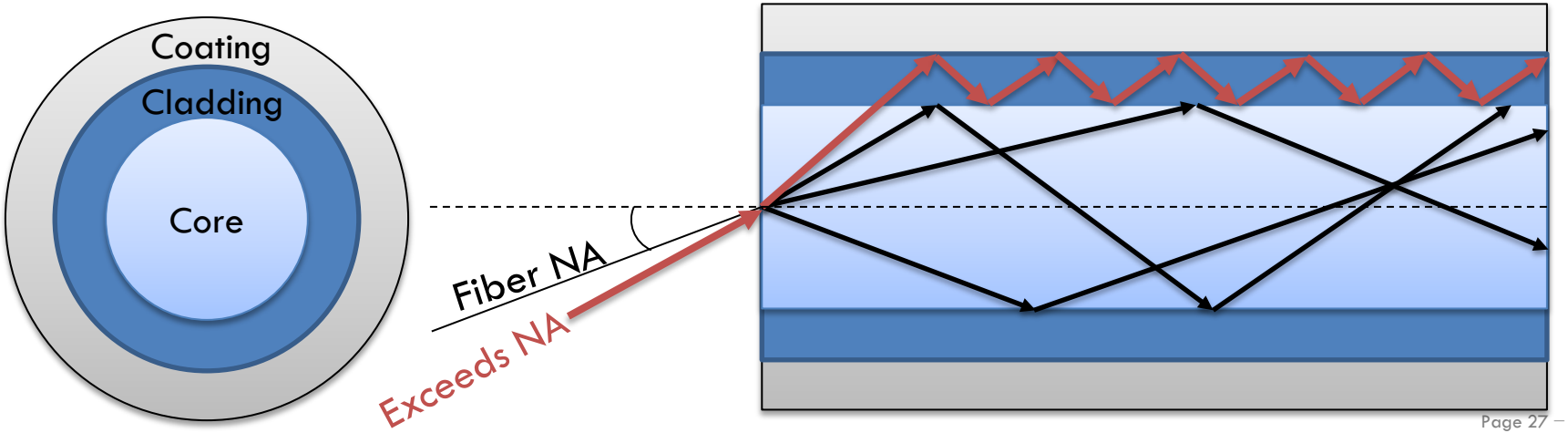
- Optical fibers contain a core for light propagation, cladding to maintain TIR, and a coating for protection
- Light can couple into the cladding, propagate as cladding modes, and worsen the output beam quality
- The two major causes of cladding modes are:
  - Exceeding the fiber NA
  - Over filling the core
- As best practice, it's recommended to stay within the fiber NA and fill up to 80% of the core to minimize coupling into cladding modes and minimize risk of damage due to high power density





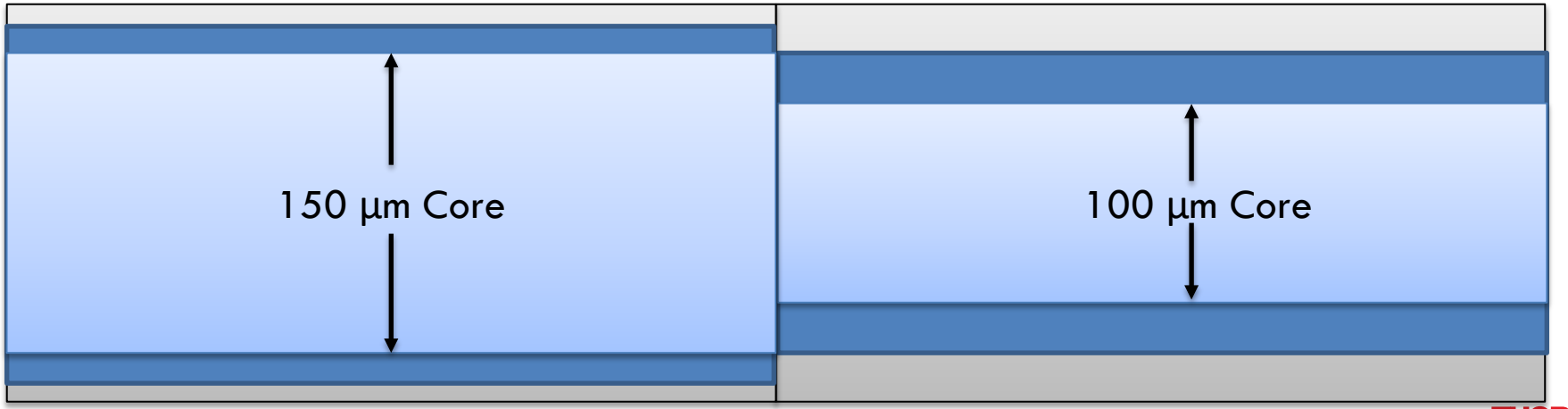
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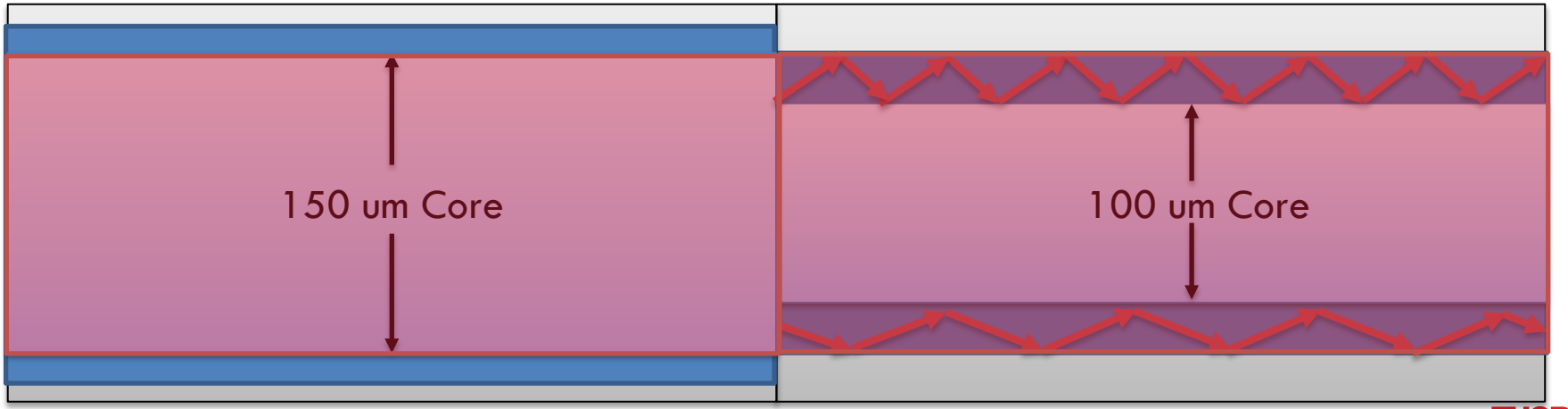
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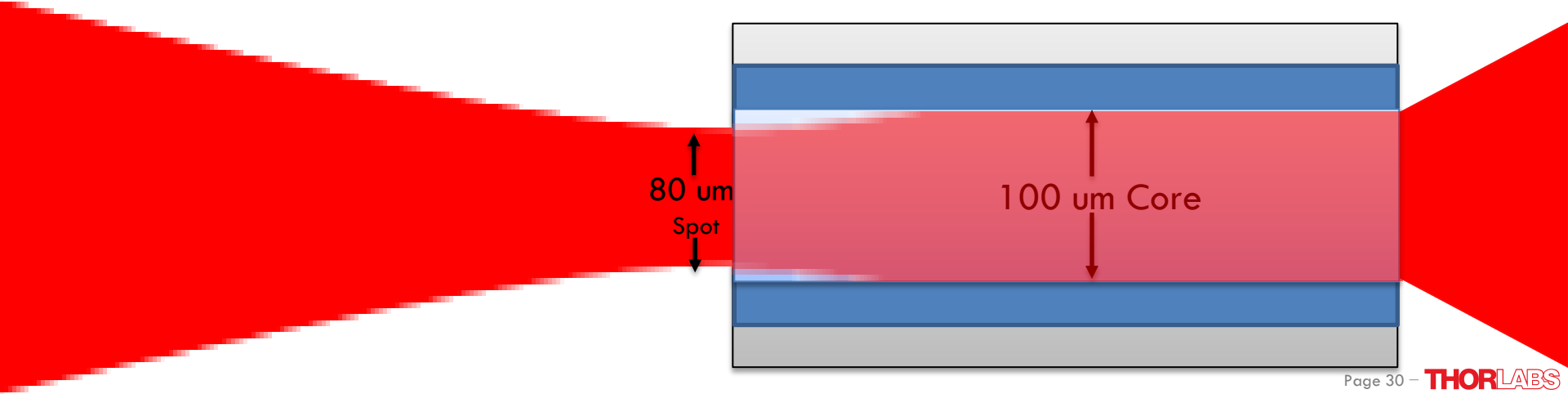
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Fiber Optic Concepts: Cladding Modes

- Optical fibers contain a core for light propagation, cladding to maintain TIR, and a coating for protection
- Light can become coupled into the cladding, propagate as cladding modes, and worse beam quality upon output
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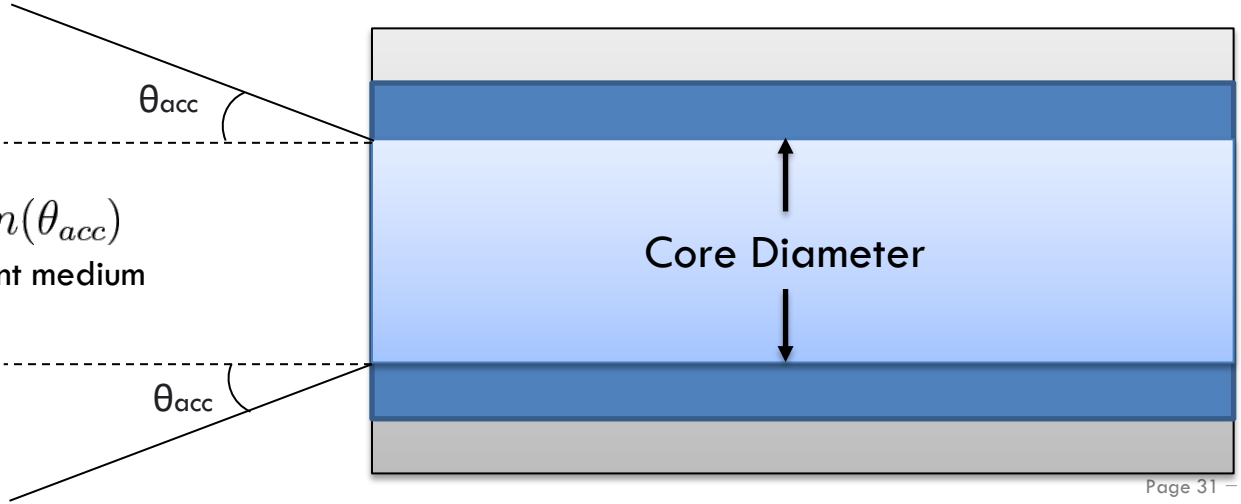


Theory: Coupling into Multimode Fiber

- The primary constraints for coupling into multimode fiber are the core diameter and NA
- Any light ray incident on the core and within the acceptance angle (NA) will couple into the fiber
- For a given collimated beam diameter (D) and fiber numerical aperture (NA), a focal length (f) must be chosen so the focused beam is within the NA of the fiber and efficiently couples
- The NA of the fiber can be used to define a minimum focal length (f_{min}) that ensures the focused light is within the fiber's NA. Focal lengths shorter than f_{min} exceed the fiber NA and won't couple all rays into the core

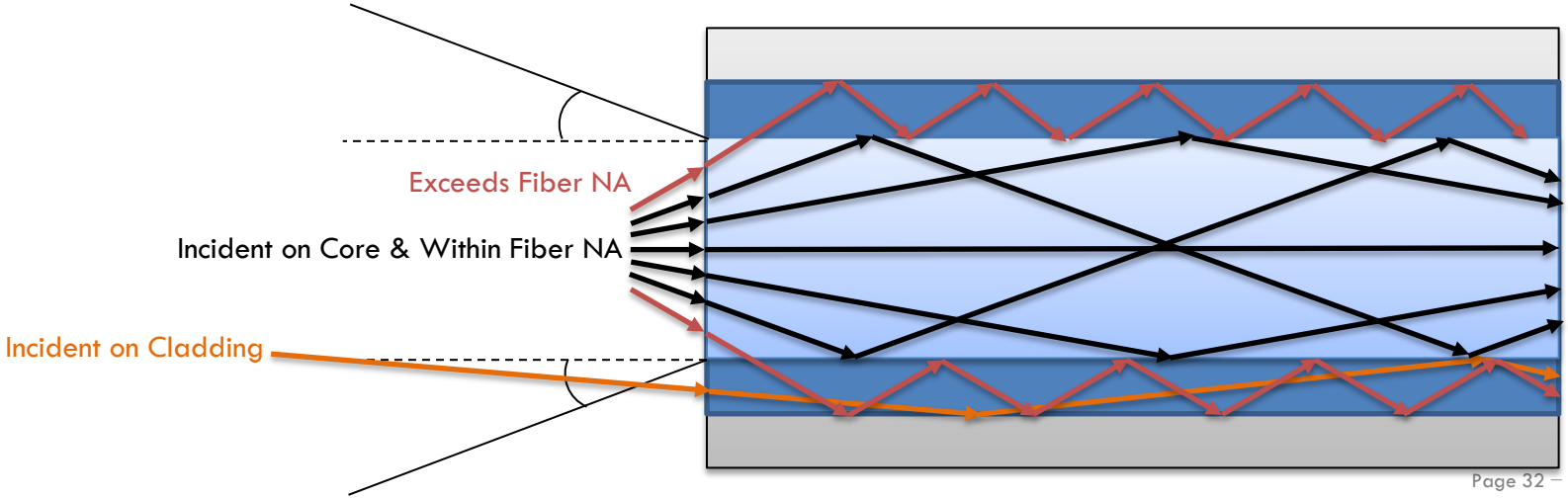
$$NA = n \cdot sin(\theta_{acc})$$

n - index of incident medium



Theory: Coupling into Multimode Fiber

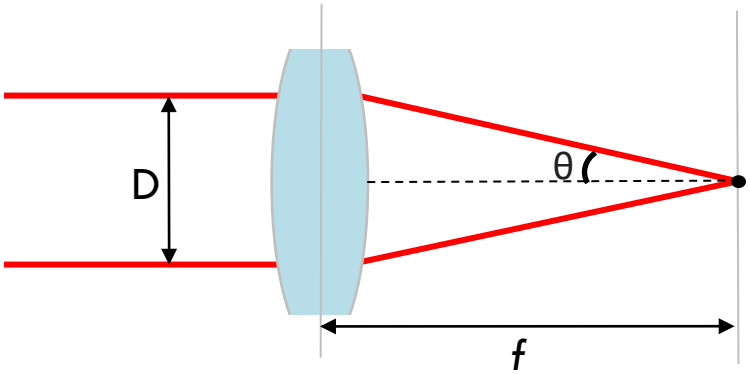
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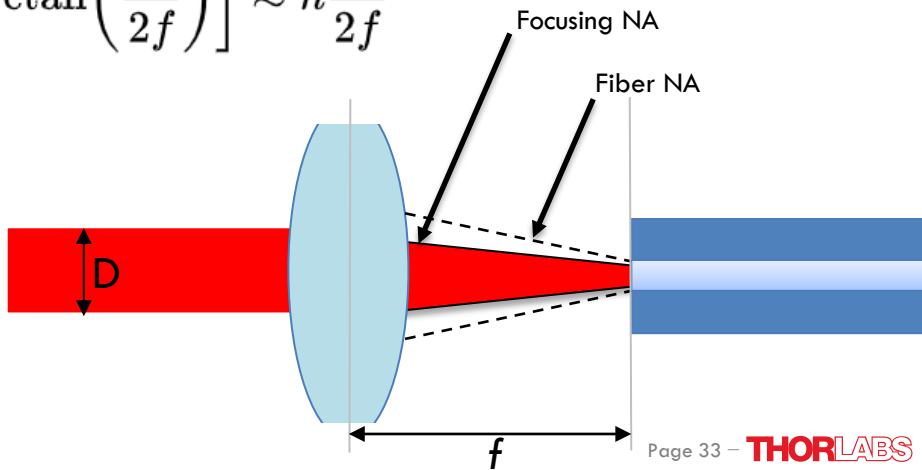


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$$\mathrm{NA} = n \sin\theta = n \sin\left[\arctan\left(\frac{D}{2f}\right)\right] \approx n\frac{D}{2f}$$

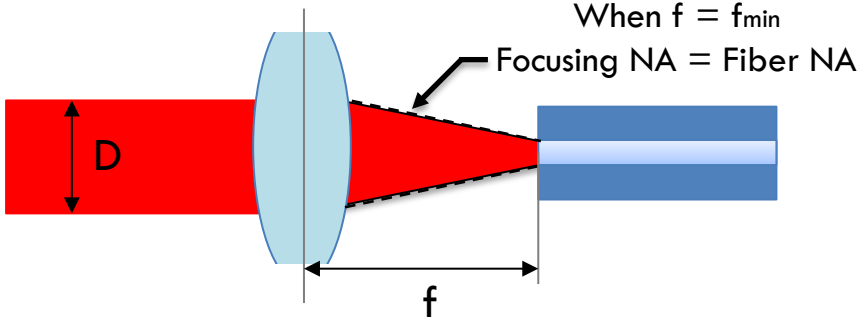




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$$\mathrm{NA} = n\sin\theta = n\sin\left[\arctan\left(\frac{D}{2f}\right)\right] \approx n\frac{D}{2f} \quad \Rightarrow \quad f_{min} \approx \frac{D}{2 \cdot NA_{fiber}}$$

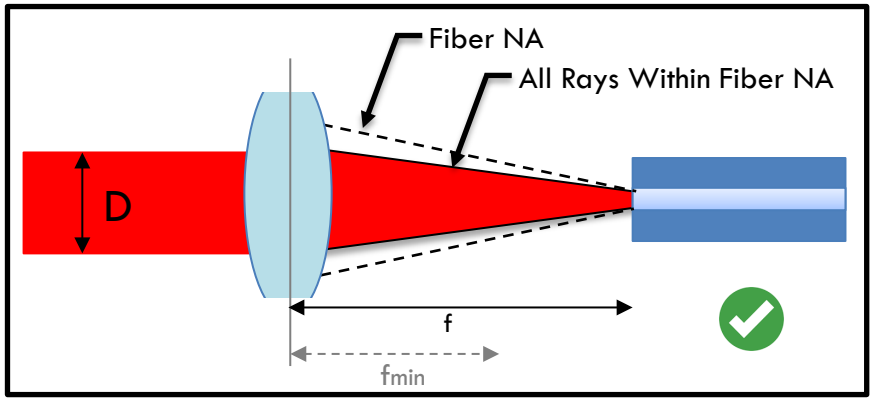


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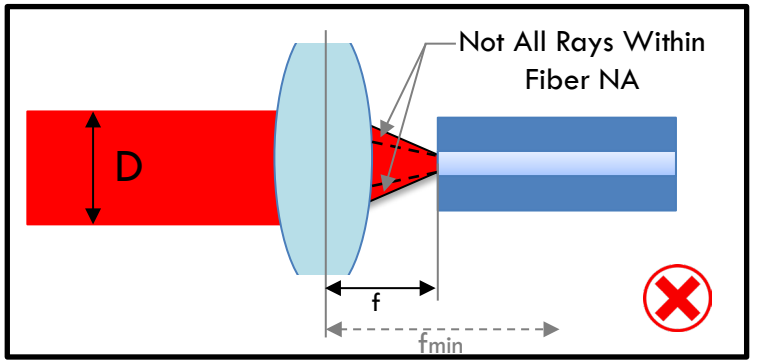
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$$f_{min} \approx \frac{D}{2 \cdot NA}$$

f > f_{min}



f < f_{min}



Theory: Multimode Fiber Coupling Efficiency

- For fiber-to-fiber mating, multimode fibers are quite forgiving
- Multimode fibers can easily mate from small to large cores and low NA to high NA fibers without loss in efficiency
- Coupling from larger to smaller cores and high NA to low NA introduces loss
- Since mode distribution can affect coupling dissimilar fibers, the below are estimates of losses based on flat top mode distribution:

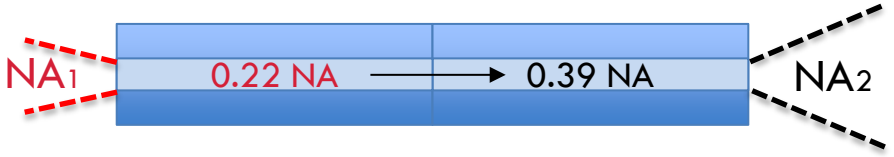
$d_1 < d_2$

All rays from a small core will be incident on a larger core fiber



$NA_1 < NA_2$

All rays from a lower NA fiber will be within the receiving fiber NA

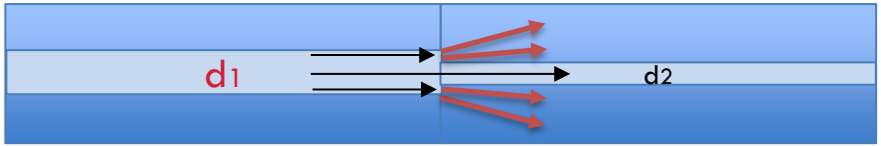


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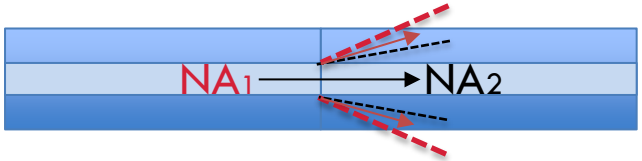
$d_1 > d_2$

Not all rays from a large core will be incident on a smaller core fiber



$NA_1 > NA_2$

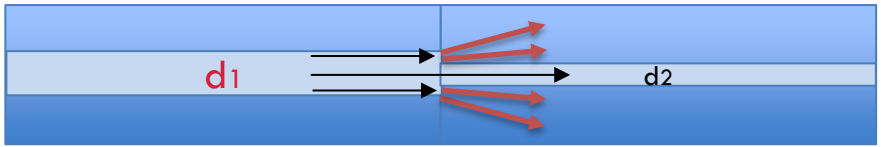
Not all rays from a high NA fiber will be within the receiving fiber NA



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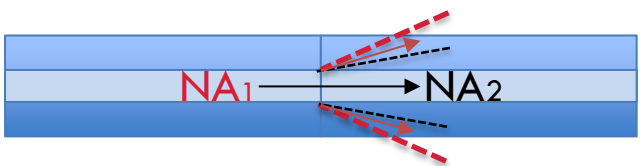
Core Mismatch Efficiency: $\eta = \frac{d_2{}^2}{d_1{}^2} \; for \; d_1 > d_2$



d_1 = 200um
d_2 = 100um

$$\eta = \frac{(100\mu m)^2}{(200\mu m)^2} = 25\%$$

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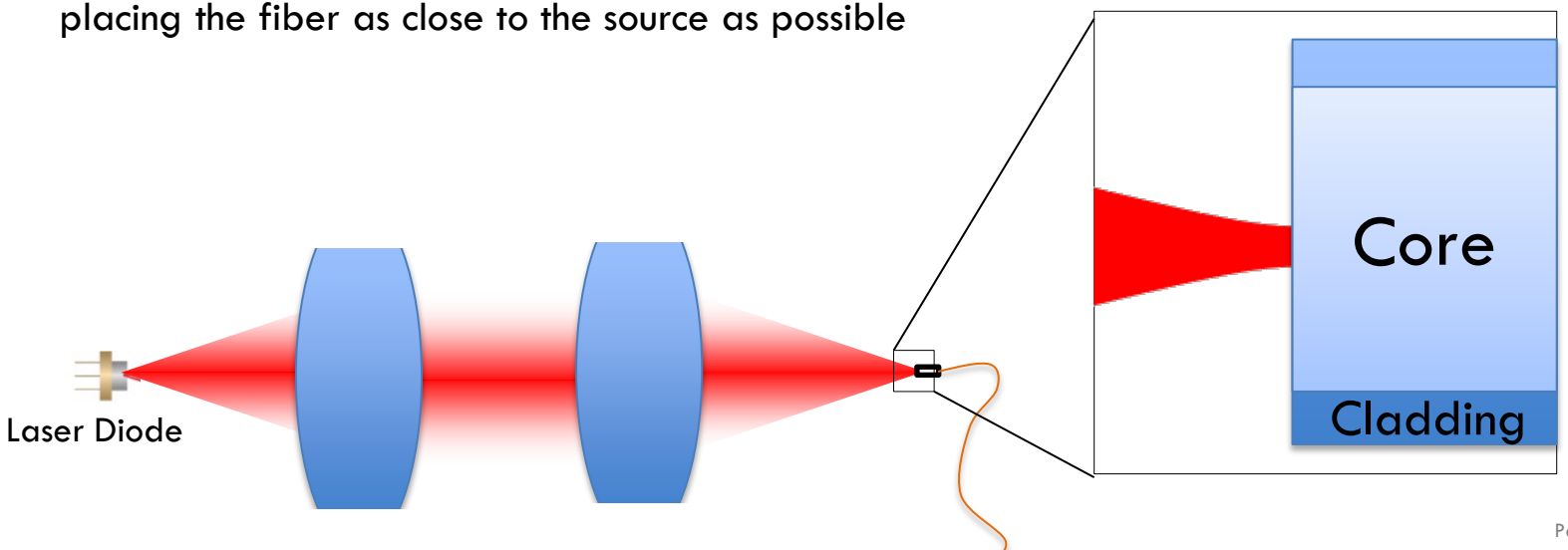


NA_1 = 0.39
NA_2 = 0.22

$$\eta = \frac{(0.22)^2}{(0.39)^2} \approx 32\%$$

Incoherent Source Coupling: LED/Lamps

- Coherent sources with Gaussian beams can focus to diffraction limited spots for coupling into single mode fiber or easily coupling into large core multimode fibers
- Incoherent sources (LEDs, lamps, etc.) are not point-like, do not produce Gaussian beams, and create images of their sources when focused, causing more difficult coupling
- Incoherent sources typically have very broad emission angles, making free-space coupling impractical
- The most efficient method of coupling large incoherent sources is by butt-coupling to multimode fibers; placing the fiber as close to the source as possible



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Image of a bulb filament formed when a lamp source is focused

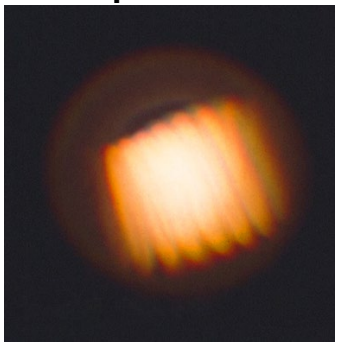
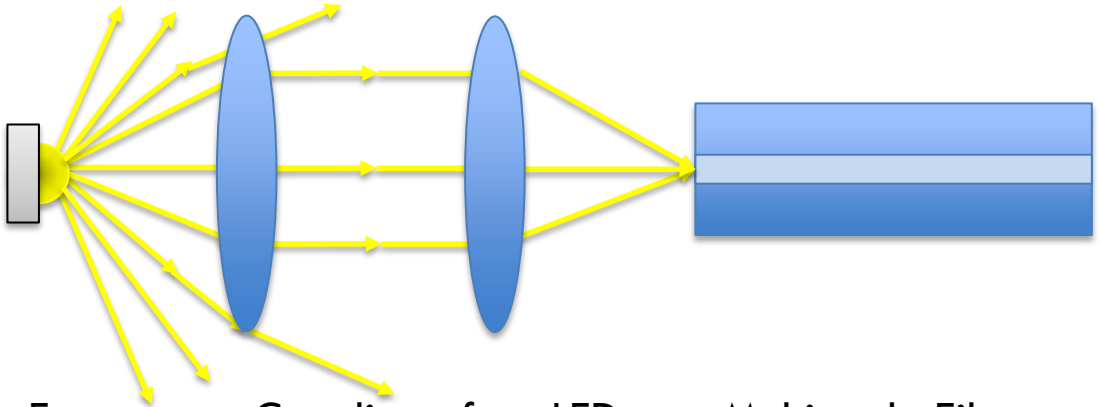


Image of square LED produced when focused



Incoherent Source Coupling: LED/Lamps

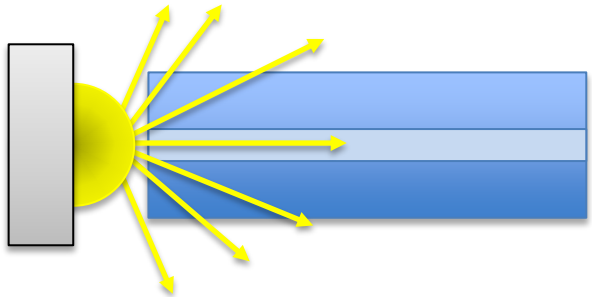
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Free-space Coupling of an LED to a Multimode Fiber

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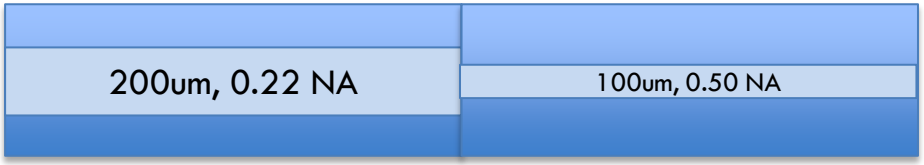


Butt-Coupling a Multimode Fiber to an Incoherent LED Source

Incoherent Source Coupling: Multimode Fiber Sources

- Coupling from a larger core to a smaller core will introduce losses when using a mating sleeve
- The optical invariance of a source allows altering its height for divergence or convergence angle since the emitter size and divergence create a fixed product
- When the smaller core receiving fiber has a larger NA than the larger core emitting fiber, optical elements can be used to achieve higher efficiency by demagnifying the emitting core diameter

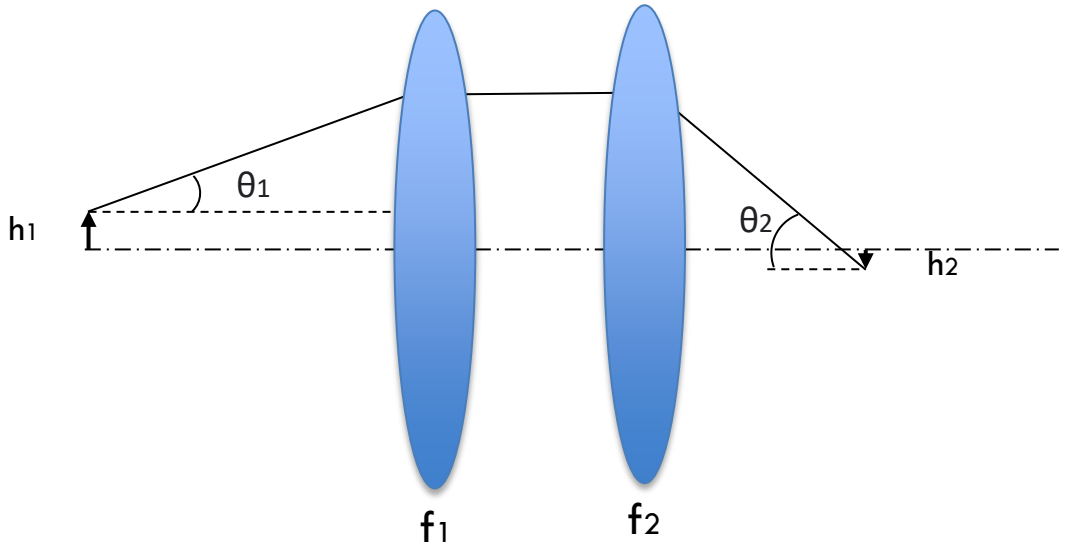
Core Mismatch



$$\eta = \frac{(100\ \mu m)^2}{(200\ \mu m)^2} = 25\%$$

Incoherent Source Coupling: Multimode Fiber Sources

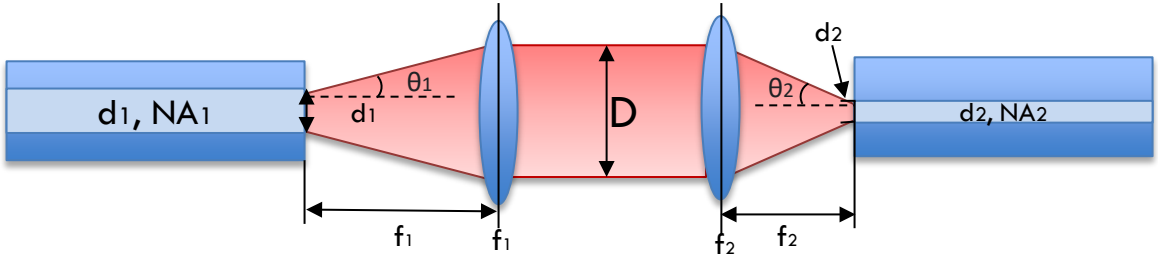
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$$h_1 \cdot \theta_1 = h_2 \cdot \theta_2$$

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$$d_1 > d_2 \qquad NA_1 < NA_2$$

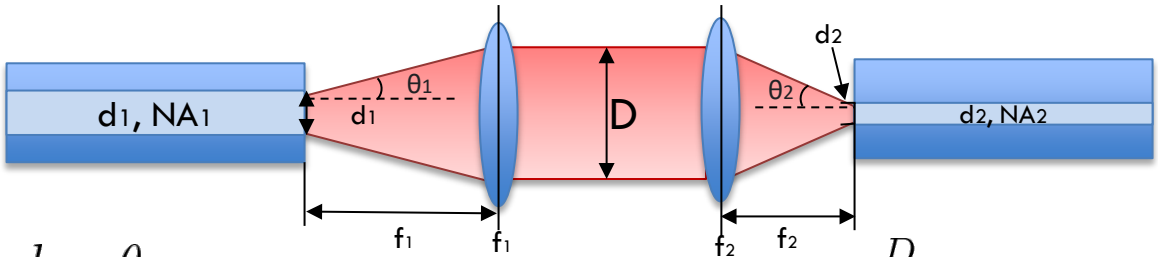
$$h_1 \cdot \theta_1 = h_2 \cdot \theta_2$$

$$h_1 = \frac{d_1}{2} \qquad h_2 = \frac{d_2}{2}$$

How should you choose the correct focal lengths?

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$$\frac{\theta_2}{\theta_1} = \frac{h_1}{h_2} = \frac{\frac{d_1}{2}}{\frac{d_2}{2}} = \frac{d_1}{d_2}$$

$$\theta_2 = \theta_1 \frac{d_1}{d_2}$$

$$NA = sin(\theta) \approx \frac{D}{2f} \Rightarrow D = 2f sin(\theta)$$

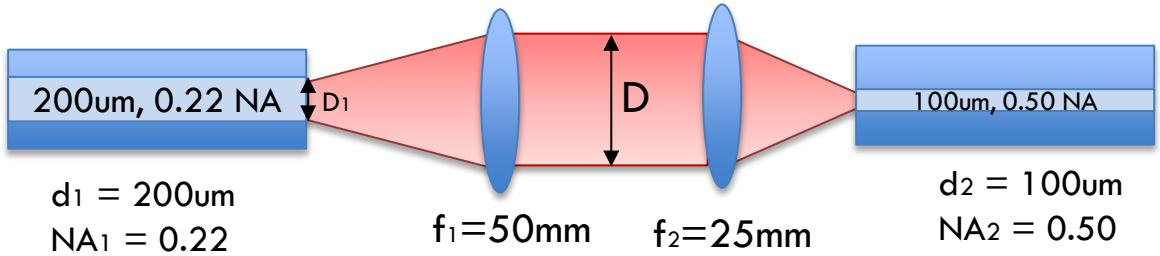
$$D = 2 \cdot f_1 \cdot sin(\theta_1) = 2 \cdot f_2 \cdot sin(\theta_2)$$

$$f_1 = f_2 \frac{sin(\theta_2)}{sin(\theta_1)} = f_2 \frac{sin(\frac{d_1}{d_2}\theta_1)}{sin(\theta_1)} \approx \frac{d_1}{d_2} f_2 \quad for\ small\ \theta_1$$

$$f_1 \approx \frac{d_1}{d_2} f_2 \quad for\ small\ NA_1$$

Incoherent Source Coupling: Multimode Fiber Sources

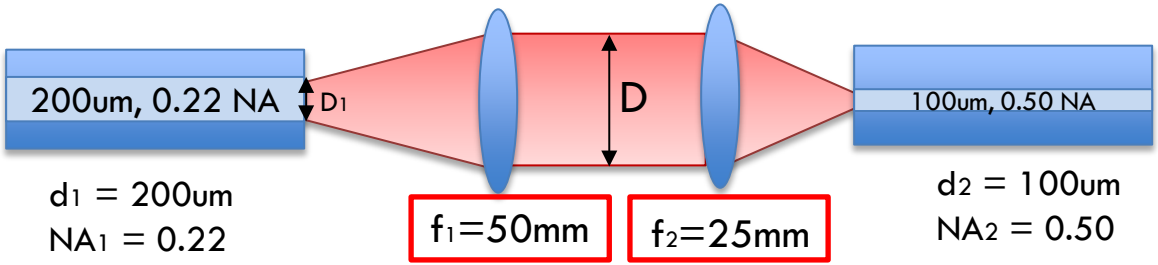
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To satisfy all light rays are incident on the core, choose the required lenses:

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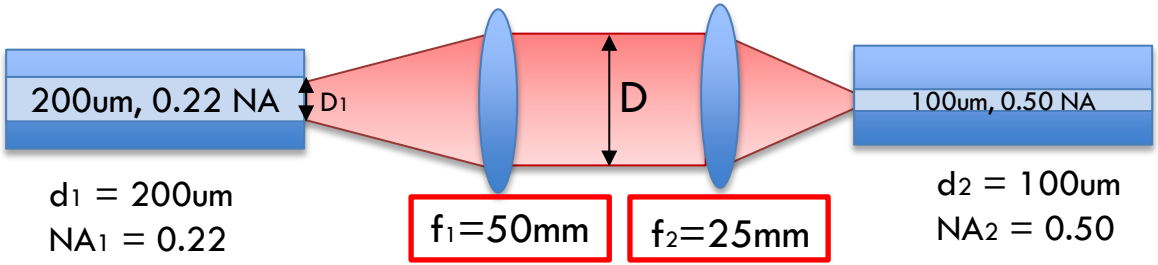


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$$f_1 \approx \frac{d_1}{d_2} f_2 = \frac{200\mu m}{100\mu m} f_2 = 2f_2$$

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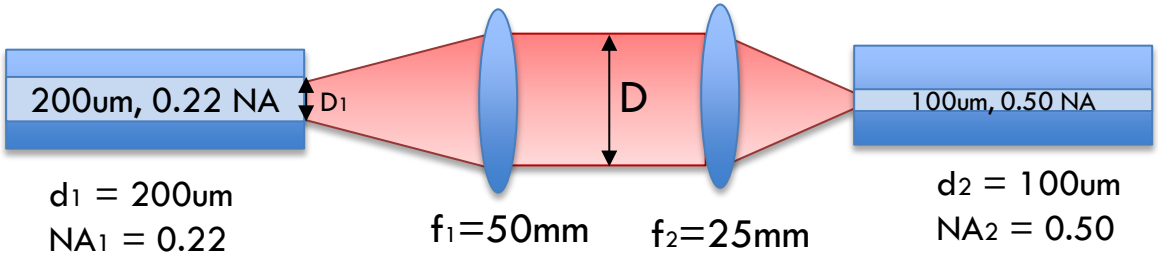


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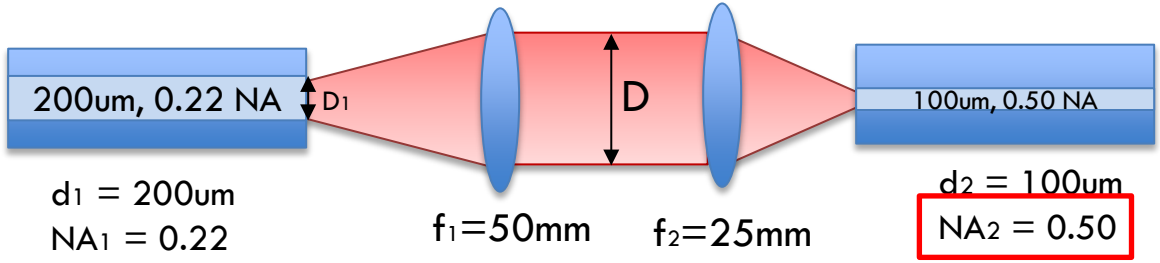
To satisfy all light rays are incident on the core, choose the required lenses:

$$f_1 \approx \frac{d_1}{d_2} f_2 = \frac{200\mu m}{100\mu m} f_2 = 2f_2$$

Verify the focusing NA does not exceed NA_2:

Incoherent Source Coupling: Multimode Fiber Sources

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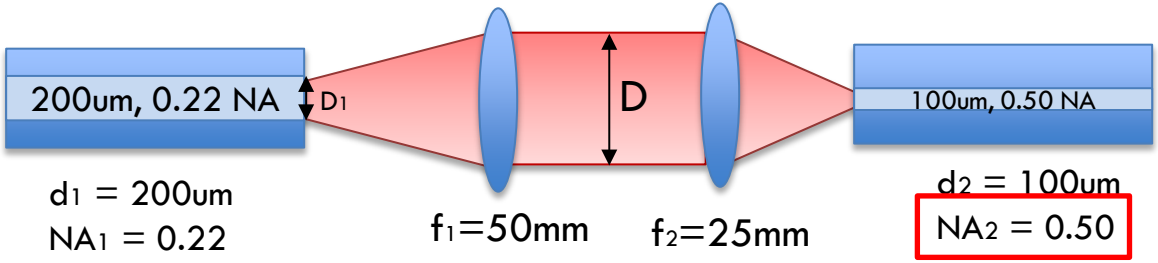
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Verify the focusing NA does not exceed NA2:

$$D = 2 \cdot f \cdot NA \qquad NA_{focus} = \frac{D}{2f_2} = \frac{2\ f_1\ NA_1}{2f_2} = \frac{100mm \cdot NA_1}{50mm} = 2 \cdot NA_1 = 0.44 < NA_2$$

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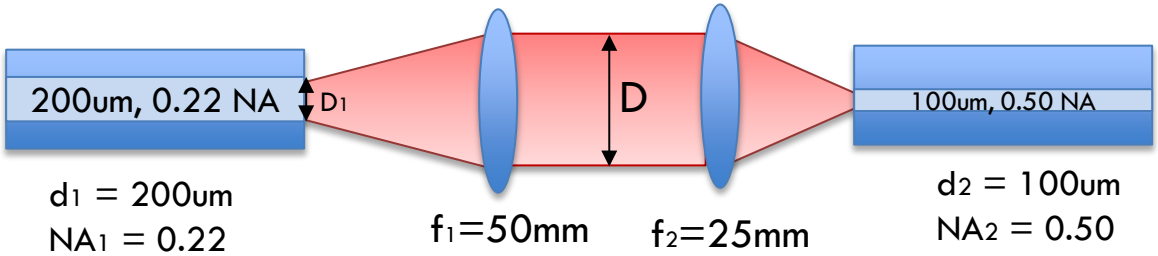
$$\eta = 100\%$$

Verify the focusing NA does not exceed NA_2:

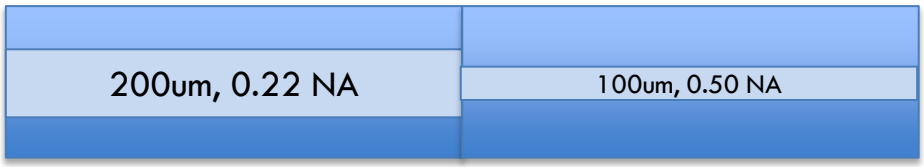
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$$\eta = 100\%$$



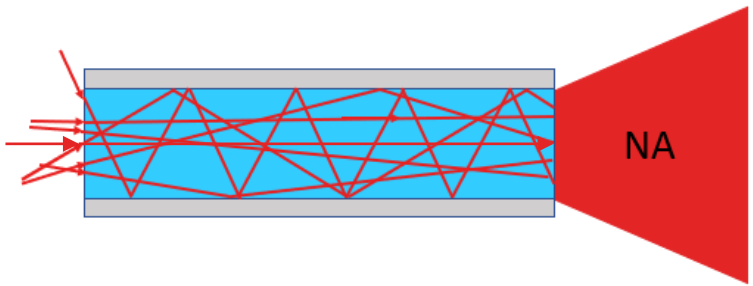
$$\eta = \frac{100um^2}{200um^2} = 25\%$$

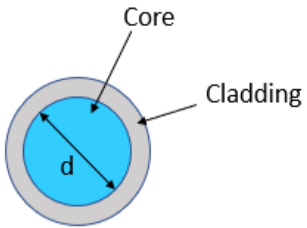
Single Mode Fibers

Fiber Optic Concepts: Single Mode Fibers

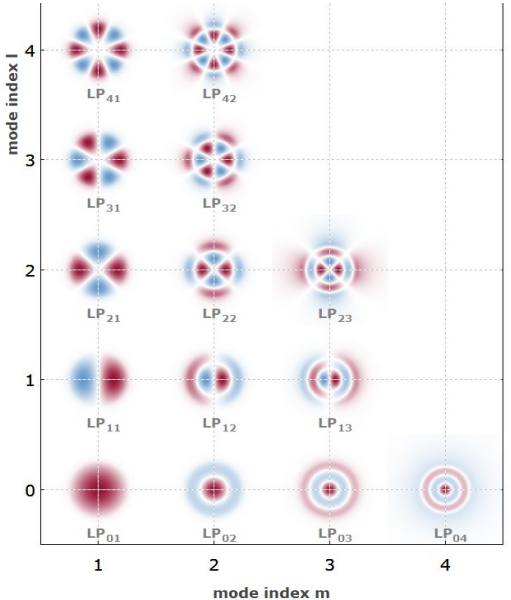
- The number of propagating modes which depend on the core diameter and the core/cladding index
- Single mode fibers have smaller core diameters, which limit propagation to only the fundamental LP01 mode
- The indices of the core and cladding can be designed to have a preferred core diameter to ensure only the propagation of the LP01 mode

$$Number\ of\ Modes \approx \frac{\pi^2 d^2 (NA)^2}{2\lambda^2} = \frac{\pi^2 d^2 (n_{core}^2 - n_{clad}^2)}{2\lambda^2}$$





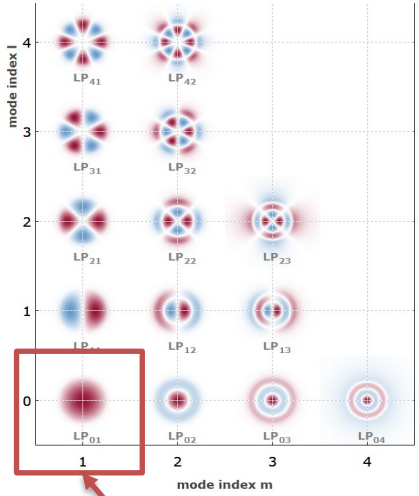
LP Modes



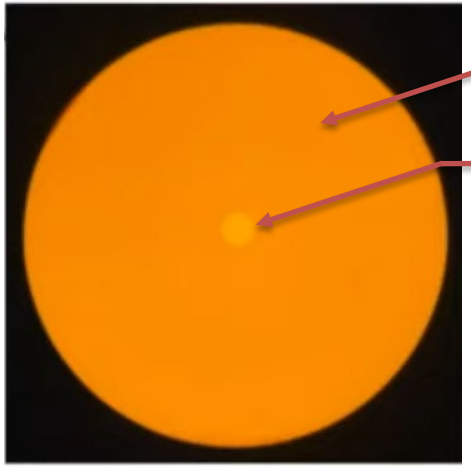
Fiber Optic Concepts: Single Mode Fibers

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LP Modes



LP01 Fundamental Mode

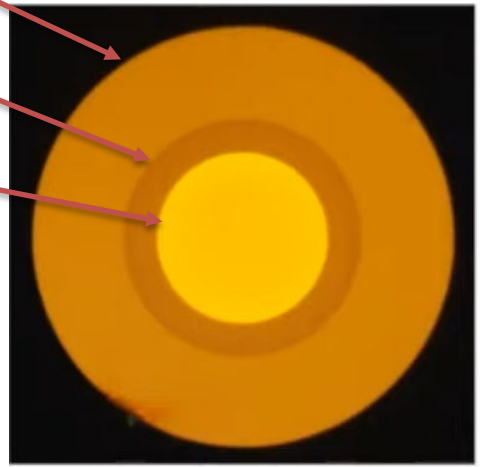


Single Mode Fiber, 9µm Core

Coating

Cladding

Core



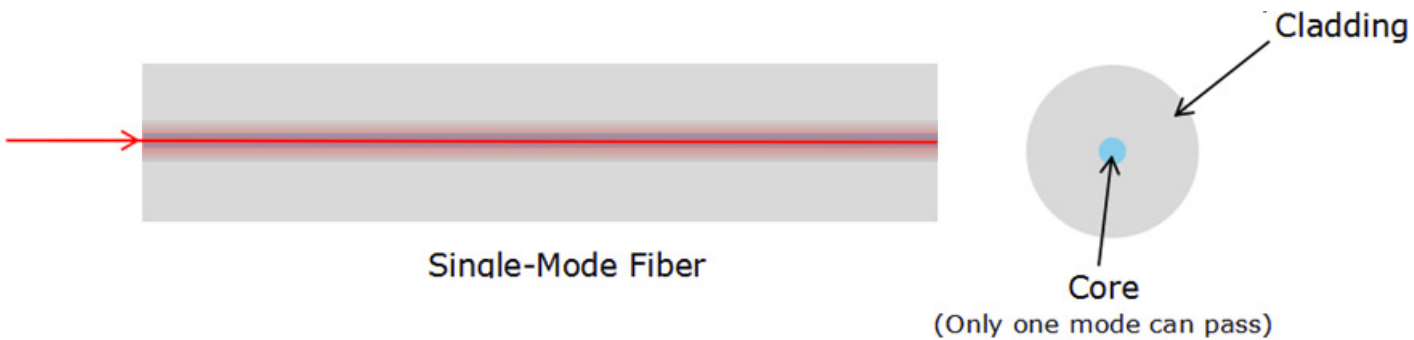
Multimode Fiber, 50µm Core

*False color has been applied to improve core/cladding contrast in images

Fiber Optic Concepts: Single Mode Fibers

- The number of propagating modes which depend on the core diameter and the core/cladding index
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- The indices of the core and cladding can be designed to have a preferred core diameter to ensure only the propagation of the LP01 mode

For single mode operation: $$Core\ Diameter < \frac{2.4\lambda}{\pi\sqrt{n_{core}^2 - n_{clad}^2}}$$

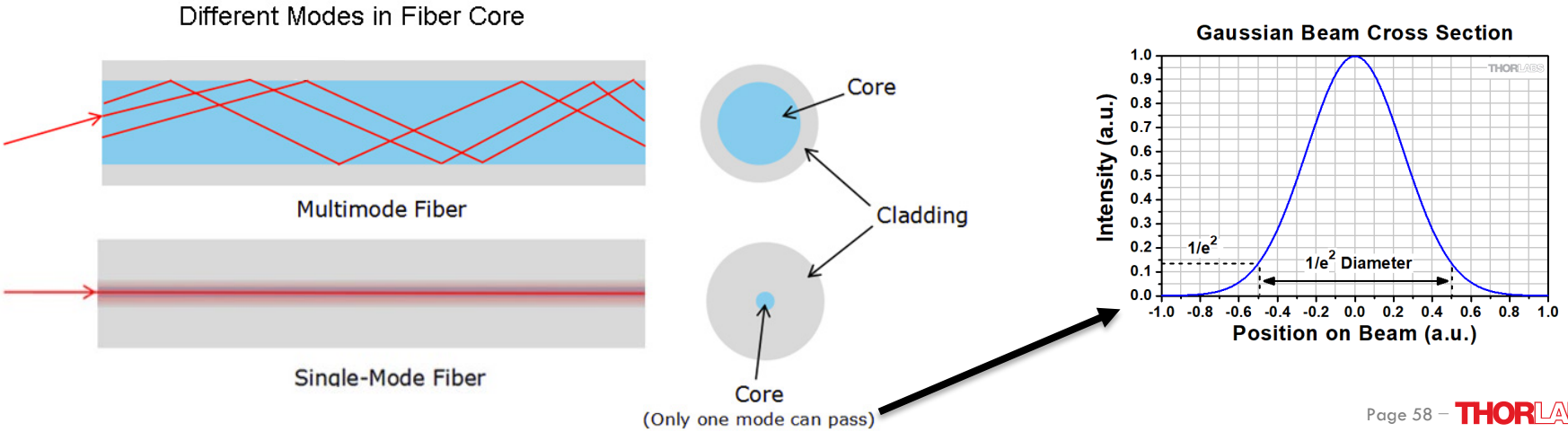


https://community.fs.com/blog/how-to-realize-multimode-to-single-mode-fiber-conversion.html

Hecht, J. (2015). Types of Optical Fibers. In *Understanding Fiber Optics* (pp. 78–79). essay, Laser Light Press.

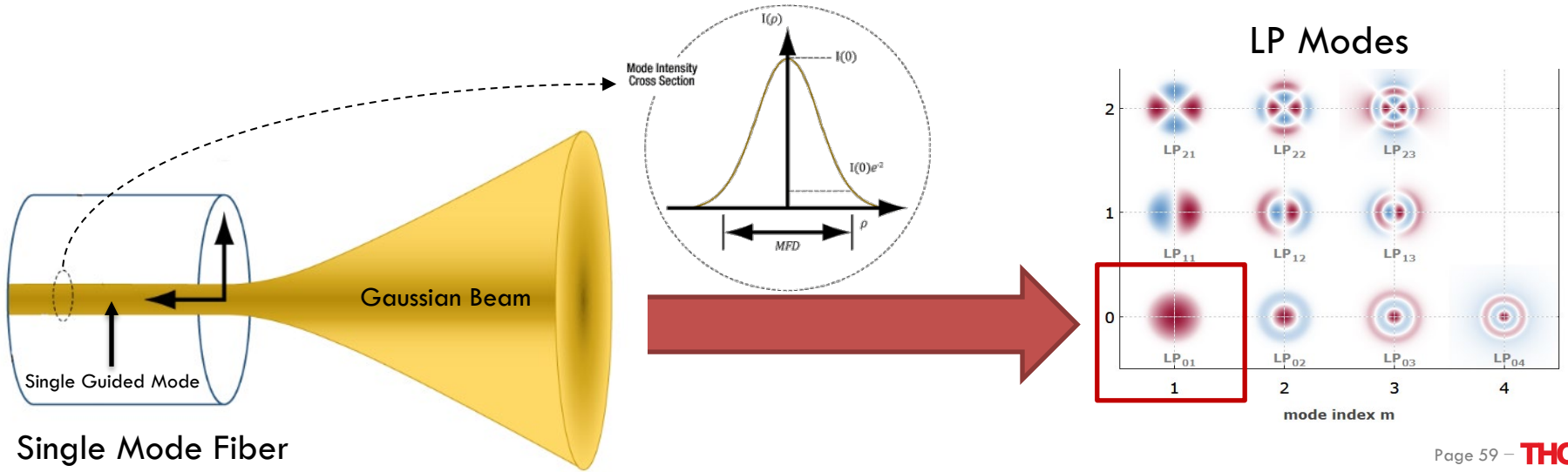
Fiber Optic Concepts: Single Mode Fibers

- While multimode fibers are often described using the ray model of light, single mode fibers are best described by the wave model of light due to their unique properties:
  - Single mode fibers only propagate a single spatial mode; the fundamental mode (LP01)
  - The LP01 mode is larger than the core and extends into the cladding based on the fiber's Mode Field Diameter (MFD)
  - The LP01 mode closely approximates a Gaussian beam and is often treated interchangeably in optical calculations regarding the beam diameter
  - The Mode Field Diameter (MFD), which also serves as the $1/e^2$ diameter of Gaussian beams, best defines the diameter of the guided mode rather than the core diameter of the fiber



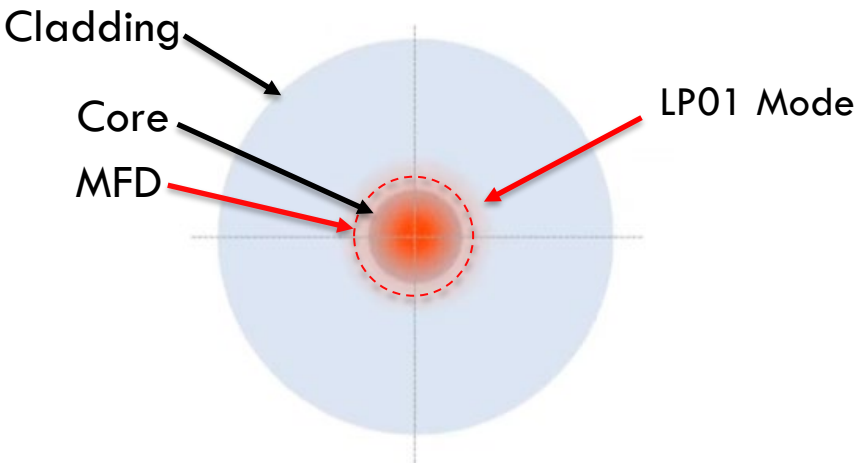
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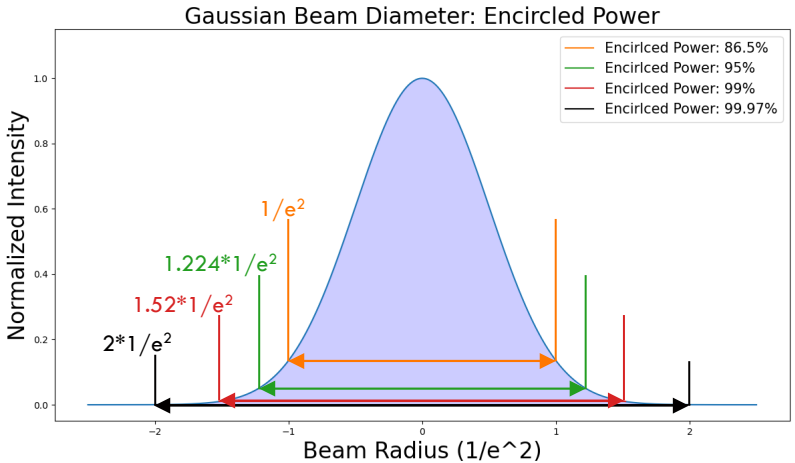


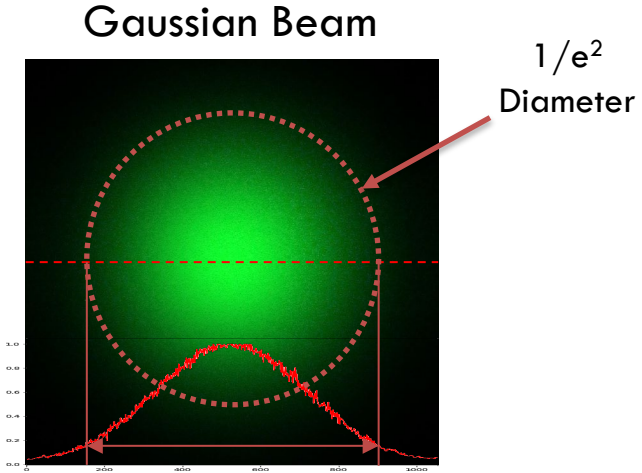
Fiber: SMF-28 Ultra
Core Diameter: 8.2 µm
MFD: 10.4 um @ 1550 nm

Power in MFD: 86.5%
Power in Core: 69.8%
Power in Cladding: 30.2%!!!

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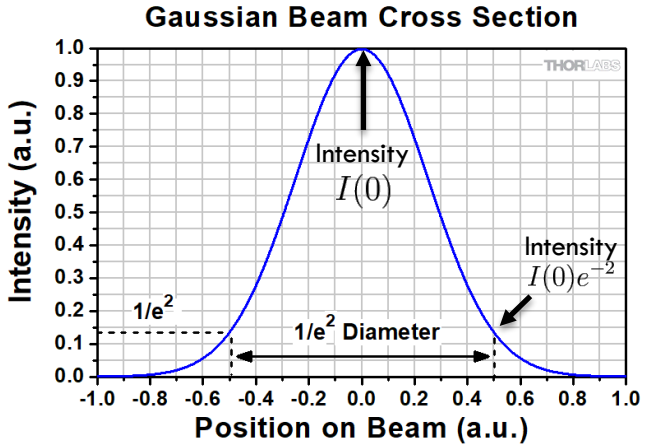


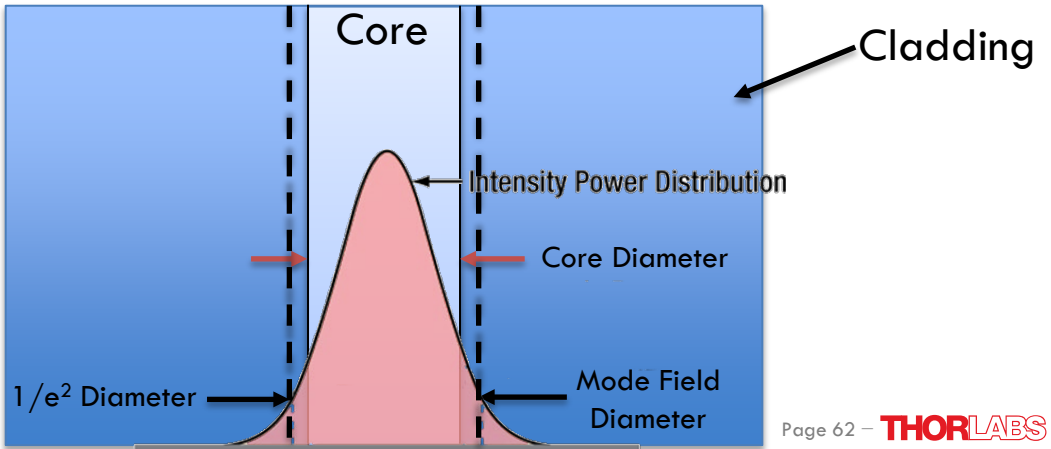


https://en.wikipedia.org/wiki/Gaussian_beam#/media/File:Green_laser_pointer_TEM00_profile.JPG

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Fiber Optic Concepts: Single Mode Fibers – Numerical Aperture

- While single mode fibers have an NA specification, this is the calculated value from the core and cladding indices and can't be treated the same as the ray model
- The constant NA ray model leads to inaccuracies in calculations since the LP01 fiber mode and Gaussian beams do not have constant angular divergence
- The Gaussian beams used for launching into single mode fibers have a flat wavefront at focus and are non-diverging when incident on the fiber
- Since the guided mode has a flat wavefront, is non-diverging in the fiber, and extends into the cladding, the total internal reflection at the core/cladding boundary does not hold, single mode fibers cannot be treated by the ray model

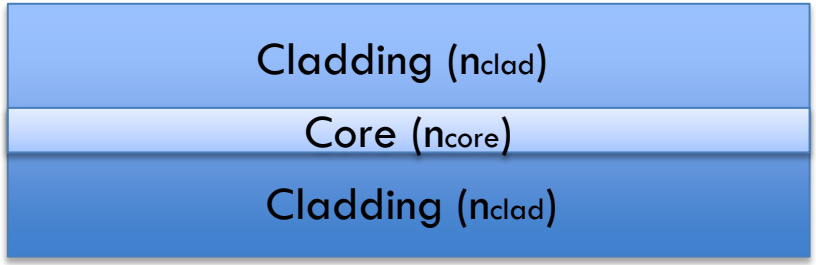
Single Mode Fiber: 400 to 680 nm

Overview | Specs | MFD Definition | Damage Threshold | Feedback

Item #	Operating Wavelength	Mode Field Diameter	Cladding	Coatin
S405-XP	400 - 680 nm	3.3 ± 0.5 µm @ 405 nm 4.6 ± 0.5 µm @ 630 nm	125.0 ± 1.0 µm	245.0 ± 15
SM400	405 - 532 nm	2.5 - 3.4 µm @ 480 nm	125 ± 1 µm	245 ± 15
SM450[a]	488 - 633 nm	2.8 - 4.1 µm[b] @ 488 nm	125 ± 1.0 µm	245 ± 15
460HP	450 - 600 nm	3.5 ± 0.5 µm @ 515 nm	125 ± 1 µm	245 ± 15

Item #	Short-Term Bend Radius	Long-Term Bend Radius	Attenuation (Max)	NA
S405-XP	≥6 mm	≥13 mm	≤30.0 dB/km @ 630 nm ≤30.0 dB/km @ 488 nm	0.12
SM400	≥10 mm	≥30 mm	≤50 dB/km @ 430 nm ≤30 dB/km @ 532 nm	0.12 - 0.14
SM450	≥5 mm	≥25 mm	≤50 dB/km @ 488 nm[c]	0.10 - 0.14
460HP	≥6 mm	≥13 mm	≤30 dB/km @ 515 nm	0.13

$$NA = \sqrt{n_{core}^2 - n_{clad}^2}$$



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Fiber: SM980-5.8-125
λ: 980 nm
NA: 0.13
MFD: 6.4 µm

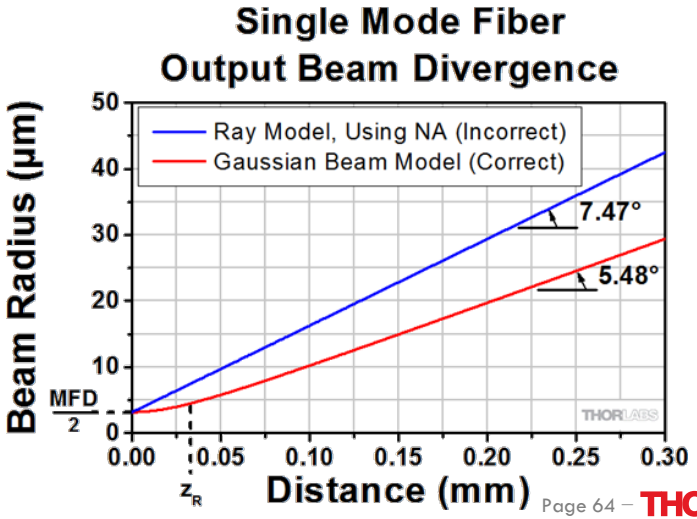
Ray Model

$$NA = sin(\theta) = \sqrt{n_{core}^2 - n_{clad}^2}$$

Gaussian Beam Model

Gaussian Rayleigh Range: $z_R = \frac{\pi}{\lambda}(\frac{MFD}{2})^2$

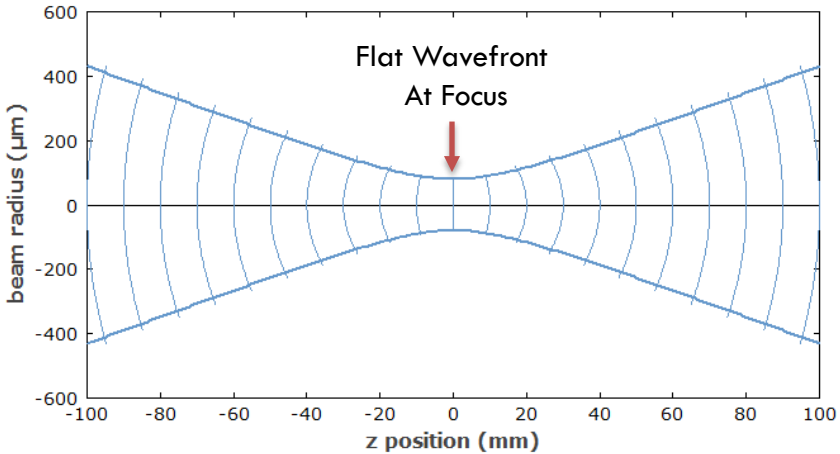
Beam Radius at Distance z: $w(z) = \frac{MFD}{2}\sqrt{1 + \left(\frac{z}{z_R}\right)^2}$



Fiber Optic Concepts: Single Mode Fibers – Numerical Aperture

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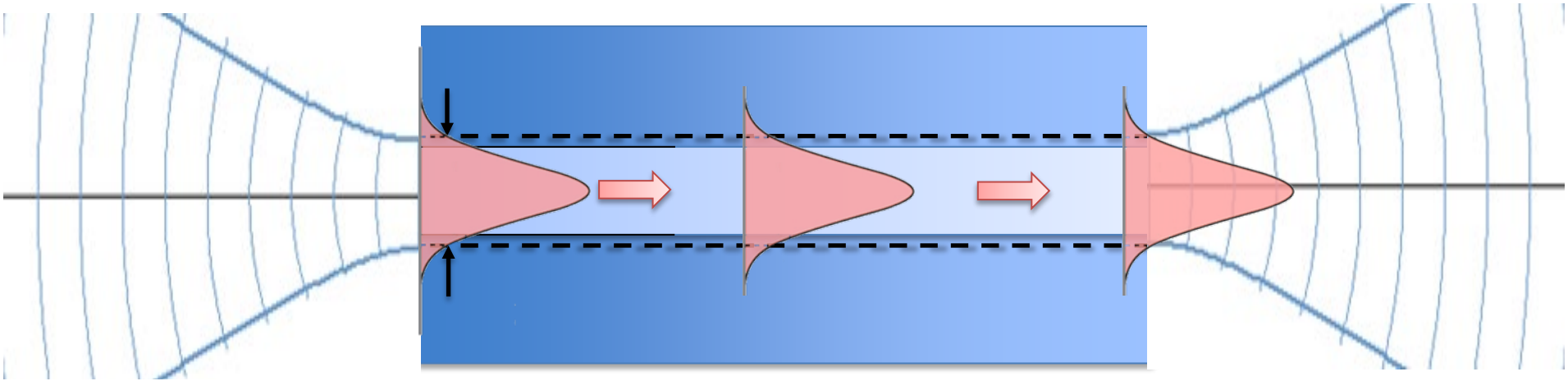
Focused Gaussian Beam Diameter and Wavefront



https://www.rp-photonics.com/gaussian_beams.html

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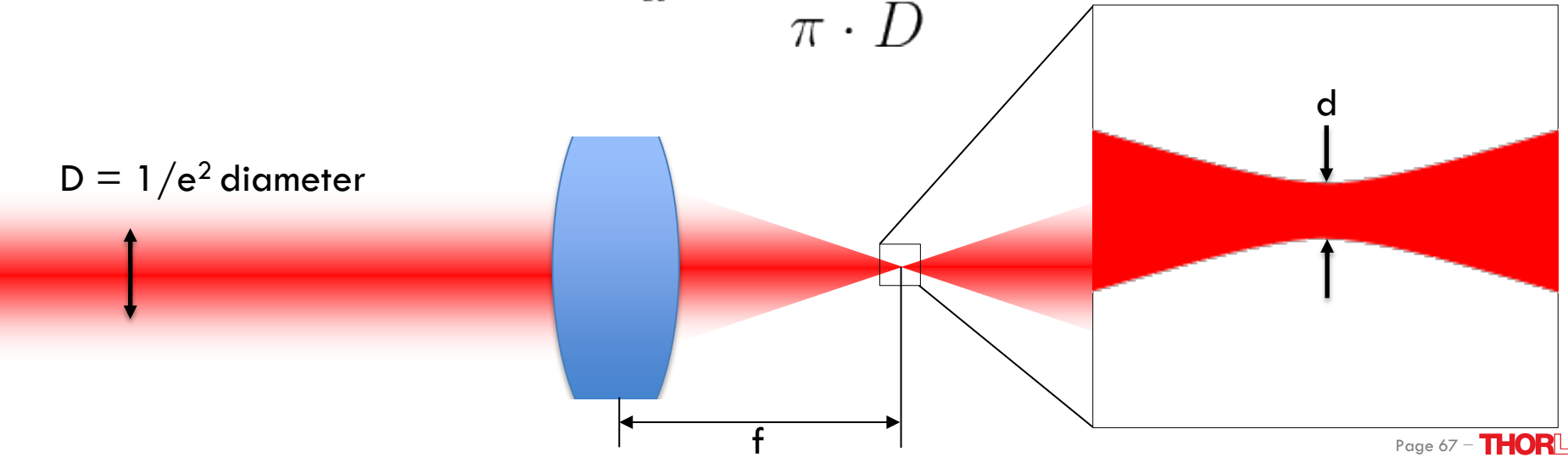


https://www.rp-photonics.com/gaussian_beams.html

Theory: Focusing Gaussian Beams

- Collimated Gaussian beams can focus to diffraction limited spot sizes when using the proper optics
- For a given collimated $1/e^2$ beam diameter (D), wavelength (λ), and focal length (f), the diffraction limited spot size (d) of the focused Gaussian beam is calculated by the following:

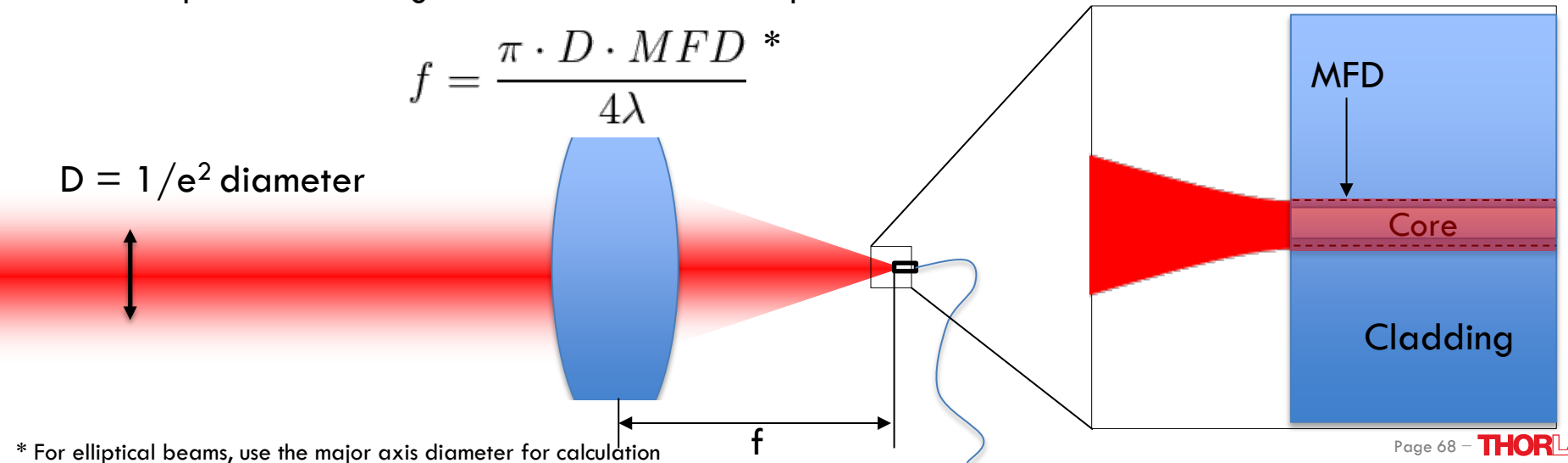
$$d = \frac{4 \cdot \lambda \cdot f}{\pi \cdot D}$$



Theory: Coupling into Single Mode Fibers

- Since the fundamental mode is Gaussian, single mode fibers will only efficiently couple light from Gaussian sources
- MFDs are designed for the diffraction limited spot sizes of Gaussian beams
- Gaussian beams provide the smallest diffraction limited spot sizes when focused, making it inefficient to focus other sources to the required MFD spot diameters
- For a given incident collimated $1/e^2$ beam diameter (D), wavelength (λ), and fiber MFD, there is a specific focal length to ensure the focused spot size matches the MFD:

$$f = \frac{\pi \cdot D \cdot MFD}{4\lambda} *$$



* For elliptical beams, use the major axis diameter for calculation

Theory: Coupling into Single Mode Fibers - Example

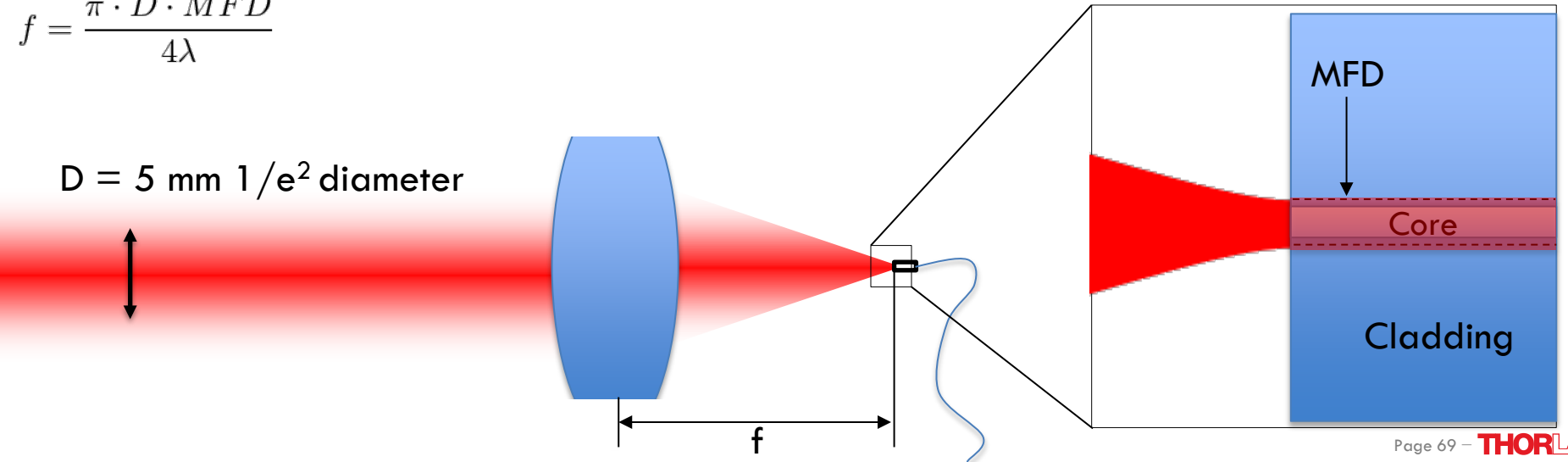
Item #	Operating Wavelength	Mode Field Diameter
SM600	633 -780 nm[a]	3.6 - 5.3 µm @ 633 nm[b]

$$\lambda = 633\ nm$$

$$D = 5\ mm$$

$$MFD_{ave} = \frac{3.6\ \mu m + 5.3\ \mu m}{2} = 4.45\ \mu m$$

$$f = \frac{\pi \cdot D \cdot MFD}{4\lambda}$$



Theory: Coupling into Single Mode Fibers - Example

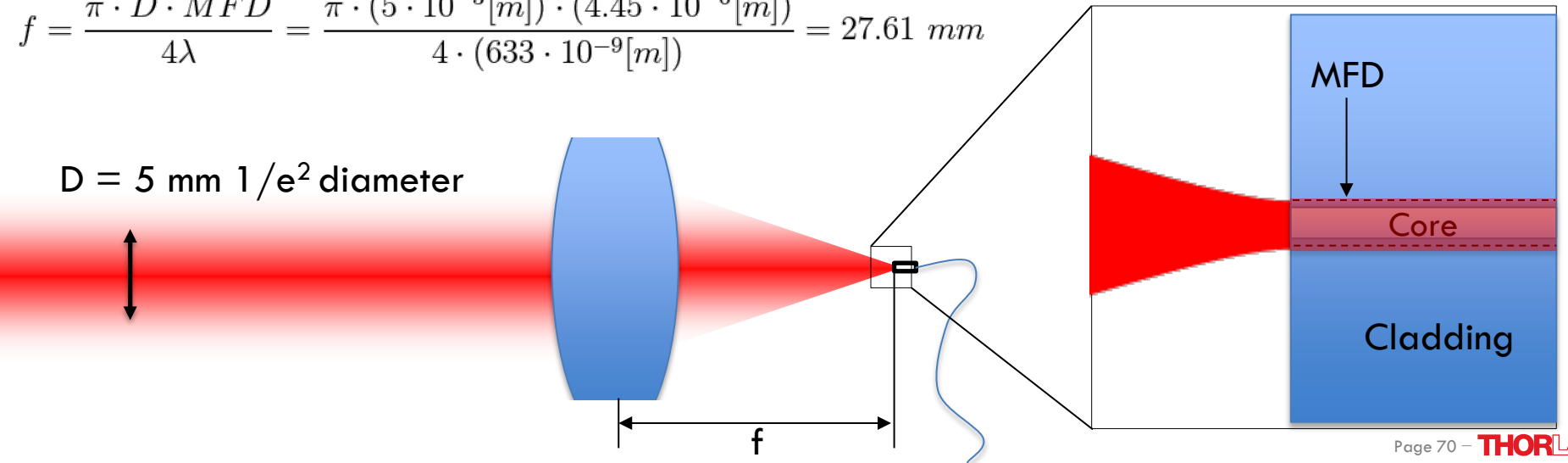
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$$f = \frac{\pi \cdot D \cdot MFD}{4\lambda} = \frac{\pi \cdot (5 \cdot 10^{-3}[m]) \cdot (4.45 \cdot 10^{-6}[m])}{4 \cdot (633 \cdot 10^{-9}[m])} = 27.61\ mm$$



Theory: Single Mode Coupling Efficiency – Mating Sleeve

- Single Mode fibers will have decreased efficiency with dissimilar MFDs, regardless of propagation direction
  - Mode Field Mismatch Efficiency:

$$\eta = \frac{4 \cdot MFD_1^2 \cdot MFD_2^2}{(MFD_1^2 + MFD_2^2)^2}$$

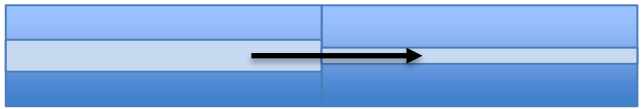


MFD_1 MFD_2

Example:



$MFD_1 = 4\ \mu m$ $MFD_2 = 6\ \mu m$



$MFD_1 = 6\ \mu m$ $MFD_2 = 4\ \mu m$

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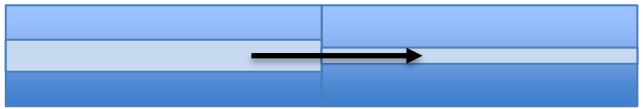
MFD1 MFD2

Example:



$MFD_1 = 4\ \mu m$ $MFD_2 = 6\ \mu m$

$$\eta = \frac{4 \cdot MFD_1^2 \cdot MFD_2^2}{(MFD_1^2 + MFD_2^2)^2} = \frac{4 \cdot (4\ \mu m)^2 \cdot (6\ \mu m)^2}{\left((4\ \mu m)^2 + (6\ \mu m)^2\right)^2} \approx 85.2\%$$



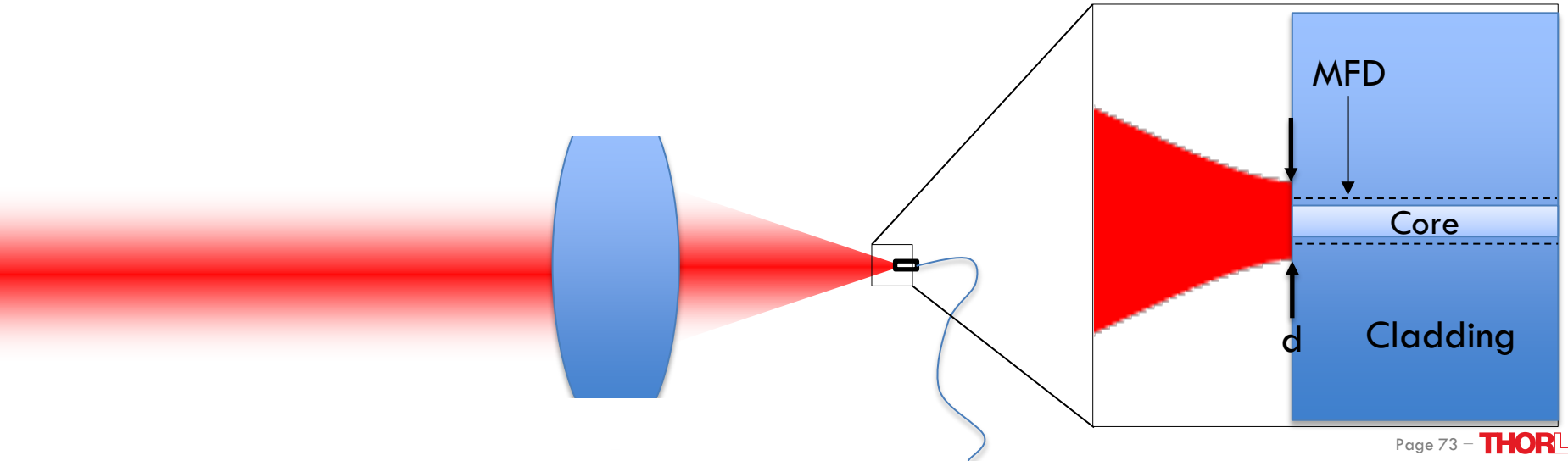
$MFD_1 = 6\ \mu m$ $MFD_2 = 4\ \mu m$

$$\eta = \frac{4 \cdot MFD_1^2 \cdot MFD_2^2}{(MFD_1^2 + MFD_2^2)^2} = \frac{4 \cdot (6\ \mu m)^2 \cdot (4\ \mu m)^2}{\left((6\ \mu m)^2 + (4\ \mu m)^2\right)^2} \approx 85.2\%$$

Theory: Single Mode Coupling Efficiency - Freespace

- When focusing a collimated Gaussian beam, the focused $1/e^2$ spot diameter (d) can substitute for MFD_1 for efficiency calculations:

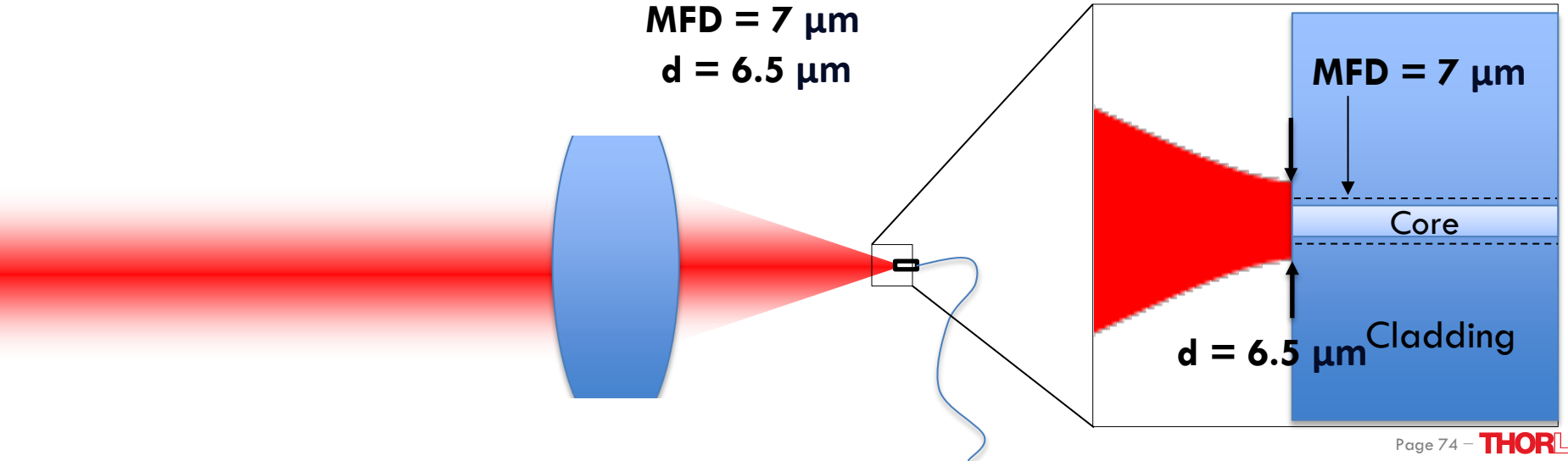
$$\eta = \frac{4 \cdot d^2 \cdot MFD^2}{(d^2 + MFD^2)^2}$$



Theory: Single Mode Coupling Efficiency - Freespace Example

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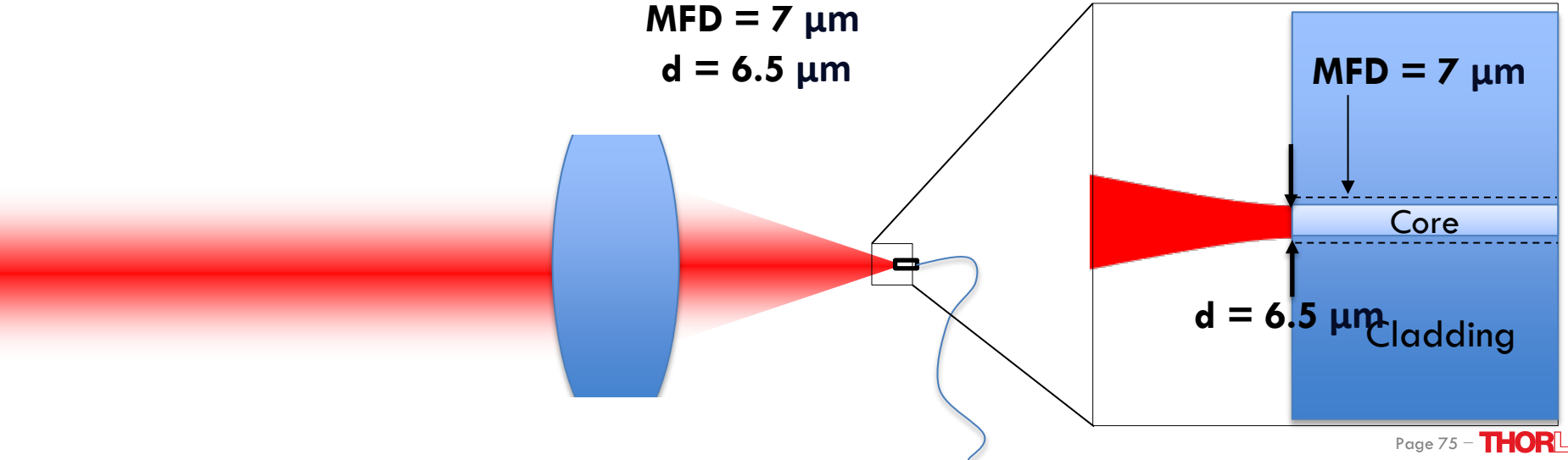
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Theory: Single Mode Coupling Efficiency - Freespace Example

- When focusing a collimated Gaussian beam, the focused $1/e^2$ spot diameter (d) can substitute for MFD_1 for efficiency calculations:

$$\eta = \frac{4 \cdot d^2 \cdot MFD^2}{(d^2 + MFD^2)^2} = \frac{4 \cdot (6.5\ \mu m)^2 \cdot (7\ \mu m)^2}{\left((6.5\ \mu m)^2 + (7\ \mu m)^2\right)^2} \approx 99.5\%$$



Single Mode Coupling Efficiency - MFD vs NA

- It was previously stated the NA should only be used for the ray like behavior of multimode fiber and not the wavelike behavior of single mode
- An ideal focal length can be calculated using the diffraction limited spot equation of Gaussian beams as well as the ray equation of multimode fibers to compare efficiency

Fiber: SM980-5.8-125
λ: 980 nm
NA: 0.13
MFD: 6.4 µm
D: 1.5 mm

Ray Model

$$f_{min} = \frac{D}{2 \cdot NA_{fiber}} = \frac{1.5\ mm}{2 \cdot 0.13} \approx \boxed{5.77mm}$$

$$d = \frac{4 \cdot \lambda \cdot f}{\pi \cdot D} = \frac{4 \cdot (980 \cdot 10^{-9}\ m) \cdot (5.77\ \cdot 10^{-3}\ m)}{\pi \cdot (1.5 \cdot 10^{-3}\ m)} = 4.80\,\mu m$$

Gaussian Beam Model

$$f = \frac{\pi \cdot D \cdot MFD}{4\lambda} = \frac{\pi \cdot (1.5 \cdot 10^{-3}\ m)) \cdot (6.4 \cdot 10^{-6}\ m)}{4 \cdot (980 \cdot 10^{-9}\ m)} = \boxed{7.69\ mm}$$

$$d = \frac{4 \cdot \lambda \cdot f}{\pi \cdot D} = \frac{4 \cdot (980 \cdot 10^{-9}\ m) \cdot (7.69\ \cdot 10^{-3}\ m)}{\pi \cdot (1.5 \cdot 10^{-3}\ m)} = 6.40\,\mu m$$

Single Mode Coupling Efficiency - MFD vs NA

- It was previously stated the NA should only be used for the ray like behavior of multimode fiber and not the wavelike behavior of single mode
- An ideal focal length can be calculated using the diffraction limited spot equation of Gaussian beams as well as the ray equation of multimode fibers to compare efficiency

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MFD: 6.4 µm
D: 1.5 mm

Ray Model

$$f_{min} = \frac{D}{2 \cdot NA_{fiber}} = \frac{1.5\ mm}{2 \cdot 0.13} \approx 5.77mm \qquad d = \frac{4 \cdot \lambda \cdot f}{\pi \cdot D} = \frac{4 \cdot (980 \cdot 10^{-9}\ m) \cdot (5.77\ \cdot 10^{-3}\ m)}{\pi \cdot (1.5 \cdot 10^{-3}\ m)} = 4.80\,\mu m$$

$$\eta = \frac{4 \cdot d^2 \cdot MFD^2}{(d^2 + MFD^2)^2} = \frac{4 \cdot (4.80\ \mu m)^2 \cdot (6.4\ \mu m)^2}{((4.80\ \mu m)^2 + (6.4\ \mu m)^2)^2} \approx 92.2\%$$



Gaussian Beam Model

$$f = \frac{\pi \cdot D \cdot MFD}{4\lambda} = \frac{\pi \cdot (1.5 \cdot 10^{-3}\ m)) \cdot (6.4 \cdot 10^{-6}\ m)}{4 \cdot (980 \cdot 10^{-9}\ m)} = 7.69\ mm$$

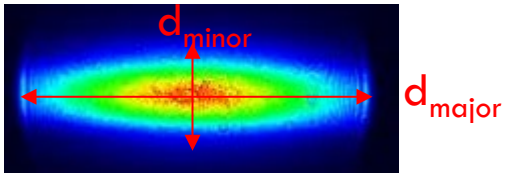
$$d = \frac{4 \cdot \lambda \cdot f}{\pi \cdot D} = \frac{4 \cdot (980 \cdot 10^{-9}\ m) \cdot (7.69\ \cdot 10^{-3}\ m)}{\pi \cdot (1.5 \cdot 10^{-3}\ m)} = 6.40\,\mu m$$

$$\eta = \frac{4 \cdot d^2 \cdot MFD^2}{(d^2 + MFD^2)^2} = \frac{4 \cdot (6.4\ \mu m)^2 \cdot (6.4\ \mu m)^2}{((6.4\ \mu m)^2 + (6.4\ \mu m)^2)^2} = 100\%$$

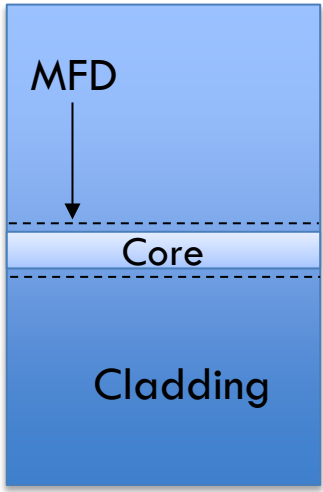


Theory: Single Mode Fiber Coupling Elliptical Spots

- Many laser diodes produce elliptical Gaussian beams, which generate elliptical spots when focused
- Elliptical spots will not perfectly match the mode field diameter and the efficiency can be calculated using the larger major(d_{major}) and smaller minor (d_{minor}) axis $1/e^2$ spot diameters:



d_{major} – Major Axis $1/e^2$ diameter
d_{minor} – Minor Axis $1/e^2$ diameter

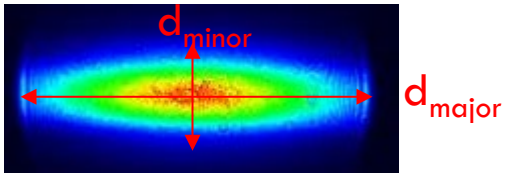


$$\eta = \frac{4 \cdot MFD^2 \cdot d_{major} \cdot d_{minor}}{(MFD^2 + d_{major}^2) \cdot (MFD^2 + d_{minor}^2)}$$

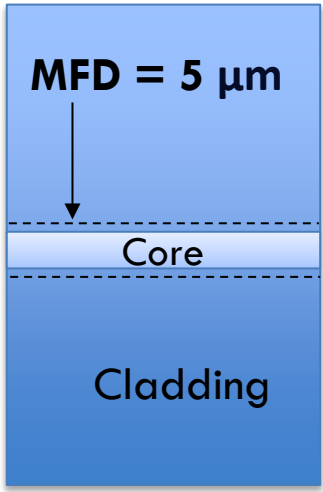
Theory: Single Mode Fiber Coupling Elliptical Spots

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- Elliptical spots will not perfectly match the mode field diameter and the efficiency can be calculated using the larger major(d_{major}) and smaller minor (d_{minor}) axis $1/e^2$ spot diameters:

EXAMPLE:



d_{major} – Major Axis $1/e^2$ diameter = 7.7 µm
d_{minor} – Minor Axis $1/e^2$ diameter = 2.3 µm

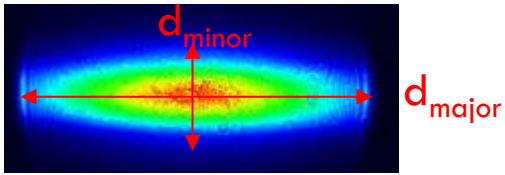


$$\eta = \frac{4 \cdot MFD^2 \cdot d_{major} \cdot d_{minor}}{(MFD^2 + d_{major}^2) \cdot (MFD^2 + d_{minor}^2)} =$$

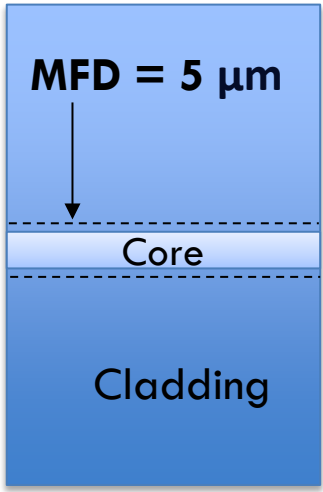
Theory: Single Mode Fiber Coupling Elliptical Spots

- Many laser diodes produce elliptical Gaussian beams, which generate elliptical spots when focused
- Elliptical spots will not perfectly match the mode field diameter and the efficiency can be calculated using the larger major(d_{major}) and smaller minor (d_{minor}) axis $1/e^2$ spot diameters:

EXAMPLE:



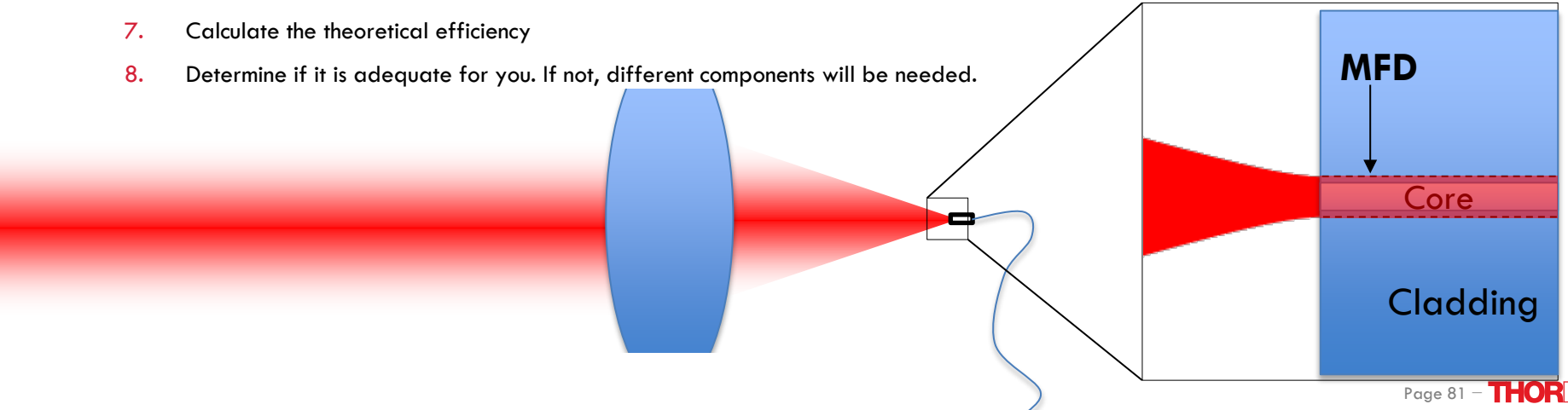
d_{major} – Major Axis $1/e^2$ diameter = 7.7 µm
d_{minor} – Minor Axis $1/e^2$ diameter = 2.3 µm



$$\eta = \frac{4 \cdot MFD^2 \cdot d_{major} \cdot d_{minor}}{(MFD^2 + d_{major}^2) \cdot (MFD^2 + d_{minor}^2)} = \frac{4 \cdot (5\ \mu m)^2 \cdot (7.7\ \mu m) \cdot (2.3\ \mu m)}{((5\ \mu m)^2 + (7.7\ \mu m)^2) \cdot ((5\ \mu m)^2 + (2.3\ \mu m)^2)} \approx 69.4\%$$

Real World Example – Single Mode Fiber Coupling

- In real world lab situations, there may not be the ideal calculated solution and must use the component with the closest match that is available
- Steps to couple a collimated Gaussian beam into a single mode fiber:
  1. Determine (find or measure) the source's wavelength (λ)
  2. Determine (find or measure) the source's collimated beam diameter (D)
  3. Determine the appropriate single mode fiber based on source wavelength
  4. Look-up the fiber MFD specification for the source wavelength (MFD)
  5. Calculate the theoretical focal length require to achieve the correct focused spot diameter (f)
  6. Find the closest focal length available
  7. Calculate the theoretical efficiency
  8. Determine if it is adequate for you. If not, different components will be needed.



Real World Example – Single Mode Fiber Coupling

- Source: CPS450 - Collimated Laser Diode Module, 450 nm, 4.5 mW
- For a collimated Gaussian beam, to couple into a single mode fiber:
  1. Determine (find or measure) the source's wavelength (λ):
  2. Determine (find or measure) the collimated beam diameter of the source (D):
  3. Determine the appropriate single mode fiber based on source wavelength:
  4. Look-up the fiber MFD specification for the source wavelength (MFD):
  5. Calculate the theoretical focal length require to achieve the correct focused spot diameter (f):
  6. Find the closest focal length coupling optic available:
  7. Calculate the theoretical efficiency and if this is adequate for your application:

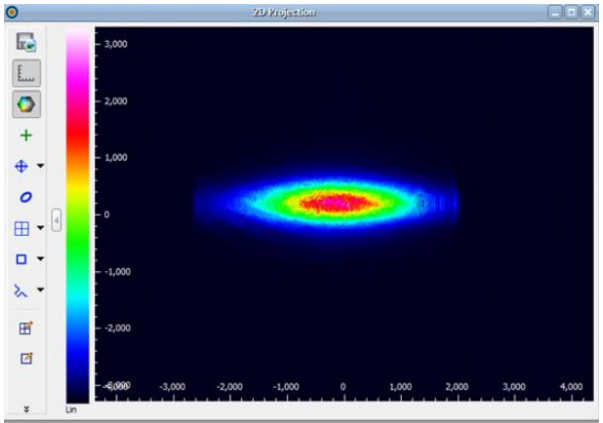


Item #	CPS450
Type	Fixed
Wavelength (Typical)	450 nm
Power (Typical)	4.5 mW
Laser Safety Class	3R
Beam Shape[C] (Click for Profile)	3.2 mm x 1.0 mm
Housing Dimensions	Ø11.0 mm x 40 mm
Specifications	

Real World Example – Single Mode Fiber Coupling

- Source: CPS450 - Collimated Laser Diode Module, 450 nm, 4.5 mW
- For a collimated Gaussian beam, to couple into a single mode fiber:
  1. Determine (find or measure) the source's wavelength (λ): **λ = 450 nm**
  2. Determine (find or measure) the collimated beam diameter of the source (D):
  3. Determine the appropriate single mode fiber based on source wavelength:
  4. Look-up the fiber MFD specification for the source wavelength (MFD):
  5. Calculate the theoretical focal length require to achieve the correct focused spot diameter (f):
  6. Find the closest focal length coupling optic available:
  7. Calculate the theoretical efficiency and if this is adequate for your application:



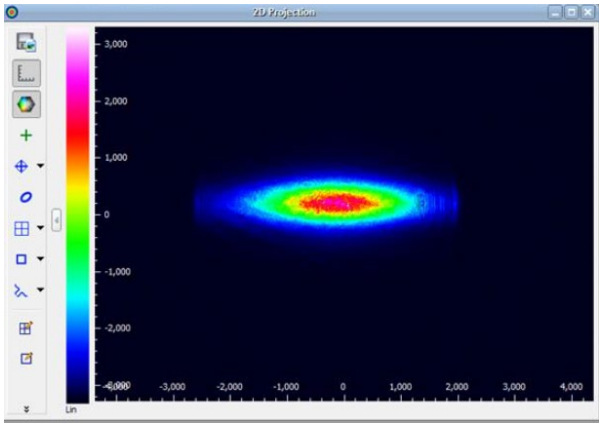


Item #	CPS450
Type	Fixed
Wavelength (Typical)	450 nm
Power (Typical)	4.5 mW
Laser Safety Class	3R
Beam Shape[C] (Click for Profile)	3.2 mm x 1.0 mm
Housing Dimensions	Ø11.0 mm x 40 mm
Specifications	

Real World Example – Single Mode Fiber Coupling

- Source: CPS450 - Collimated Laser Diode Module, 450 nm, 4.5 mW
- For a collimated Gaussian beam, to couple into a single mode fiber:
  1. Determine (find or measure) the source's wavelength (λ): **λ = 450 nm**
  2. Determine (find or measure) the collimated beam diameter of the source (D): **D_{major}= 3.2 mm; D_{minor} = 1.0 mm**
  3. Determine the appropriate single mode fiber based on source wavelength:
  4. Look-up the fiber MFD specification for the source wavelength (MFD):
  5. Calculate the theoretical focal length require to achieve the correct focused spot diameter (f):
  6. Find the closest focal length coupling optic available:
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Item #	CPS450
Type	Fixed
Wavelength (Typical)	450 nm
Power (Typical)	4.5 mW
Laser Safety Class	3R
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Real World Example – Single Mode Fiber Coupling

- Source: CPS450 - Collimated Laser Diode Module, 450 nm, 4.5 mW
- For a collimated Gaussian beam, to couple into a single mode fiber:
  1. Determine (find or measure) the source's wavelength (λ): **λ = 450 nm**
  2. Determine (find or measure) the collimated beam diameter of the source (D): **D_{major}= 3.2 mm; D_{minor} = 1.0 mm**
  3. Determine the appropriate single mode fiber based on source wavelength: **SM400**
  4. Look-up the fiber MFD specification for the source wavelength (MFD):
  5. Calculate the theoretical focal length require to achieve the correct focused spot diameter (f):
  6. Find the closest focal length coupling optic available:
  7. Calculate the theoretical efficiency and if this is adequate for your application:

+1	Qty	Docs	Part Number - Universal	
+1			S405-XP	Single Mode Optical Fiber, 400 - 680 nm, Ø125 µm Cladding
+1			SM400	Single Mode Optical Fiber, 405 - 532 nm, Ø125 µm Cladding
+1			SM450	Single Mode Optical Fiber, 488 - 633 nm, Ø125 µm Cladding
+1			460HP	Single Mode Optical Fiber, 450 - 600 nm, Ø125 µm Cladding

Real World Example – Single Mode Fiber Coupling

- Source: CPS450 - Collimated Laser Diode Module, 450 nm, 4.5 mW
- For a collimated Gaussian beam, to couple into a single mode fiber:
  1. Determine (find or measure) the source's wavelength (λ): **λ = 450 nm**
  2. Determine (find or measure) the collimated beam diameter of the source (D): **D_{major}= 3.2 mm; D_{minor} = 1.0 mm**
  3. Determine the appropriate single mode fiber based on source wavelength: **SM400**
  4. Look-up the fiber MFD specification for the source wavelength (MFD): **MFD_{ave} ≈ 3 µm**
  5. Calculate the theoretical focal length require to achieve the correct focused spot diameter (f):
  6. Find the closest focal length coupling optic available:
  7. Calculate the theoretical efficiency and if this is adequate for your application:

Item #	Operating Wavelength	Mode Field Diameter
S405-XP	400 - 680 nm	3.3 ± 0.5 µm @ 405 nm 4.6 ± 0.5 µm @ 630 nm
SM400	405 - 532 nm	2.5 - 3.4 µm @ 480 nm
SM450[a]	488 - 633 nm	2.8 - 4.1 µm[b] @ 488 nm
460HP	450 - 600 nm	3.5 ± 0.5 µm @ 515 nm

Real World Example – Single Mode Fiber Coupling

- Source: CPS450 - Collimated Laser Diode Module, 450 nm, 4.5 mW
- For a collimated Gaussian beam, to couple into a single mode fiber:
  1. Determine (find or measure) the source's wavelength (λ): **λ = 450 nm**
  2. Determine (find or measure) the collimated beam diameter of the source (D): **D_{major}= 3.2 mm; D_{minor} = 1.0 mm**
  3. Determine the appropriate single mode fiber based on source wavelength: **SM400**
  4. Look-up the fiber MFD specification for the source wavelength (MFD): **MFD_{ave} ≈ 3 µm**
  5. Calculate the theoretical focal length require to achieve the correct focused spot diameter (f): **f = 16.8 mm**
  6. Find the closest focal length coupling optic available:
  7. Calculate the theoretical efficiency and if this is adequate for your application:

$$f = \frac{\pi \cdot D_{major} \cdot MFD}{4\lambda} = \frac{\pi \cdot (3.2 \cdot 10^{-3}\ m) \cdot (3 \cdot 10^{-6}\ m)}{4 \cdot (450 \cdot 10^{-9}\ m)} \approx 16.8\ mm$$

Real World Example – Single Mode Fiber Coupling

- Source: CPS450 - Collimated Laser Diode Module, 450 nm, 4.5 mW
- For a collimated Gaussian beam, to couple into a single mode fiber:
  1. Determine (find or measure) the source's wavelength (λ): **λ = 450 nm**
  2. Determine (find or measure) the collimated beam diameter of the source (D): **D_{major}= 3.2 mm; D_{minor} = 1.0 mm**
  3. Determine the appropriate single mode fiber based on source wavelength: **SM400**
  4. Look-up the fiber MFD specification for the source wavelength (MFD): **MFD_{ave} ≈ 3 µm**
  5. Calculate the theoretical focal length require to achieve the correct focused spot diameter (f): **f = 16.8 mm**
  6. Find the closest focal length coupling optic available: **f = 15.3 mm**
  7. Calculate the theoretical efficiency and if this is adequate for your application:

Aspheric FiberPorts for FC/PC Connectors (EFL 11 mm or Greater)

Item #	EFL	Input MFD[a,b]	Output $1/e^2$ Waist Diameter[a]	Max Waist Distance[a,c]	Divergence[a]	Lens Characteristics				Length L[f]
						CA[d]	NA	AR Range[e]	Material	
PAF2P-15A	15.3 mm	3.5 µm	2.52 mm	5490 mm	0.227 mrad	5.0 mm	0.16	350 - 700 nm	D-ZK3	0.87" (22.0 mm)
PAF2P-15B	15.3 mm	5.0 µm	3.33 mm	5082 mm	0.325 mrad	5.0 mm	0.16	600 - 1050 nm	D-ZK3	0.87" (22.0 mm)
PAF2P-15C	15.3 mm	10.4 µm	2.92 mm	2151 mm	0.675 mrad	5.0 mm	0.16	1050 - 1620 nm	D-ZK3	0.87" (22.0 mm)
PAF2P-15D	15.3 mm	13.0 µm	3.00 mm	1775 mm	0.850 mrad	5.0 mm	0.16	1.8 - 2.4 µm	D-ZK3	0.87" (22.0 mm)
PAF2P-18A	18.4 mm	3.5 µm	3.01 mm	7936 mm	0.190 mrad	5.5 mm	0.15	350 - 700 nm	D-ZK3	0.87" (22.0 mm)
PAF2P-18B	18.4 mm	5.0 µm	3.98 mm	7347 mm	0.272 mrad	5.5 mm	0.15	600 - 1050 nm	D-ZK3	0.87" (22.0 mm)
PAF2P-18C	18.4 mm	10.4 µm	3.49 mm	3107 mm	0.565 mrad	5.5 mm	0.15	1050 - 1620 nm	D-ZK3	0.87" (22.0 mm)
PAF2P-18D	18.4 mm	13.0 µm	3.60 mm	2564 mm	0.707 mrad	5.5 mm	0.15	1.8 - 2.4 µm	D-ZK3	0.87" (22.0 mm)

a. These values are calculated using the following fibers at the specified wavelengths: -A: 460HP at 450 nm, -B: 780HP at 850 nm, -C: SMF-28-J9 at 1550 nm, -D: SM2000 at 1996 nm, -E: ZrF_4 Fiber at 2.30 µm

Real World Example – Single Mode Fiber Coupling

- Source: CPS450 - Collimated Laser Diode Module, 450 nm, 4.5 mW
- For a collimated Gaussian beam, to couple into a single mode fiber:
  1. Determine (find or measure) the source's wavelength (λ): **λ = 450 nm**
  2. Determine (find or measure) the collimated beam diameter of the source (D): **D_{major}= 3.2 mm; D_{minor} = 1.0 mm**
  3. Determine the appropriate single mode fiber based on source wavelength: **SM400**
  4. Look-up the fiber MFD specification for the source wavelength (MFD): **MFD_{ave} ≈ 3 µm**
  5. Calculate the theoretical focal length require to achieve the correct focused spot diameter (f): **f = 16.8 mm**
  6. Find the closest focal length coupling optic available: **f = 15.3 mm**
  7. Calculate the theoretical efficiency and if this is adequate for your application: **η =**

Real World Example – Single Mode Fiber Coupling

- Source: CPS450 - Collimated Laser Diode Module, 450 nm, 4.5 mW
- For a collimated Gaussian beam, to couple into a single mode fiber:
  1. Determine (find or measure) the source's wavelength (λ): **λ = 450 nm**
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  5. Calculate the theoretical focal length require to achieve the correct focused spot diameter (f): **f = 16.8 mm**
  6. Find the closest focal length coupling optic available: **f = 15.3 mm**
  7. Calculate the theoretical efficiency and if this is adequate for your application: **η = 61%**

$$d_{major} = \frac{4 \cdot \lambda \cdot f}{\pi \cdot D_{major}} = \frac{4 \cdot (450 \cdot 10^{-9}\ m) \cdot (15.3\ \cdot 10^{-3}\ m)}{\pi \cdot (3.2\ \cdot 10^{-3}\ m)} = 2.74\ \mu m$$

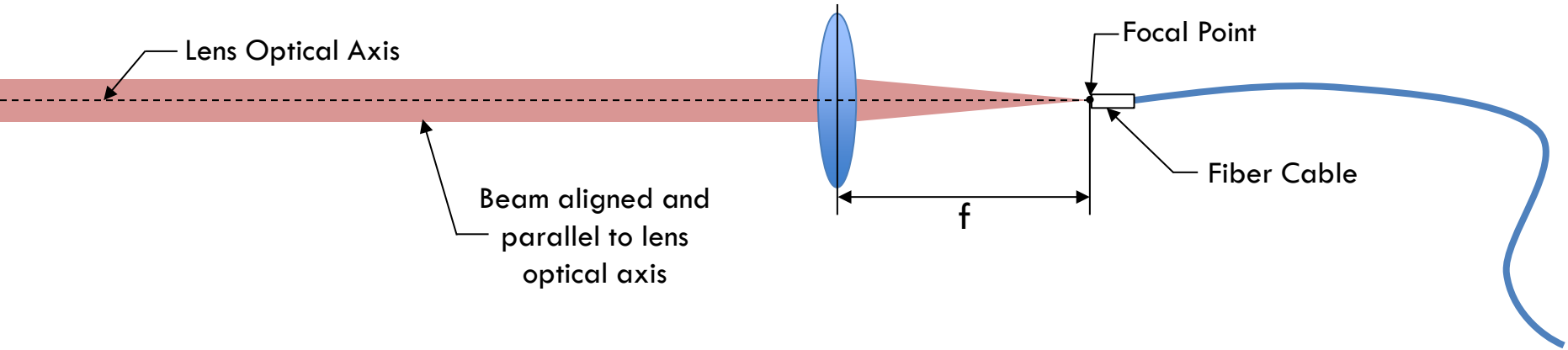
$$d_{minor} = \frac{4 \cdot \lambda \cdot f}{\pi \cdot D_{minor}} = \frac{4 \cdot (450 \cdot 10^{-9}\ m) \cdot (15.3\ \cdot 10^{-3}\ m)}{\pi \cdot (1\ \cdot 10^{-3}\ m)} = 8.77\ \mu m$$

$$\eta = \frac{4 \cdot MFD^2 \cdot d_{major} \cdot d_{minor}}{(MFD^2 + d_{major}^2) \cdot (MFD^2 + d_{minor}^2)} = \frac{4 \cdot (3\ \mu m)^2 \cdot (2.74\ \mu m) \cdot (8.77\ \mu m)}{((3\ \mu m)^2 + (2.74\ \mu m)^2) \cdot ((3\ \mu m)^2 + (8.77\ \mu m)^2)} \approx 61\%$$

Fiber Alignment
&
Degrees of Freedom

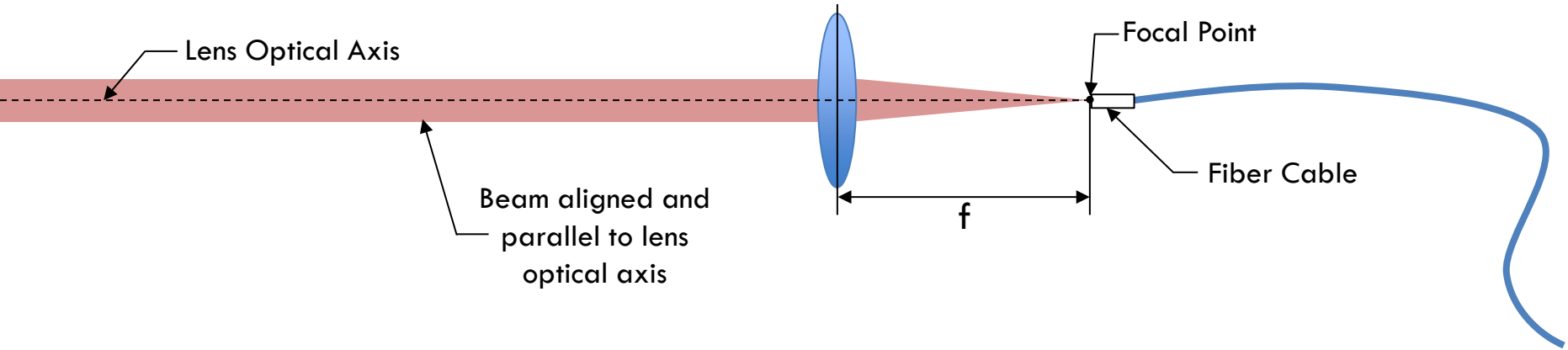
Freespace Optical Alignment: Tilt & Displacement

- Beams parallel to a lens's optical axis will focus to the on-axis focal point of the lens and is the desired alignment for fiber coupling
- When the beam lies on the optical axis, it creates a symmetrically focused beam, which couples light with high efficiency
- A beam parallel to, --but offset from-- the optical axis will still focus to the lens's on-axis focal point
- When the beam is offset from the optical axis it may still focus to the fiber, but will alter the beam spot geometry, reducing efficiency



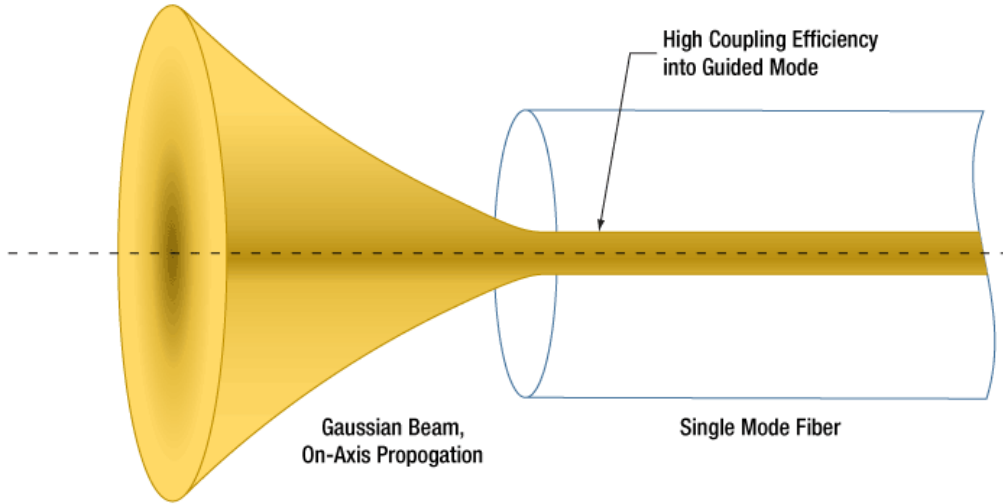
Freespace Optical Alignment: Tilt & Displacement

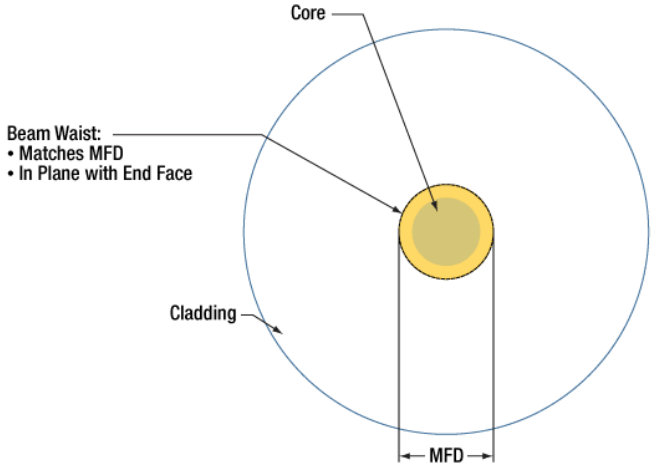
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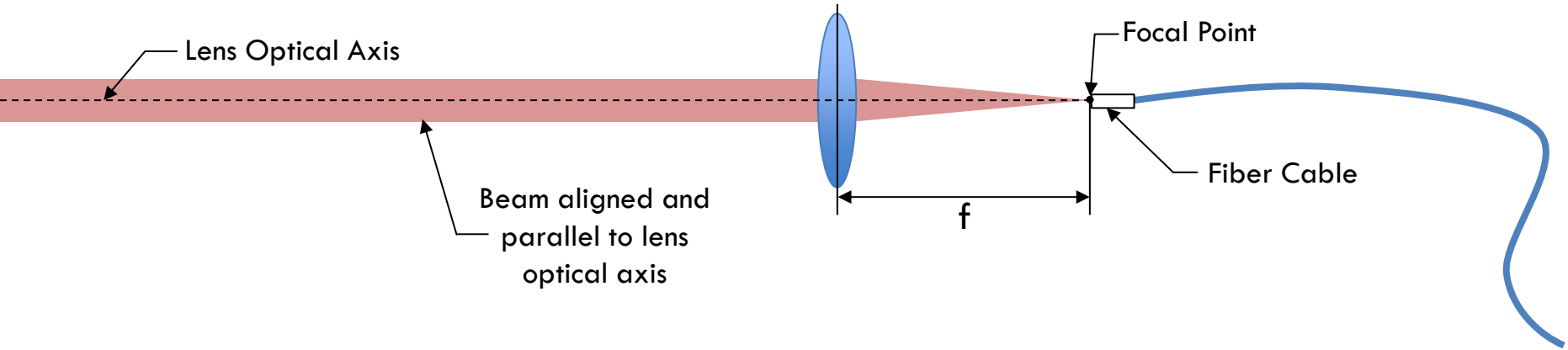
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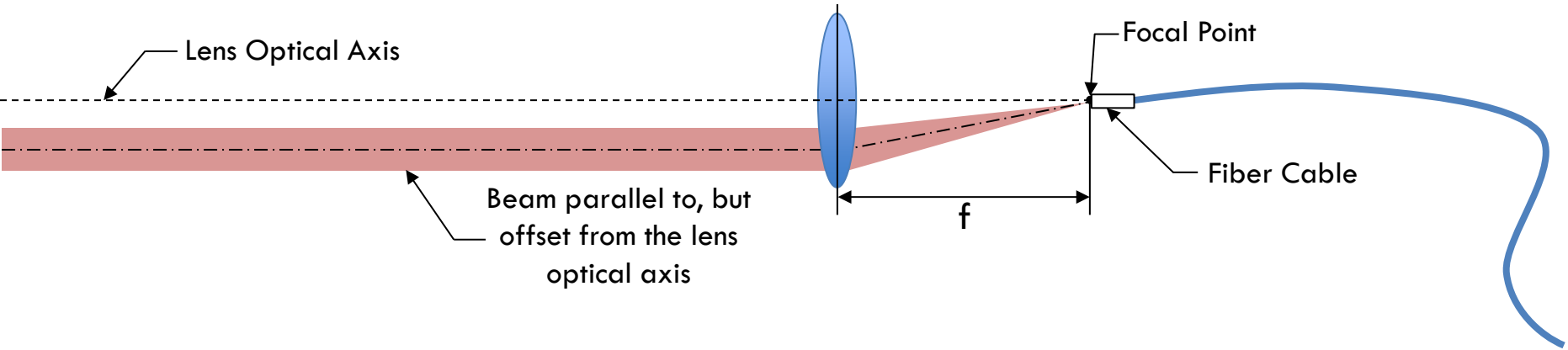
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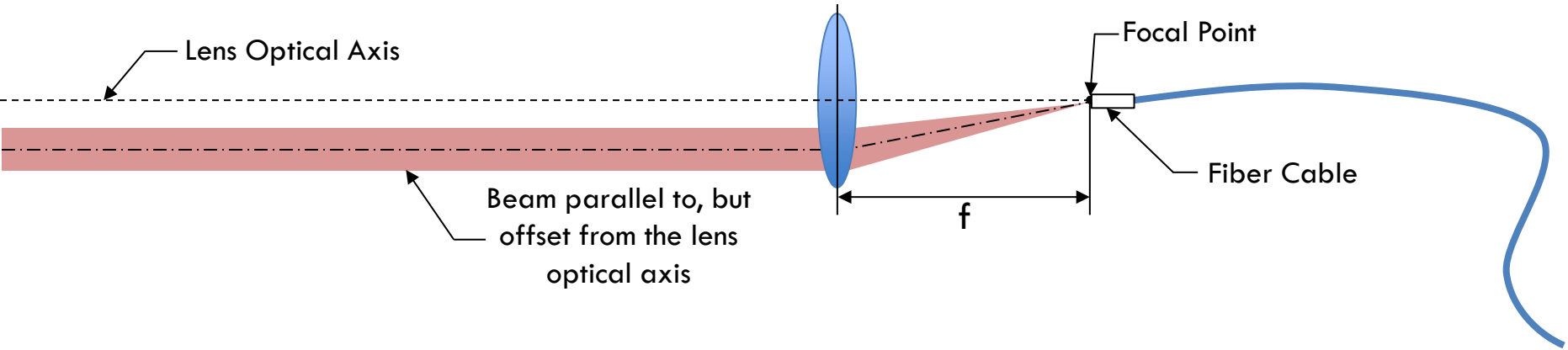
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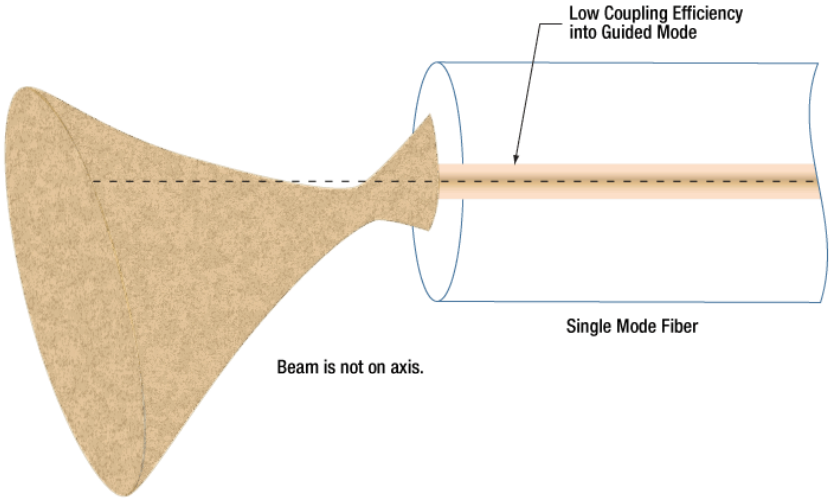
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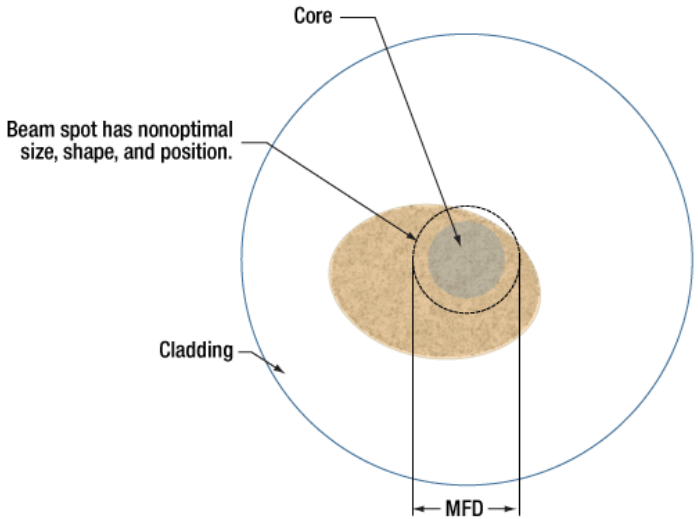
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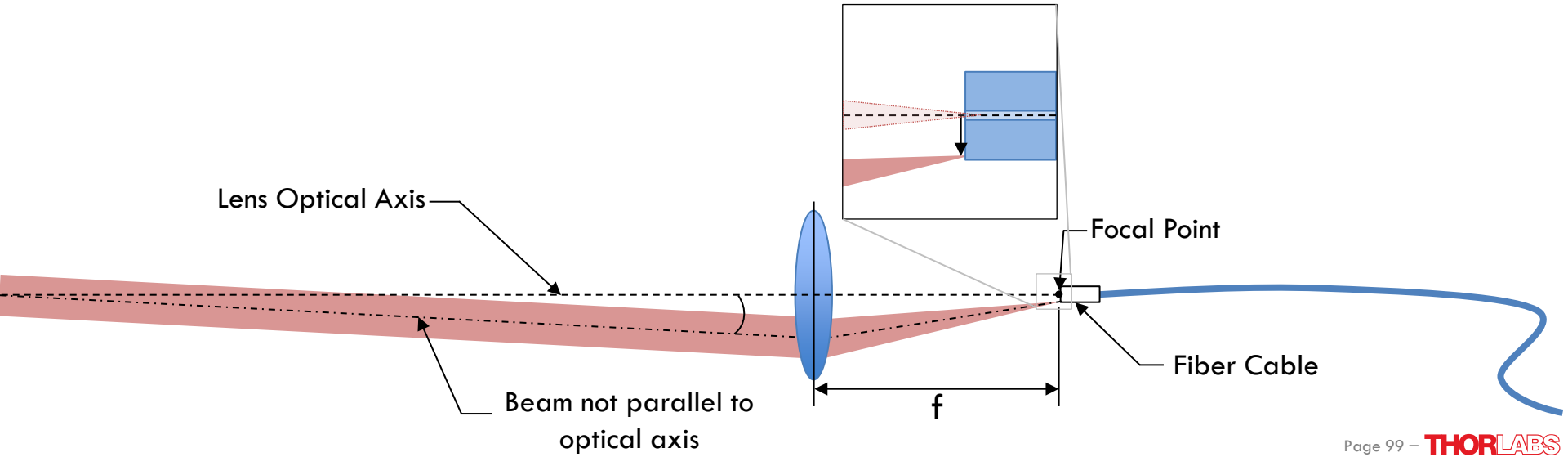
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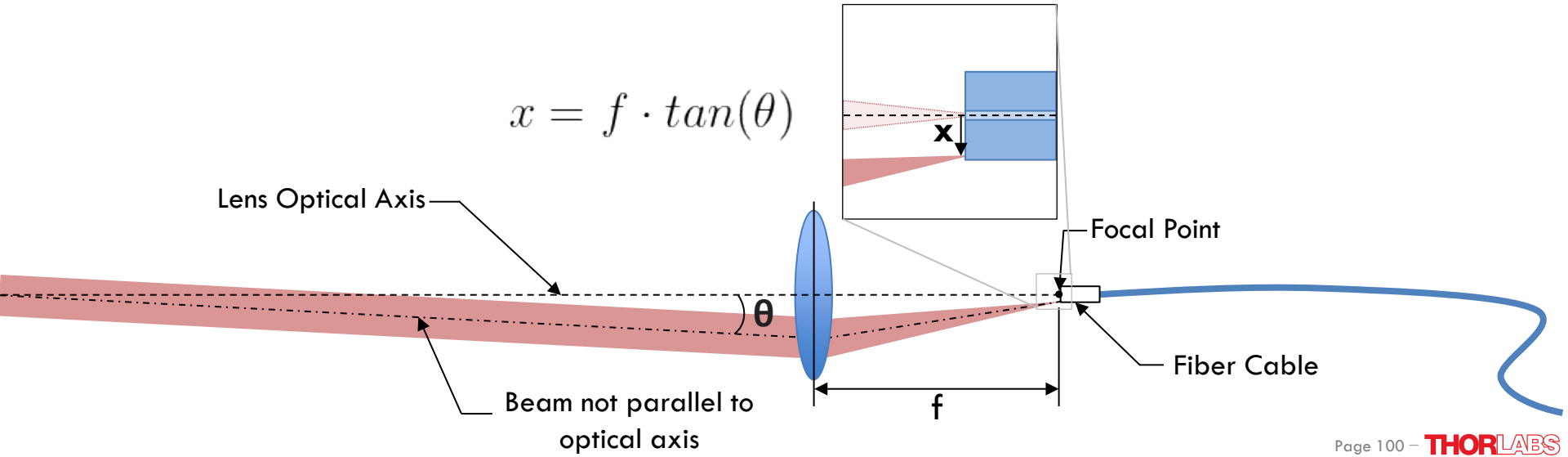
- Optical beams skewed from the optical axis will focus to spots offset from the lens's focal point
- For small angles, this translates the focal spot position without introducing much aberration
- At large angles, the beam can suffer from field curvature, distorting the focused spot and displacing it from the focus point plane
- Fine angular adjustment of the incident beam or the lens can be used to optimize alignment for coupling light into fiber



Freespace Optical Alignment: Tilt & Displacement

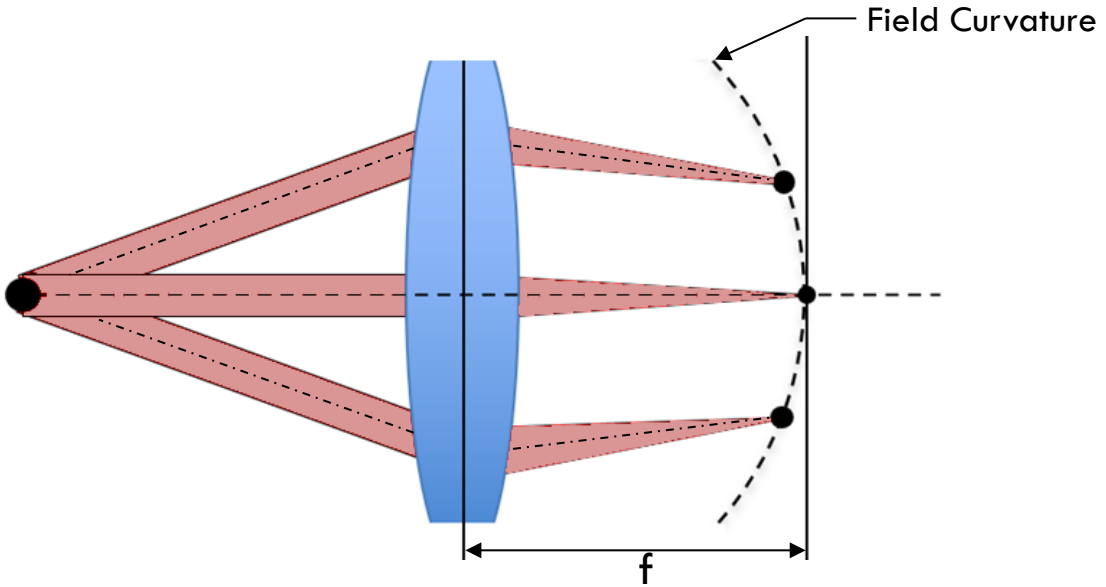
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$$x = f \cdot tan(\theta)$$



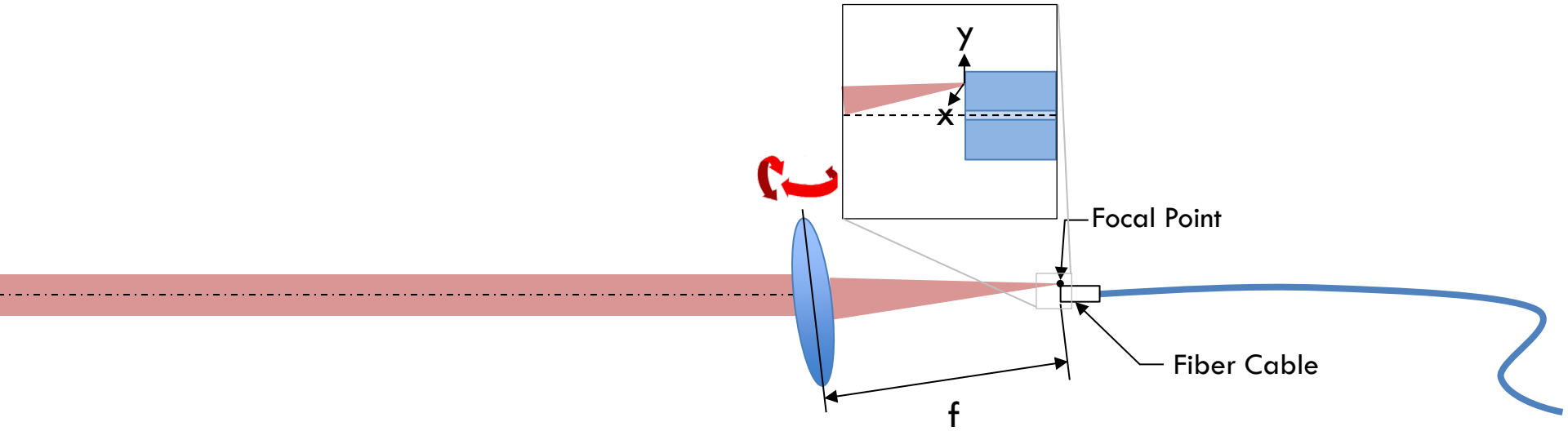
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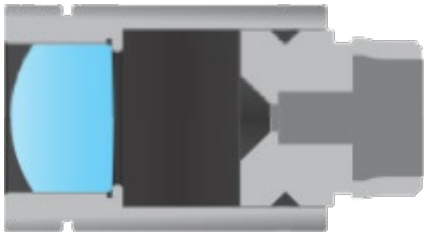
Freespace Optical Alignment: Degrees of Freedom

- In many systems, fiber collimators may be used for coupling light into fiber
  - Since fixed fiber collimators are aligned to the correct fiber to lens distance to collimate, they are also pre-aligned at the distance to accept a collimated beam and focus into the fiber
  - Additionally, using a collimator doesn't require independent adjustment of the fiber and lens, which simplifies alignment compared to other methods:
    - Fixed Source with Independent Coupling Components
    - Fixed Source with Collimator

Freespace Coupling
Pigtailed Fiber Collimator Pair

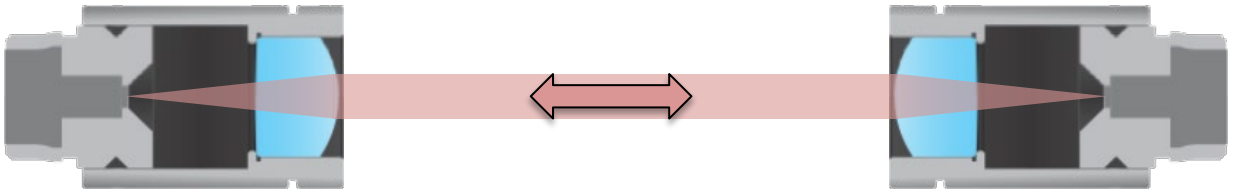


Fixed-Focus Aspheric Fiber Collimator



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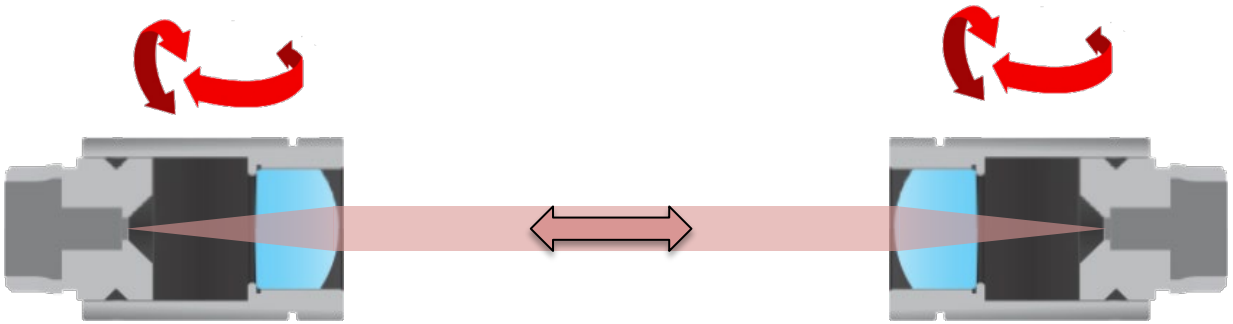


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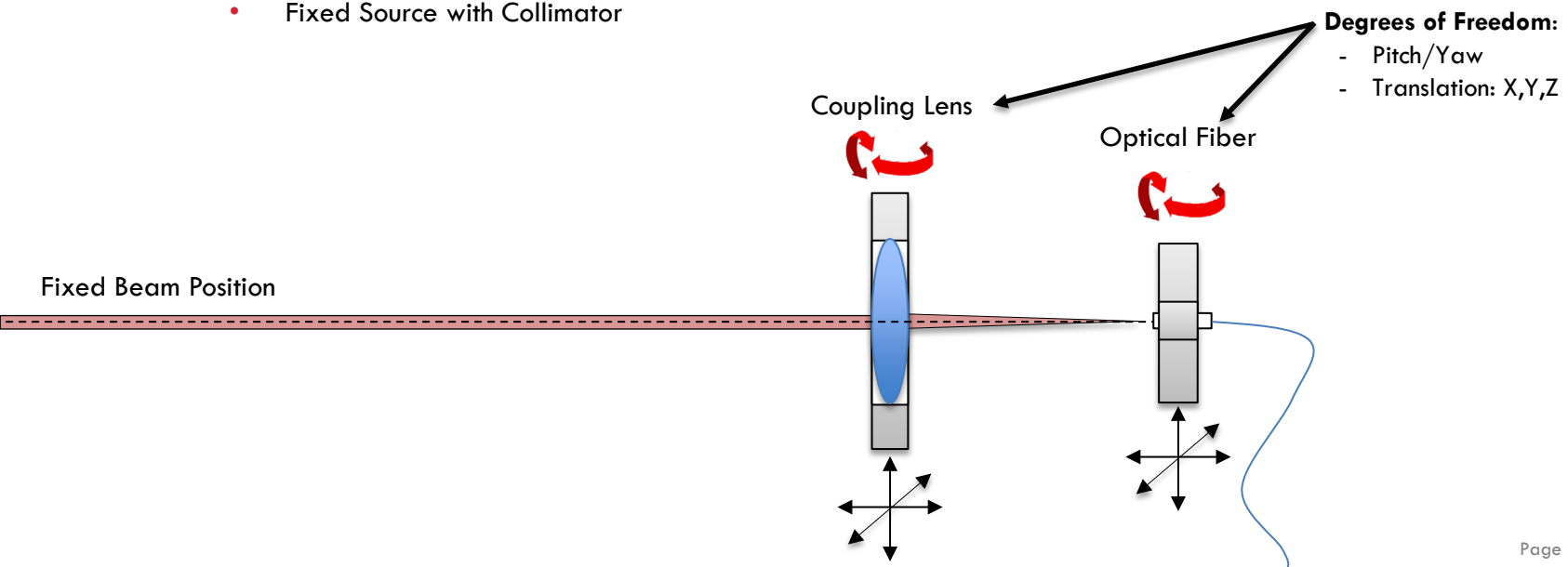
Degrees of Freedom:
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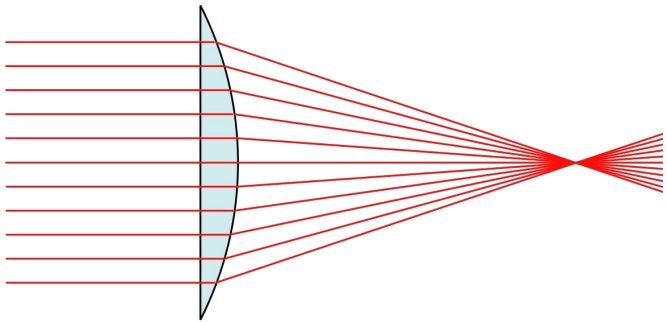


Optical Aberration
&
Optic Selection

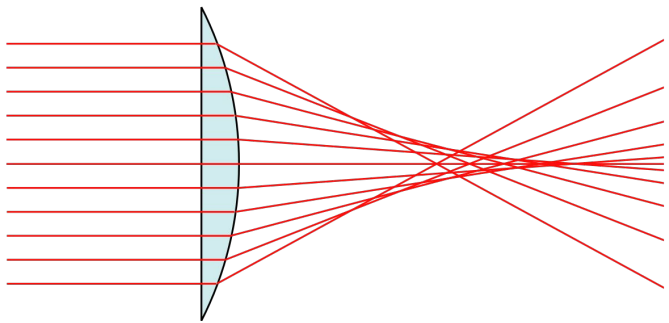
Optic Selection & Aberration

- For single-mode-coupling, achieving the diffraction limited focused spot diameter is critical
- Until this point, we've calculated the ideal focal length required to achieve the correct spot size, but have not considered the performance and aberrations present with different lens types
- Using an incorrect lens can introduce aberrations that prevent achieving the calculated spot size and reduces the coupling efficiency

Diffraction Limited Focus

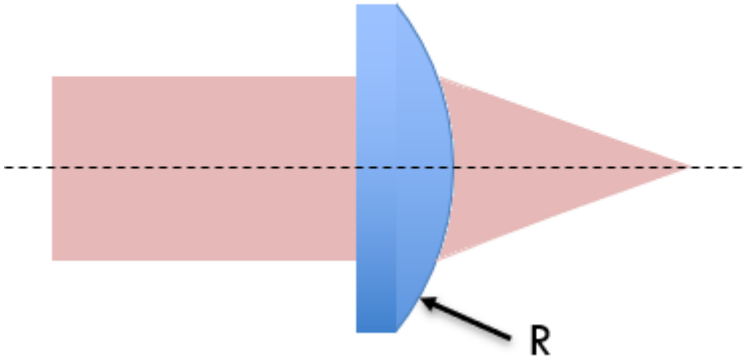


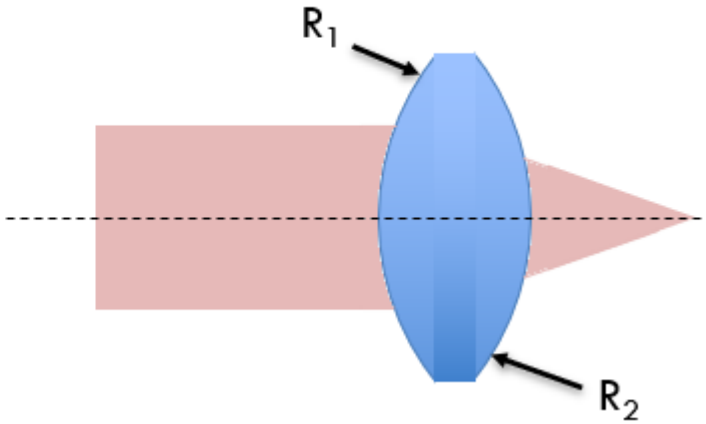
Aberration Limited Focus



Optic Selection & Aberration: Spherical Aberrations

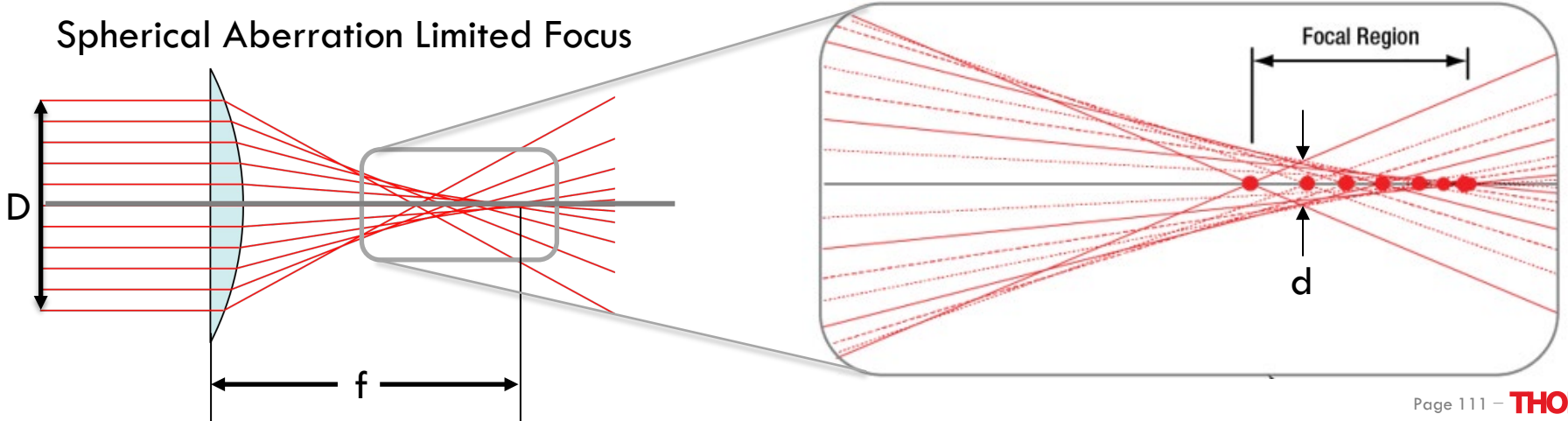
- Plano-Convex and Bi-Convex lens's spherical surfaces focus light rays to a spot
- Spherical surfaces are not the perfect shape to focus light rays and introduce spheric aberrations
- Spherical aberrations are a result of rays at different distances from the optical axis focusing to different focal lengths, introducing a focal region that produces a larger focused spot (d)
- For spherical singlet lenses (single element), the larger the focal ratio (f/D), the greater the spherical aberrations and the more the focused spot size deviates from the calculated value





Optic Selection & Aberration: Spherical Aberrations

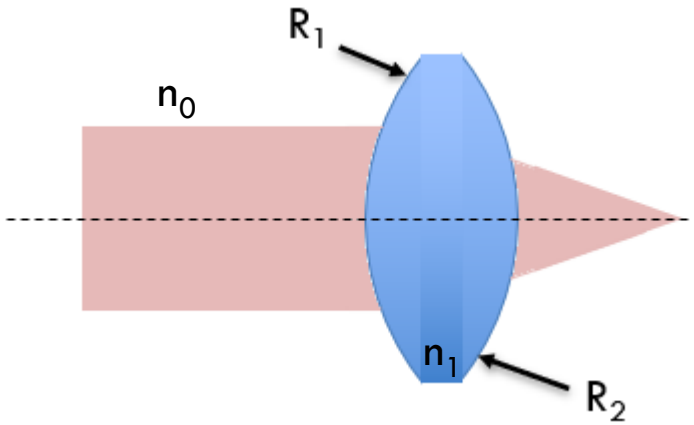
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Optic Selection & Aberration: Chromatic Aberrations

- A lens's focal length is dependent on its shape as well as the material's refractive index
- Optical materials have chromatic dispersion; they do not have a constant refractive index for all wavelengths in their operating range, resulting in a wavelength (λ) dependent focal length (chromatic aberration)
- Broad wavelength sources are susceptible to chromatic aberration, introducing a spectral focal region, larger apparent focused spot, and wavelength dependent coupling efficiency
- The greater the chromatic dispersion, the greater introduced chromatic aberrations

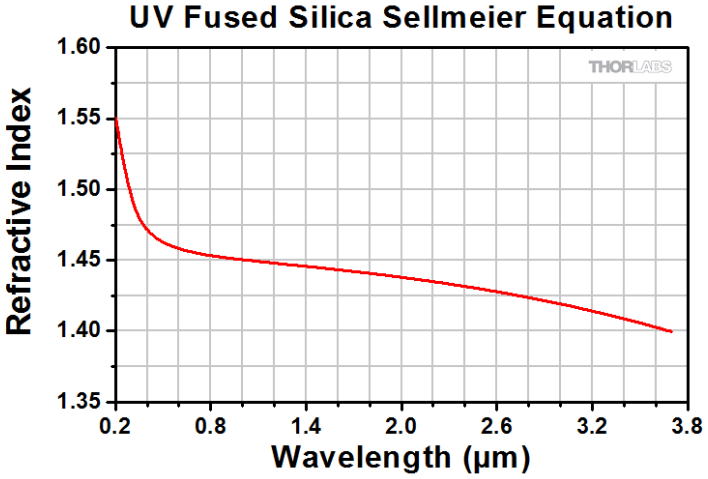
$$\frac{1}{f} = \left(\frac{n_1 - n_0}{n_0}\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$



Optic Selection & Aberration: Chromatic Aberrations

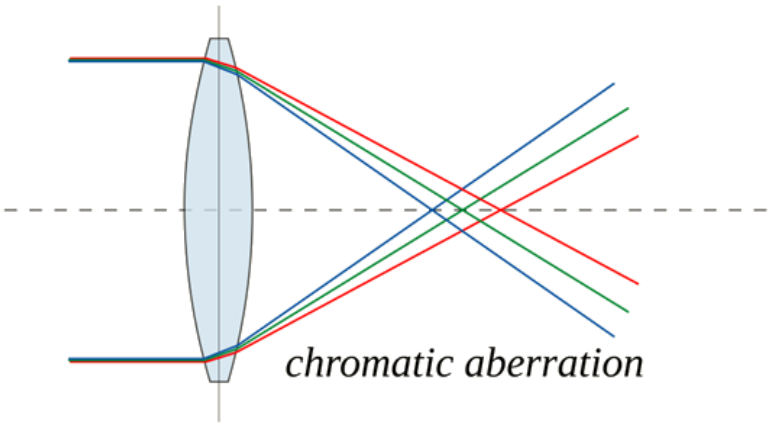
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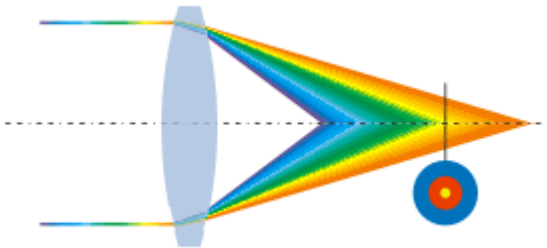


https://en.wikipedia.org/wiki/Chromatic_aberration#/media/File:Chromatic_aberration_lens_diagram.svg

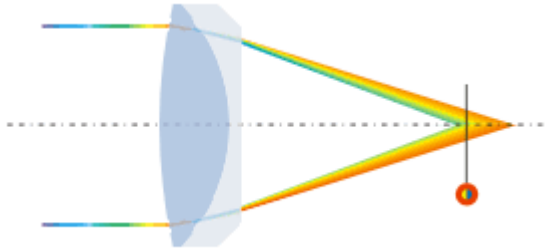
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Lens with High Chromatic Dispersion and Chromatic Aberration



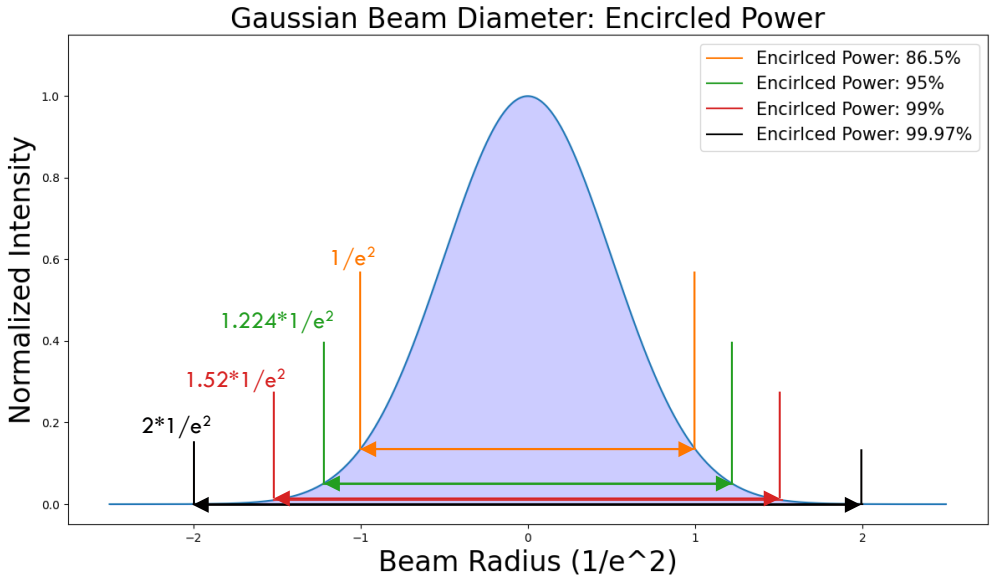
Lens with Low Chromatic Dispersion and Chromatic Aberration

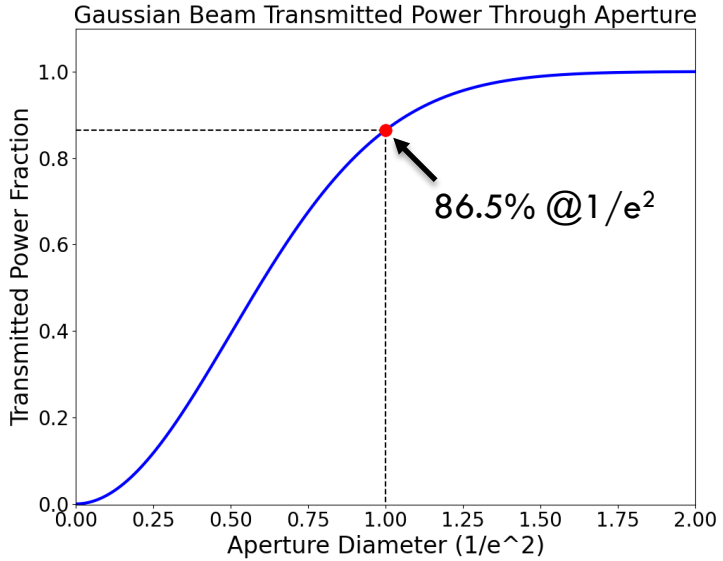


https://en.wikipedia.org/wiki/Optical_aberration#/media/File:Lens_chromatic_aberration.png

Optic Selection: Clear Aperture

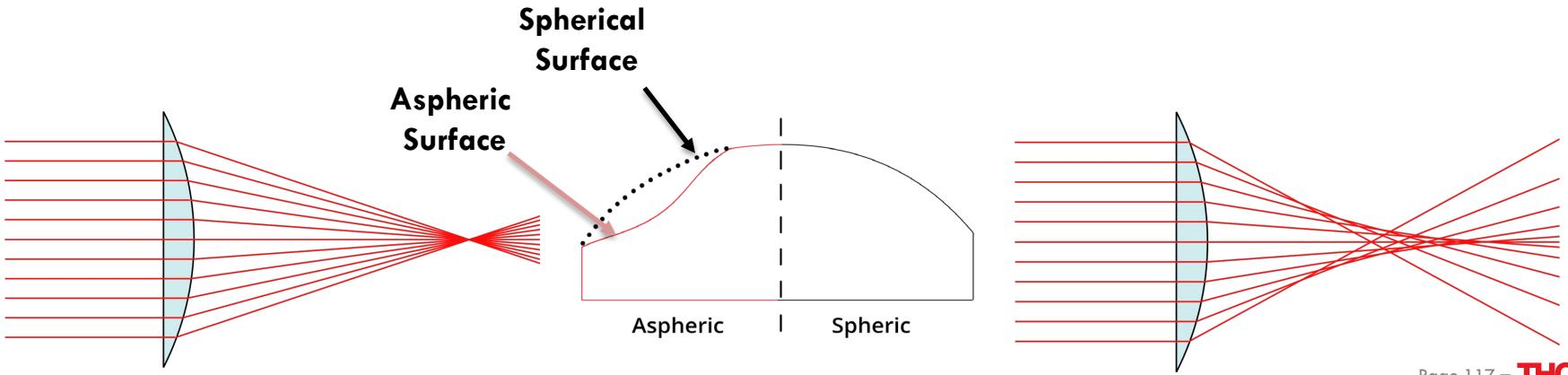
- Gaussian beams used in single mode fiber coupling are defined by their $1/e^2$ diameter, however only 86.5% of a Gaussian beam's power is within this diameter
- To avoid reduced efficiency, it is critical to use a sufficiently large clear aperture to capture the beam's power
- A clear aperture >1.52x the $1/e^2$ diameter ensures >99% of the beam's power





Optic Selection & Aberration: Spherical & Aspheric Lenses

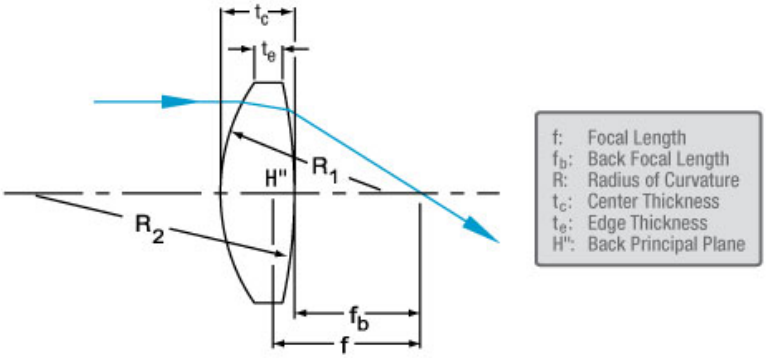
- Spherical singlet lenses (plano convex, bi-convex) introduce both spherical and chromatic aberrations and are generally not suggested for applications requiring diffraction limited performance
- Aspheric lenses have single or multiple surfaces that deviate from a spheres to prevent the introduction of spherical aberration
- Aspheric surfaces can be designed to provide spherical-aberration-free performance over narrow or broad bands, but are still subject to chromatic aberration, making them the most ideal for coupling monochromatic sources



Optic Selection & Aberration: Best Form Spherical Lenses

- Best form lenses are single element lenses designed to minimize spherical aberration, while still using spherical surfaces
- For a given material, two dissimilar radius-of-curvature values are determined to compensate for spherical aberration, providing diffraction limited performance with small incident beams
- Since they are made of a single optical material, best form lenses will have some amount of chromatic dispersion, introducing chromatic aberrations and are best for coupling monochromatic sources

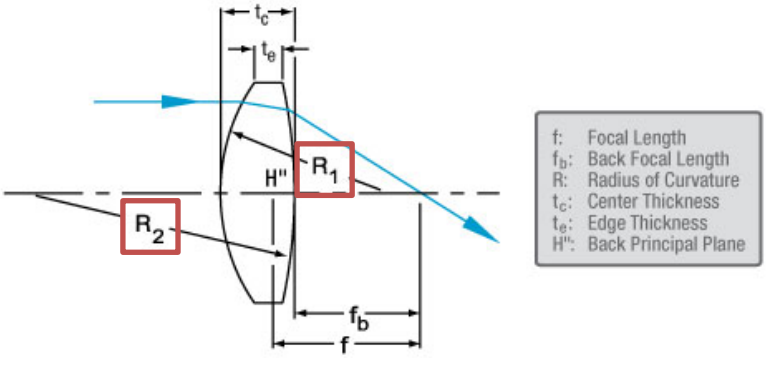
Item #	Input Diameter for Diffraction-Limited Performance @ 632.8 nm	Input Diameter for Diffraction-Limited Performance @ 1064 nm
LBF254-040	5.0 mm	6.0 mm
LBF254-050	6.0 mm	7.0 mm
LBF254-075	8.0 mm	9.5 mm
LBF254-100	10.0 mm	11.5 mm
LBF254-150	13.5 mm	16.0 mm
LBF254-200	17.0 mm	20.0 mm



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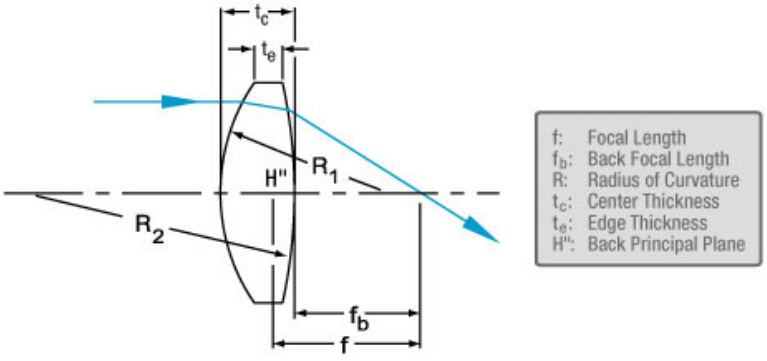
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LBF254-200	17.0 mm	20.0 mm



Optic Selection & Aberration: Best Form Spherical Lenses

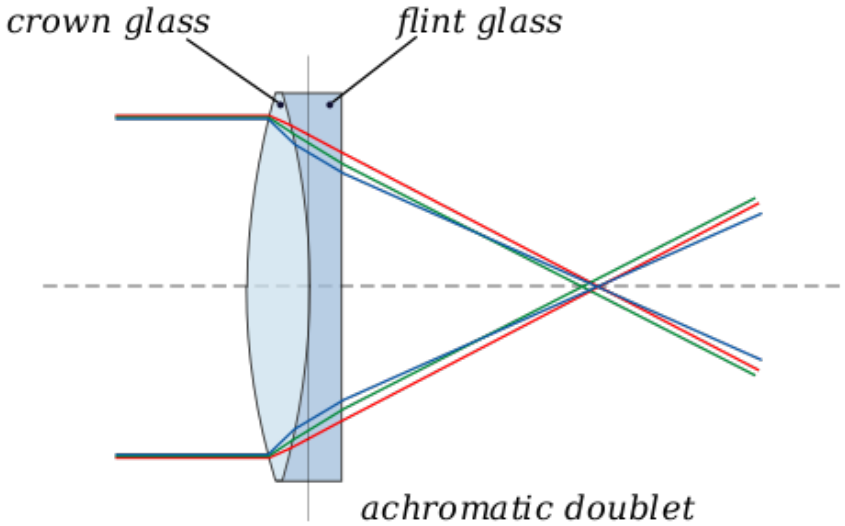
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- For a given material, two dissimilar radius of curvature values are determined to compensate for spherical aberration, providing diffraction limited performance with small incident beams
- Since they are made of a single optical material, best form lenses will have some amount of chromatic dispersion, introducing chromatic aberrations and are best for coupling monochromatic sources

Item #	Input Diameter for Diffraction-Limited Performance @ 632.8 nm	Input Diameter for Diffraction-Limited Performance @ 1064 nm
LBF254-040	5.0 mm	6.0 mm
LBF254-050	6.0 mm	7.0 mm
LBF254-075	8.0 mm	9.5 mm
LBF254-100	10.0 mm	11.5 mm
LBF254-150	13.5 mm	16.0 mm
LBF254-200	17.0 mm	20.0 mm



Optic Selection & Aberration: Achromatic Lenses

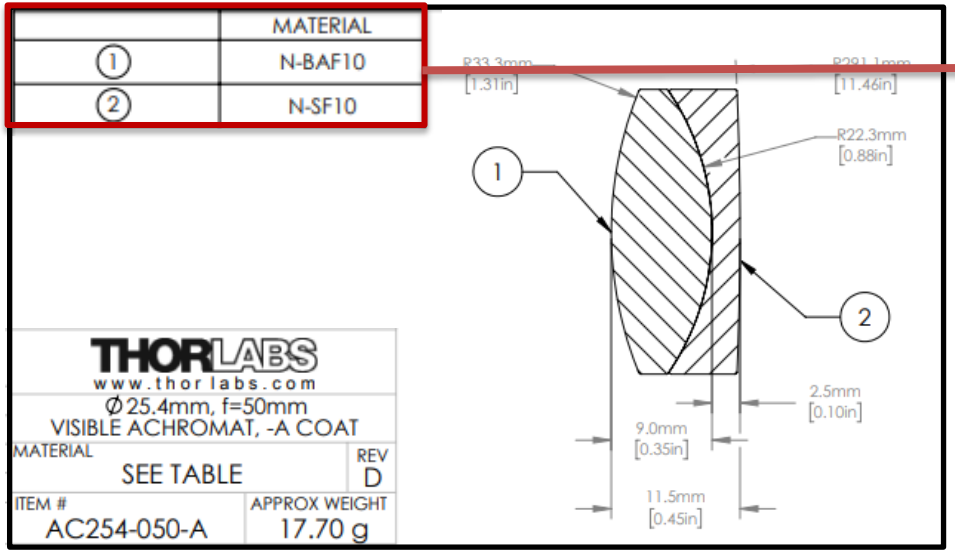
- Achromatic lenses are spherical multi-element lens designs which use differing materials for each element to reduce both spherical aberrations and chromatic aberrations
- Using a high chromatic dispersion negative element and a low dispersion positive element, the optical path difference between wavelengths can reduce chromatic aberration while also compensating spherical aberration
- Designs of achromats will not completely remove chromatic aberration, but often bring two design wavelengths into a common focus. They also reduce chromatic aberration outside the common-focus-wavelengths better than their singlet counterparts (plano-convex, best form, asphere, etc.)

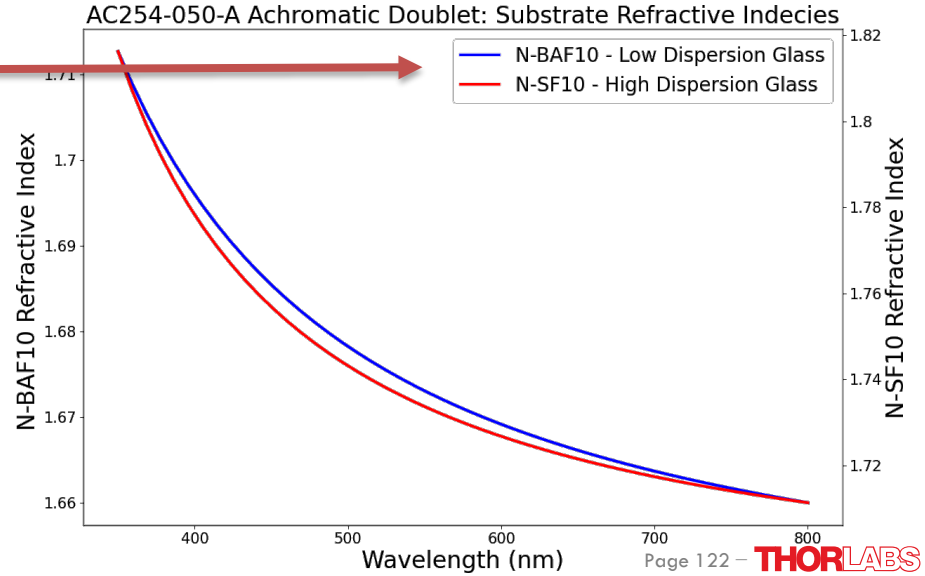


https://en.wikipedia.org/wiki/File:Lens6b-en.svg

Optic Selection & Aberration: Achromatic Lenses

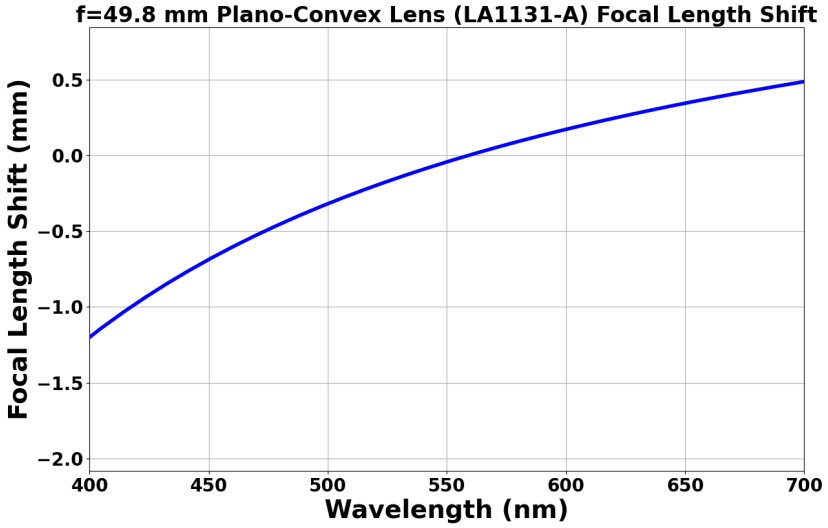
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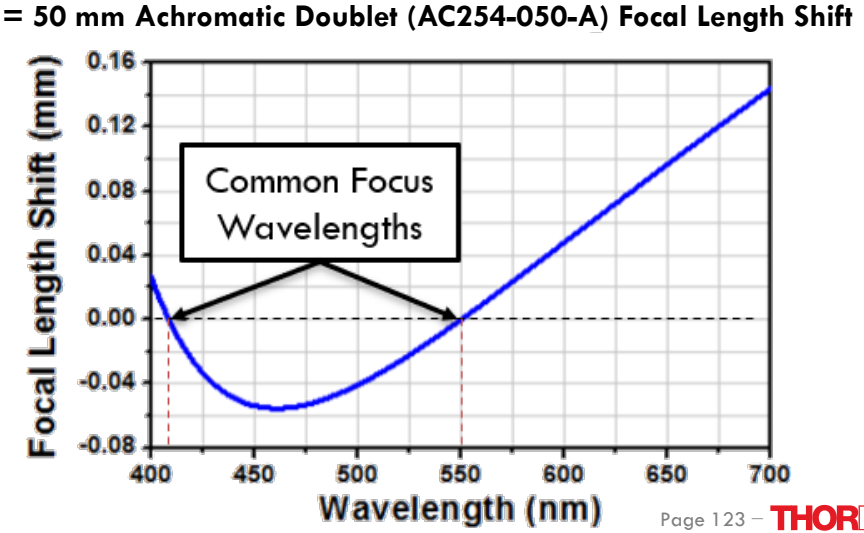




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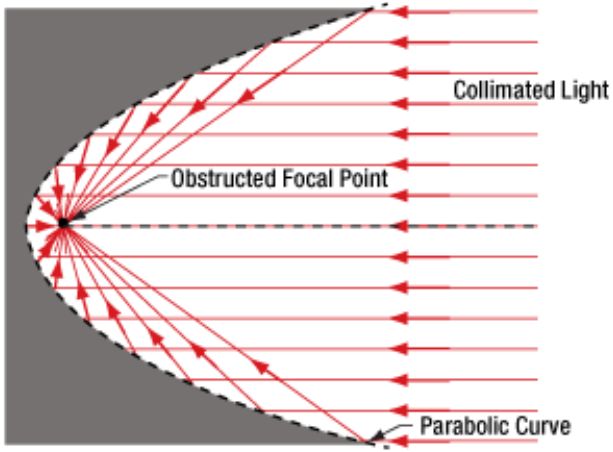




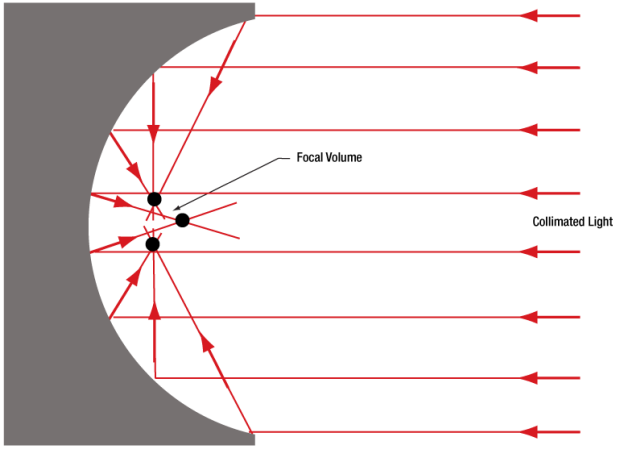
Optic Selection & Aberration: Parabolic Mirrors

- While spherical mirrors are subject to spherical aberration, parabolic mirrors focus with diffraction limited performance when incident beams are parallel to their optical axis
- Off-axis parabolic (OAP) sections of the parent parabolic mirrors can be used to avoid obstructing the aperture of the parent parabola, while maintaining on-axis performance
- Since chromatic aberrations are caused by wavelength dependent refractive index in refractive optics, OAP mirrors can be used for fiber coupling, free of both chromatic and spherical aberration

Parabolic Mirror

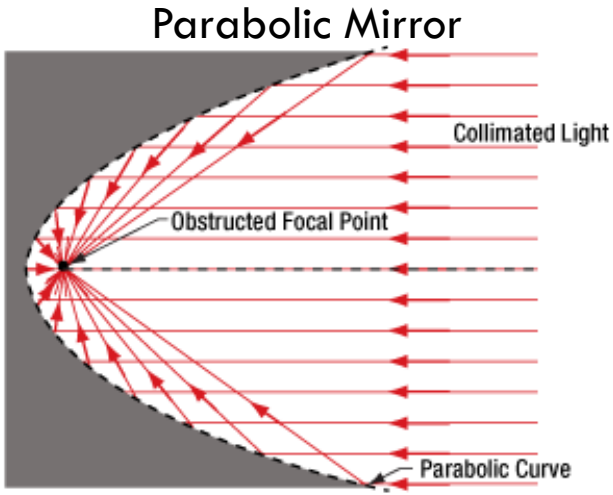


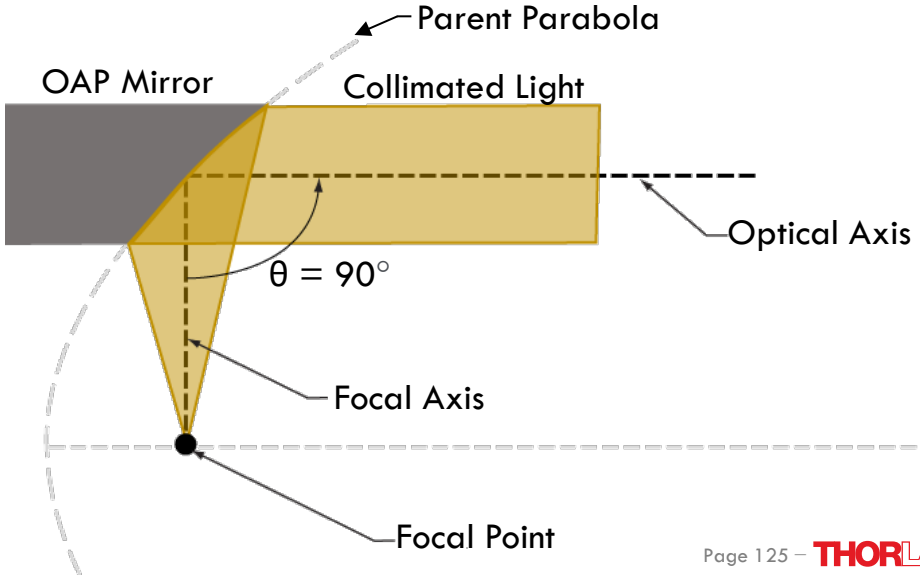
Spherical Mirror



Optic Selection & Aberration: Parabolic Mirrors

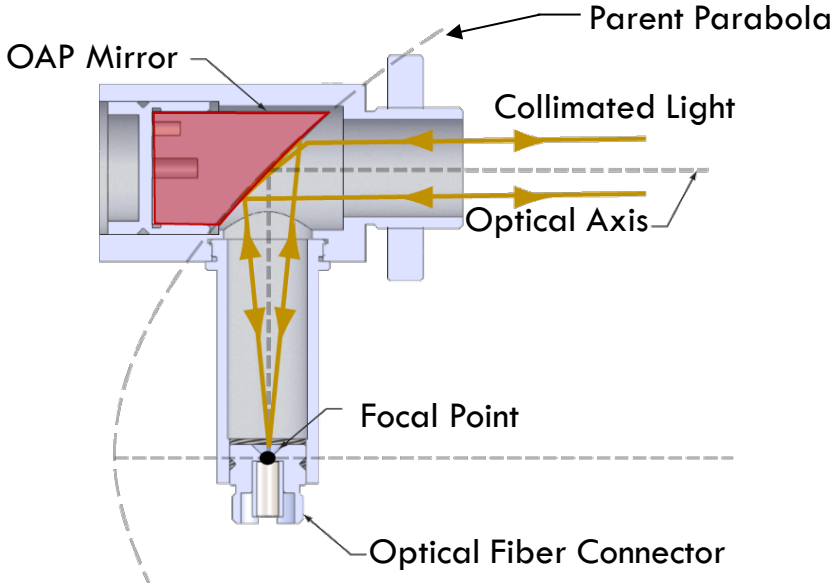
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Optic Selection Guide – Single Mode Fiber Coupling

Source	Spherical Lens	Aspheric Lens	Best Form Lens	Achromatic Lens	OAP Mirror
Laser Diode					
SM Coupled Source					
MM Coupled Source*					
Tunable Laser (Narrow)					
Tunable Laser (Broad)					
Multiple Laser Lines					
LED*					
Supercontinuum					
Superluminescent Diodes (SLD)					
Broadband Lamp*					

Recommended

Not Always Recommended

Not Recommended

* non-Gaussian sources cannot efficiently couple into single mode fibers

Optic Selection Guide – Multimode Fiber Coupling

Source	Spherical Lens	Aspheric Lens	Best Form Lens	Achromatic Lens	OAP Mirror
Laser Diode					
SM Coupled Source					
MM Coupled Source					
Tunable Laser (Narrow)					
Tunable Laser (Broad)					
Multiple Laser Lines					
LED*					
Supercontinuum					
Superluminescent Diodes (SLD)					
Broadband Lamp*					

Recommended

Not Always Recommended

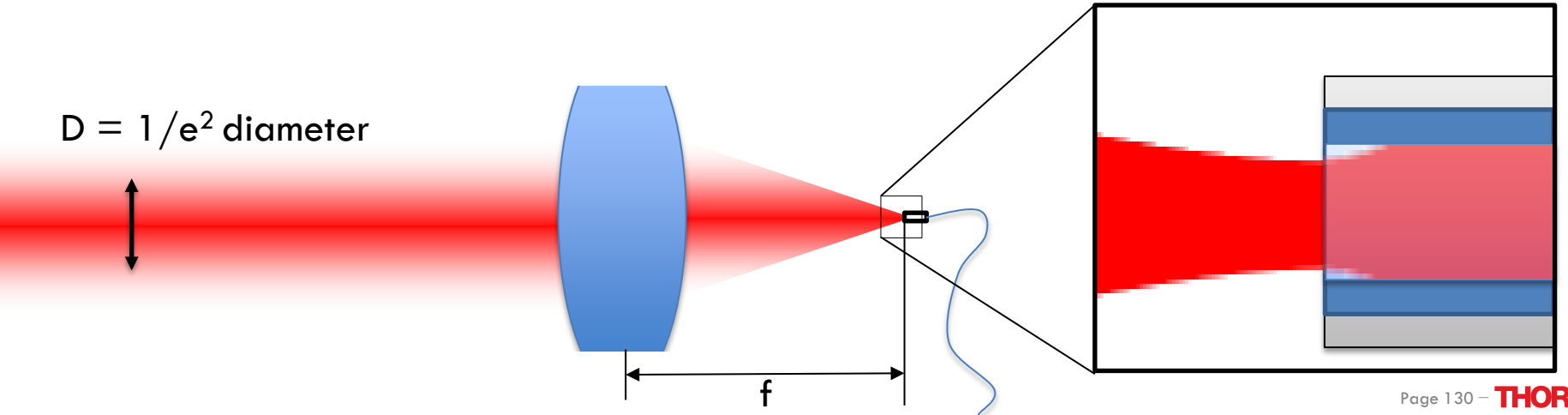
Not Recommended

*Lage incoherent sources should be butt-coupled and do not gain efficiency by using optics

Lab Work

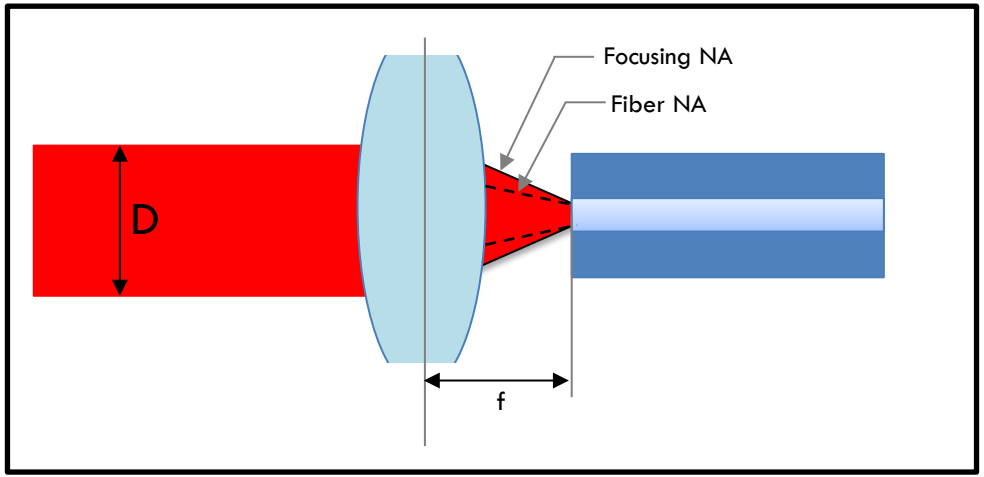
Lab Work - Part 1: Freespace SM to MM Coupling - Background

- The primary requirements for efficiently coupling into multimode fiber are:
  - The incident beam focuses onto the core of the fiber with a spot smaller than the core diameter
  - The incident beam focuses with an NA that does not exceed the fiber's numerical aperture (NA_{fiber})
- The calculated focused spot size (d) for a Gaussian beam with $1/e^2$ diameter (D), wavelength (λ), and lens focal length (f) can be calculated to ensure it is smaller than the core diameter
- The focusing NA (NA_{focus}) can also be calculated to ensure the fiber NA is not exceeded, as well as the shortest usable focal length (f_{min}) before light exceeds the receiving fiber's NA (NA_{fiber})



Lab Work - Part 1: Freespace SM to MM Coupling - Background

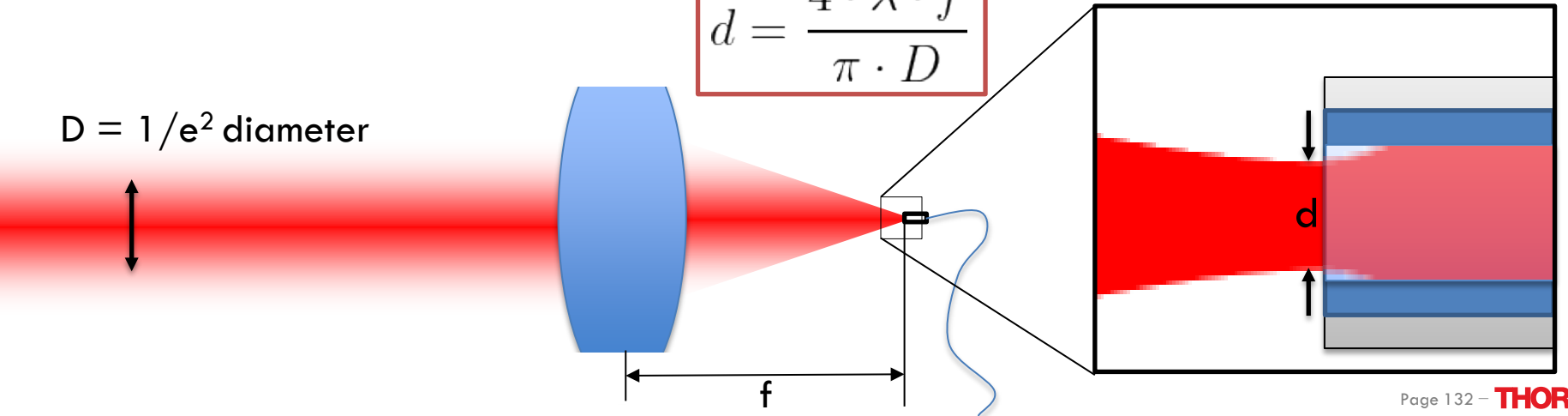
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$$d = \frac{4 \cdot \lambda \cdot f}{\pi \cdot D}$$

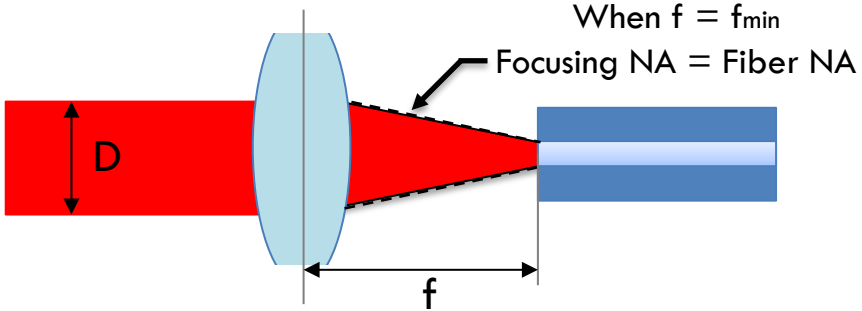


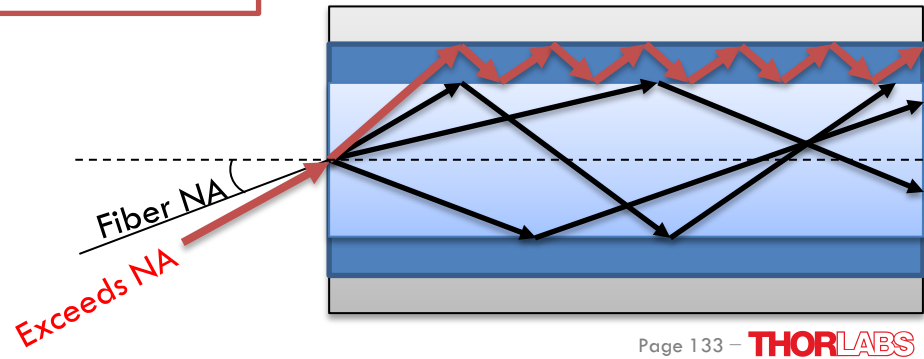
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$$NA_{focus} \approx \frac{D}{2 \cdot f}$$

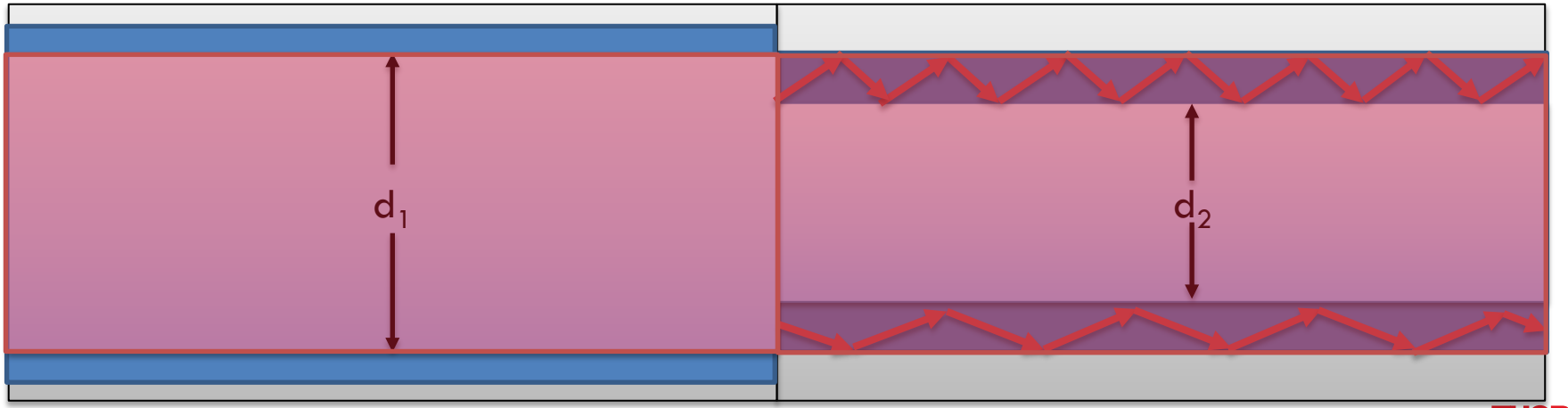
$$f_{min} \approx \frac{D}{2 \cdot NA_{fiber}}$$





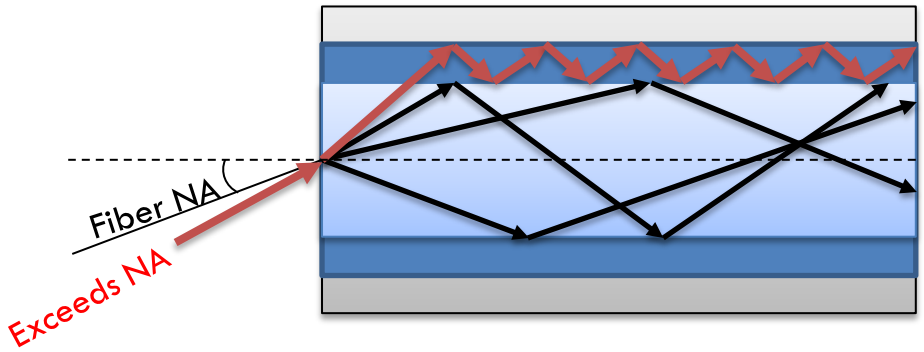
Lab Work - Part 1: Freespace SM to MM Coupling - Background

- When a mated fiber core or focused spot exceeds the core diameter of a receiving fiber, the light not incident on the core will not couple into the fiber, resulting in loss or propagate of cladding modes
- When a mated fiber's NA or focusing NA exceeds the NA of a receiving fiber, the light incident on the core will not couple into the fiber, resulting in loss or propagation of cladding modes
- For well mode mixed fibers, these coupling efficiencies can be approximated:



Lab Work - Part 1: Freespace SM to MM Coupling - Background

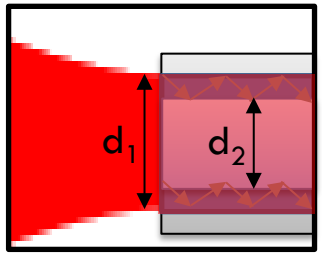
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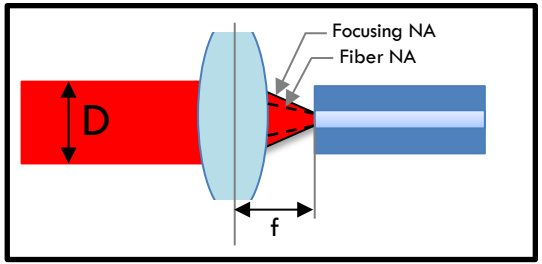
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Core/Spot Mismatch Efficiency: $\eta = \frac{d_2{}^2}{d_1{}^2} \; for \; d_1 > d_2$



NA Mismatch Efficiency: $\eta = \frac{(NA_{fiber})^2}{(NA_{focus})^2} \; for \; NA_{focus} > NA_{fiber}$



Lab Work - Part 1: Freespace SM to MM Coupling - Background

- This lab explores collimator to collimator freespace coupling from single mode to multimode fibers which is viewable in the course alignment section of this Insight

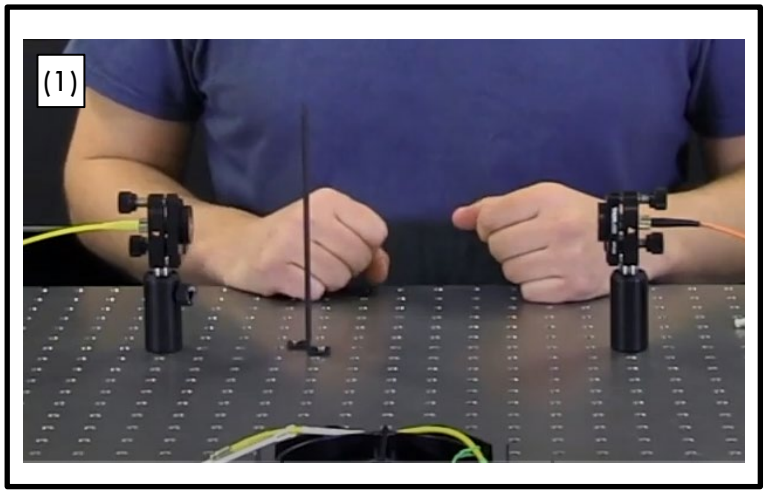
PHOTO NEEDED



Multimode Fiber
- 200 µm, 0.22 NA
- 200 µm, 0.39 NA
- 200 µm, 0.50 NA

Lab Work - Part 1: Freespace SM to MM Coupling

- Freespace Fiber Coupling Single Mode into Multimode Fiber: **Setup**
  - Mount the F280FC-B collimator into the AD11F adapter and KM100T kinematic mount onto a post and post holder (1)
  - Mount the F230SMA-B collimator into the AD11F adapter and KM100T kinematic mount onto a post and post holder in the same row of mounting holes as the F280FC-B and at least 6" inches apart (1)
  - Connect the single mode patch cable from the S1FC635 to the F280FC-B collimator (1)



Mount the collimators and connect the source fiber

Lab Work - Part 1: Freespace SM to MM Coupling

- Freespace Fiber Coupling Single Mode into Multimode Fiber: **Setup**
  - Connect the S120C power sensor to the PM400 Power Meter Console and turn on the PM400 (1)
  - Hit the wavelength setting (λ) and set the laser's wavelength (635 nm). (2)
  - Change the PM400's Bandwidth Setting to HIGH and power range to auto range, if not already set (2)
  - With the laser off, Zero the PM400 to offset the dark current and ambient coupled light (2)



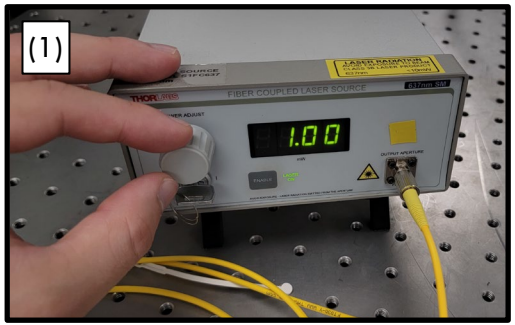
Connect the S120C to the PM400



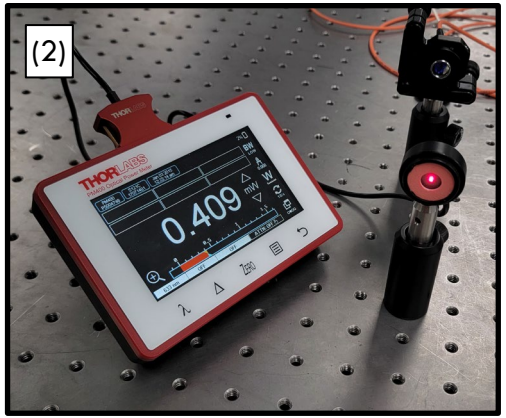
Change the PM400 settings

Lab Work - Part 1: Freespace SM to MM Coupling

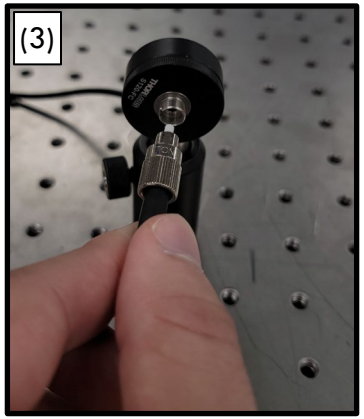
- Freespace Fiber Coupling Single Mode into Multimode Fiber: **Setup**
  - Turn on the S1FC635 benchtop laser source and turn on to 1 mW setting (1)
  - Place the S120C power sensor in the beam path to measure the output power (2)
  - Record the power measured by the PM400 as the baseline incident power
  - Remove the S120C from the beam path, thread on the S120-SMA adapter to the S120C, and connect the other end of the 200 µm 0.22 NA fiber to the S120-SMA (3)
  - Re-Zero the PM400 to account for reduced ambient light



Enable the S1FC635 to ~1 mW



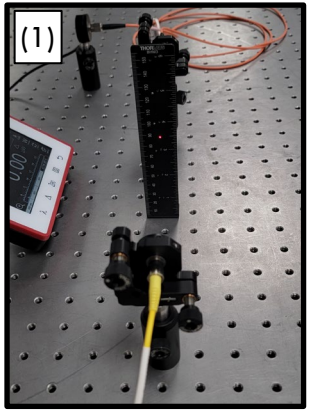
Measured the collimated beams' power



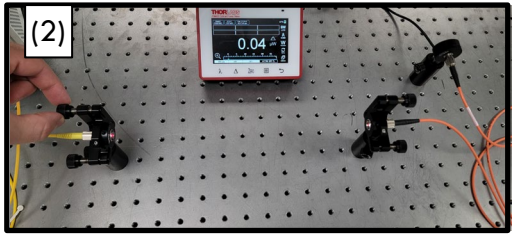
Connect the multimode patch cable to the S120C

Lab Work - Part 1: Freespace SM to MM Coupling

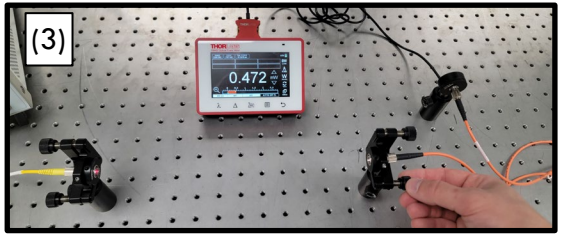
- Freespace Fiber Coupling Single Mode into Multimode Fiber: **Test Procedure**
  - Enable the S1FC635 benchtop laser source and observe collimated light from the F280FC-B collimator (1)
  - Rough align the adjusters of the F280FC-B's kinematic mount to point the collimator at the F230SMA-B coupling collimator (2)
  - Rough align the adjusters of the F230SMA-B kinematic mount to point the collimator towards the source F280FC-B (3)
  - Monitor the PM400 and make alignment adjustments between the two collimators until coupled power is maximized. Use the multimode course alignment techniques of the Insight to optimize alignment.



Turn on the S1FC635 and observe the collimated beam



Rough align the source collimator



Rough align the coupling collimator

Lab Work - Part 1: Freespace SM to MM Coupling

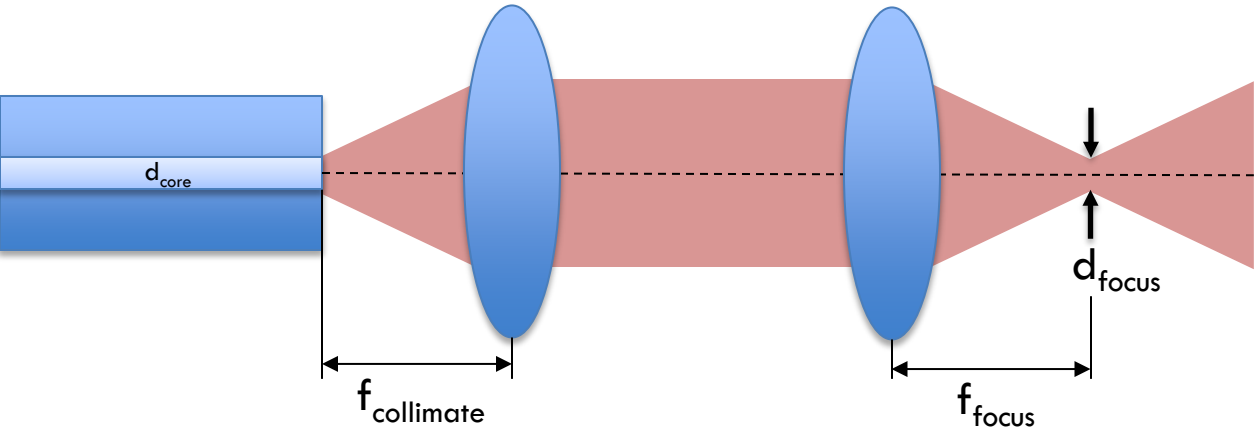
- Freespace Fiber Coupling Single Mode into Multimode Fiber: **Test Procedure**
  - Record the maximized power achieved from alignment
  - Disable the laser and replace the 200 µm, 0.22 NA patch cable with the 200 µm 0.39 NA fiber
  - Since an alignment was already performed, it should not take much alignment adjustment to maximize the coupling efficiency with the new fiber
  - Enable the laser and repeat the same alignment procedure to maximize coupling and record the final power
  - Repeat this whole process one last time with the 200 µm 0.50 NA fiber and record the final maximized power

Lab Work - Part 1: Freespace SM to MM Coupling

- Freespace Fiber Coupling Single Mode into Multimode Fiber: **Analysis**
  - **Below are useful specifications of the hardware used for this lab:**
    - **F280FC-B**
      - **Focal Length: 18.24 mm**
      - **Collimated Beam Diameter ($1/e^2$): 3.4 mm**
    - **F230SMA-B**
      - **Focal Length: 4.43 mm**
  - Calculate the efficiency achieved for each of the 200 µm core fibers based on the baseline power from the measurement that was taken with the S1FC635 connected to the power sensor
  - Calculate the Focusing NA based on the $1/e^2$ beam diameter and the focal length of the coupling collimator
    - If the Focusing NA exceeds the NA of any of the coupling fibers, calculate the theoretical efficiency from exceeding the NA
  - Calculate the diffraction limited spot size using the collimated $1/e^2$ beam diameter and the focal length of the coupling collimator
    - If the spot size is larger than the core, calculate the theoretical efficiency from the spot/core diameter mismatch
  - How do the measured coupling efficiencies compare with the theoretical values?

Lab Work - Part 2: Freespace MM to MM Coupling - Background

- Single mode Gaussian beams have very good focusing performance and provide diffraction limited spots when using the appropriate focusing optic
- Multimode coupled sources are inherently non-point-like, which will not achieve diffracted limited spot sizes upon focusing
- If using low aberration lenses, using the same focal length to both collimate and focus from a multimode fiber will focus to the same diameter as the emitting core
- Using different focal lengths introduce a magnification that will changes the focused spot size and focusing-NA depending on the focal lengths used to collimate and focus

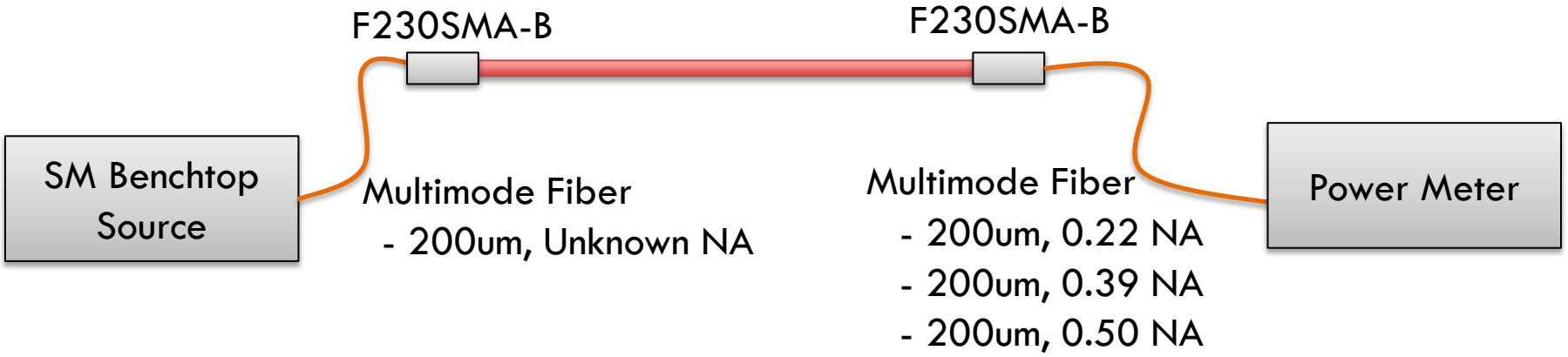


$$d_{focus} = d_{core}\left(\frac{f_{focus}}{f_{collimate}}\right)$$

Lab Work - Part 2: Freespace MM to MM Coupling - Background

- When the source fiber of a collimated multimode beam is unknown, the spot diameter and Numerical Aperture focus is also unknown, making optical selection more difficult and can introduce higher losses
- This lab investigates freespace coupling between multimode fibers and dependence of the collimated fiber's numerical aperture on coupling efficiency

Photo Needed



Lab Work - Part 2: Freespace MM to MM Coupling

- Freespace Fiber Coupling Multimode into Multimode Fiber: **Setup**
  - Use the same setup as Freespace Fiber Coupling Single Mode into Multimode Fiber setup
  - Remove the F280FC-B collimator from the source's kinematic mount and replace it with the F230FC-B so the collimating and coupling collimators have the same focal length
  - Disconnect the single mode fiber from the S1FC635 benchtop laser source and connect the cable labeled "200um Core, Unknown NA" to the S1FC635 and the collimating F230FC-B
  - Disconnect the S120-SMA adapter and coupling fiber from the power sensor and move the power sensor into the beam path
  - Enable the power of the S1FC635, re-zero the PM400, and take a baseline power measurements reading of the collimated beam
  - Disable the laser source, reconnect the S120-SMA, and reconnect the 200 µm 0.22 NA fiber back to the power sensor
  - With the fiber reattached and the source off, re-Zero the PM400 one last time

Lab Work - Part 2: Freespace MM to MM Coupling

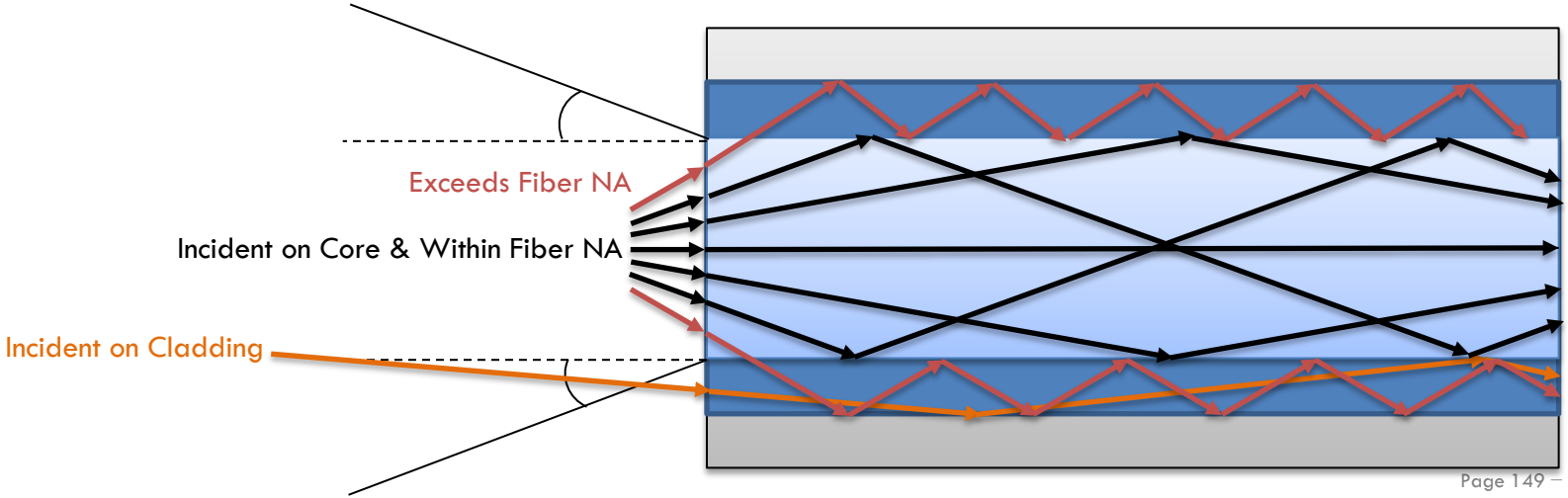
- Freespace Fiber Coupling Multimode into Multimode Fiber: **Test Procedure**
  - For each of the following fibers:
    - 200 µm, 0.22 NA
    - 200 µm, 0.39 NA
    - 200 µm, 0.50 NA
  - Freespace fiber couple the collimated multimode coupled source and align to maximize the coupling efficiency
  - Record the maximum power achieved for each

Lab Work - Part 2: Freespace MM to MM Coupling

- Freespace Fiber Coupling Multimode into Multimode Fiber: **Analysis**
  - Calculate the coupling efficiency using the baseline freespace power and the maximized powers measured with each of the fibers:
    - 200 µm, 0.22 NA
    - 200 µm, 0.39 NA
    - 200 µm, 0.50 NA
  - The collimated multimode fiber is labeled as 200 µm core, but with an unknown NA
    - Based on the efficiencies measured for each fiber above, what might you be able to determine about the NA of the source fiber?
  - How did optimizing the alignment coupling efficiency of the multimode coupled source compare with the single mode coupled source of Lab part 1?

Lab Work - Part 3: Coupling Conditions & Mode Mixing - Background

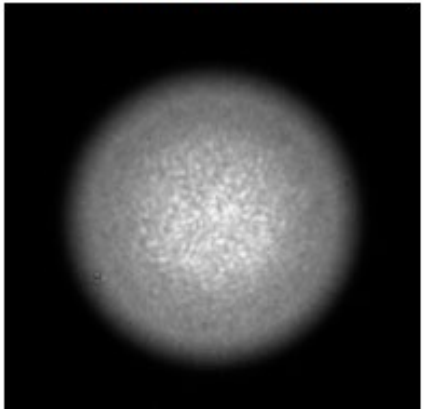
- It was previously stated, the primary requirements for efficiently coupling into multimode fiber are:
  - The incident beam focuses onto the core of the fiber with a spot smaller than the core diameter
  - The incident beam focuses with an NA that does not exceed the fiber's numerical aperture (NA_{fiber})
- While this is true, the intensity distribution of light in a multimode fiber is dependent on the launch conditions launching into the fiber
- The intensity distribution and output profile will only be a flat top profile when light is distributed equally among all modes of a multimode fiber will



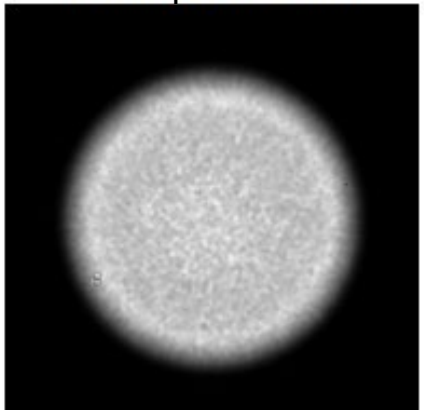
Lab Work - Part 3: Coupling Conditions & Mode Mixing - Background

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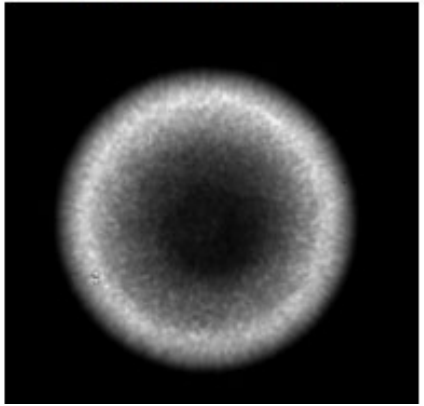
Lower Order Modes
Gaussian Distribution



Well Mode Mixed
Flat Top Distribution

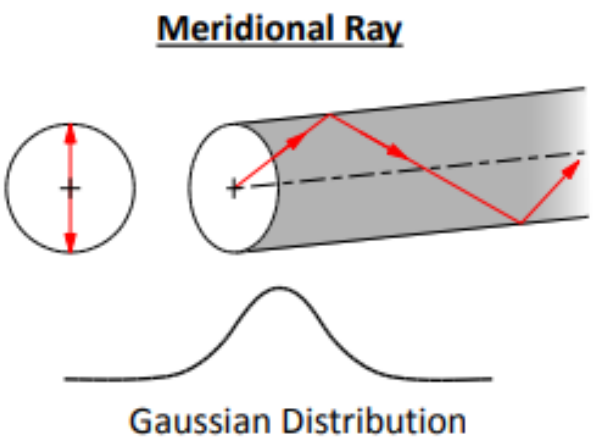


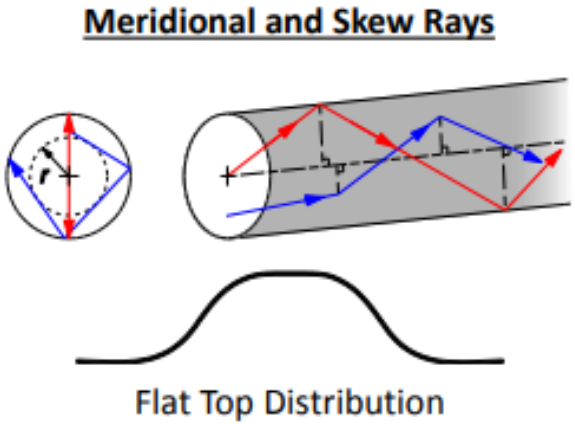
Higher Order Modes
"Donut" Distribution

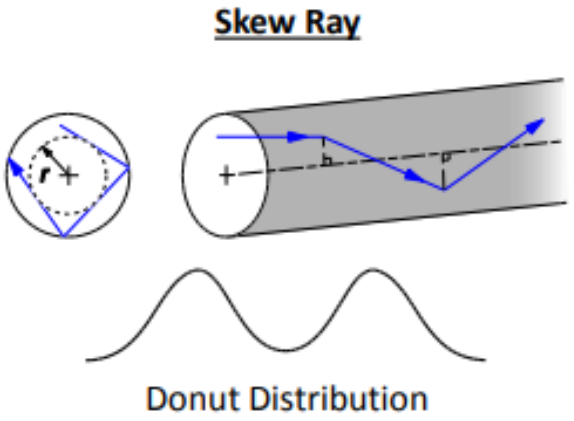


Lab Work - Part 3: Coupling Conditions & Mode Mixing - Background

- Meridional rays pass through the central axis of the core and tend to fill the lower order modes
- Skewed rays travel in helical paths near the edges of the core and tend to fill higher order modes
- By launching into the fiber off-axis, the relative proportion of meridional rays to skewed rays can be adjusted to influence the fiber's mode distribution as well as its resulting output profile

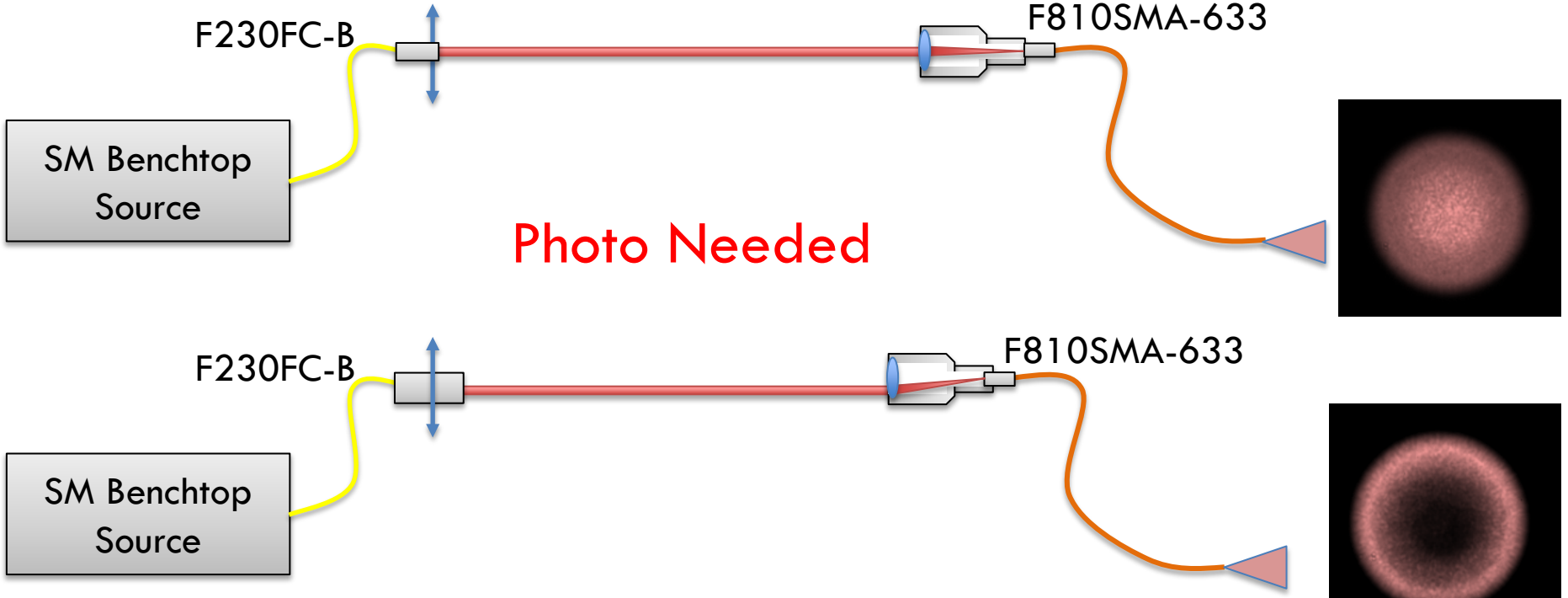






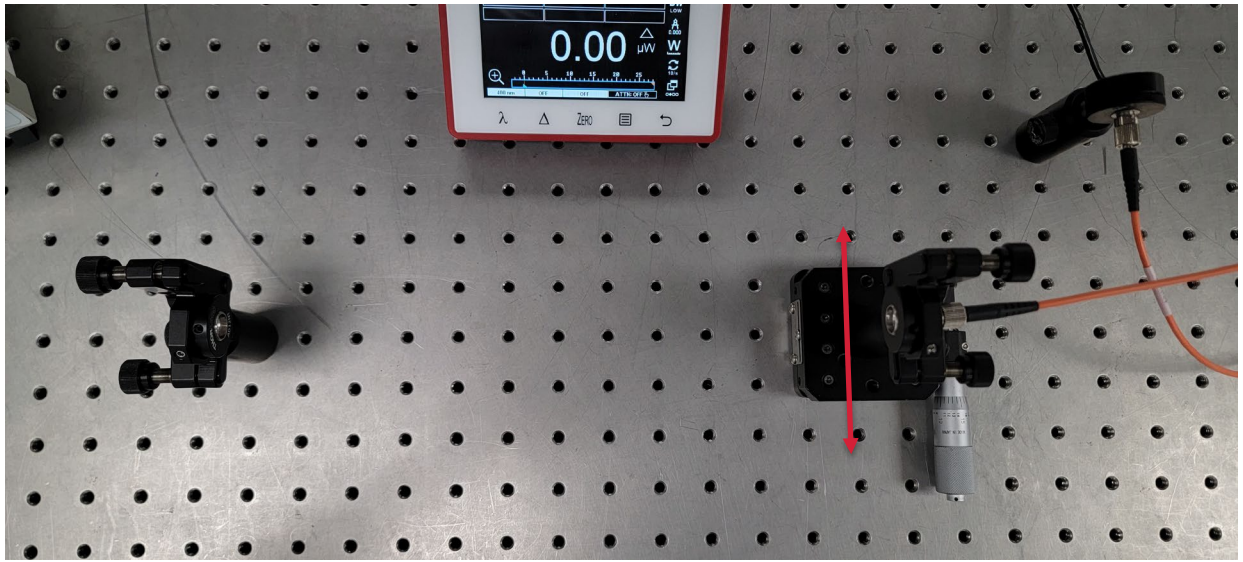
Lab Work - Part 3: Coupling Conditions & Mode Mixing - Background

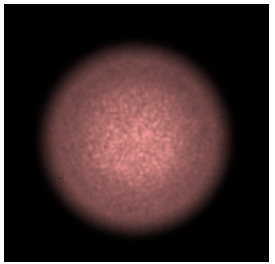
- This lab investigates the coupling launch condition dependence on the modal distribution and intensity distribution of multimode fibers:

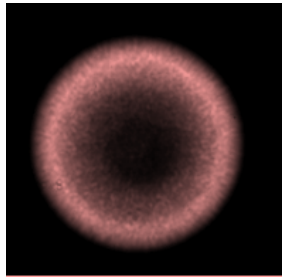


Lab Work - Part 3: Coupling Conditions & Mode Mixing - Background

- This lab investigates the coupling launch condition dependence on the modal distribution and intensity distribution of multimode fibers:

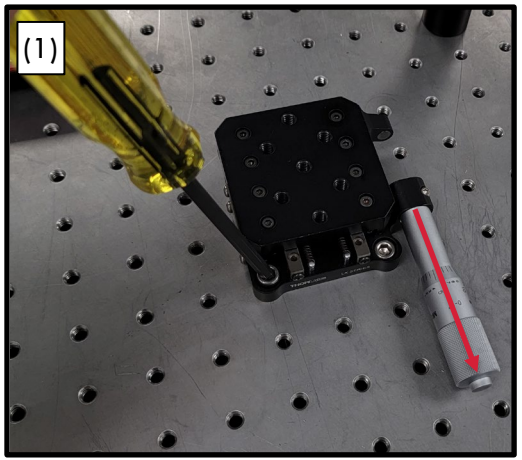




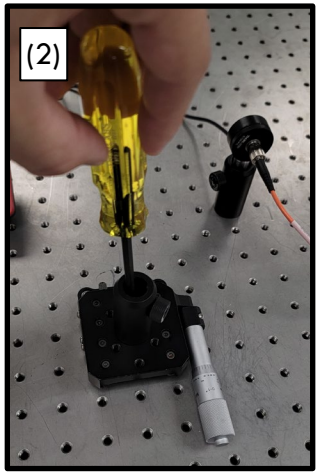


Lab Work - Part 3: Coupling Conditions & Mode Mixing

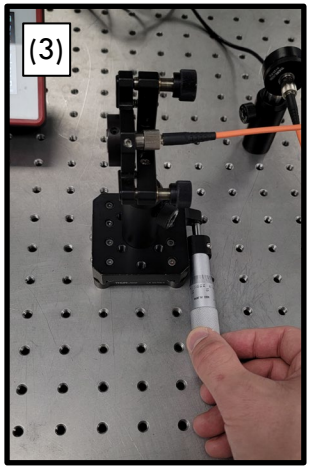
- Coupling Conditions & Mode Mixing: **Setup**
  - Mount the translation stage on the optical table so that the adjuster thumbscrew is facing towards you [1]
  - Mount a post holder in the center tapped hole of the translation stage [2]
  - Mount the F810SMA-635 coupling collimator in its KM100T with the AD15F adapter onto the post and post holder on the translation mount connect the 400um 0.22 NA patch cable[3]
  - Align the translation mount so it's in its middle neutral position at the center [3]



Mount the stage to the table with adjuster towards you



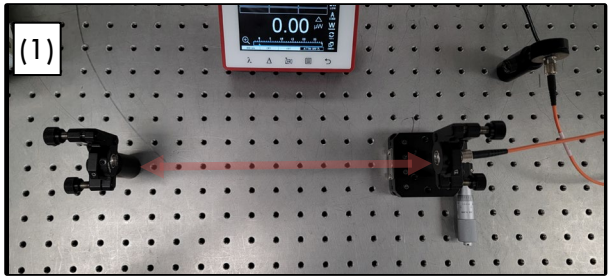
Mount the post holder on the stage's center tap



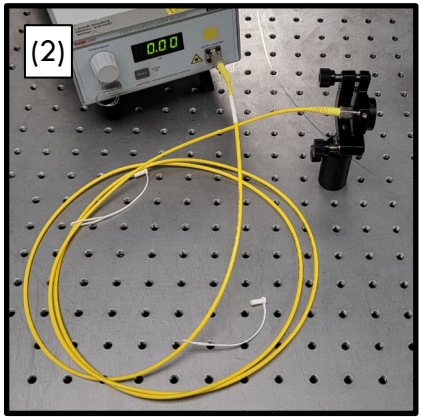
Mount the collimator and adjust the stage position to center

Lab Work - Part 3: Coupling Conditions & Mode Mixing

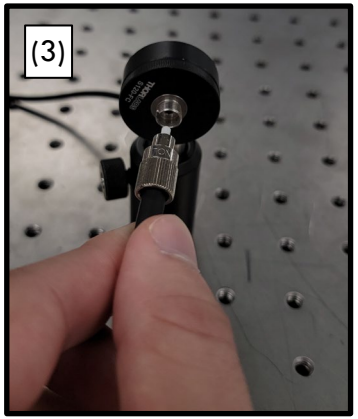
- Coupling Conditions & Mode Mixing: **Setup**
  - Mount the F230FC-B coupling collimator in its KM100T with the AD11F [1]
  - Mount this on a post and in post holder at least 6” away from the F810SMA-635 collimator on a row a holes that aligns with the F810SMA-635 collimator in its middle neutral stage position [1]
  - Connect the single mode patch cable from the S1FC635 to the F230FC-B collimator [2]
  - Connect the other end of the 400 µm, 0.22 NA fiber to the S120-SMA and S120C sensor [3]



Mount the F230FC-B on the same row as the F810SMA-635



Connect the single mode cable to the S1FC635 and F230FC-B



Connect the multimode cable from the F810SMA-635 to the power sensor

Lab Work - Part 3: Coupling Conditions & Mode Mixing

- Coupling Conditions & Mode Mixing: **Setup**
  - Connect the S120C power sensor to the PM400 Power Meter Console and turn on the PM400 (1)
  - Hit the wavelength setting (λ) and set the laser's wavelength (635 nm) is not already set (2)
  - Change the PM400's Bandwidth Setting to HIGH and power range to auto range, if not already set (2)
  - With the laser off, Zero the PM400 to offset the dark current and ambient coupled light (2)



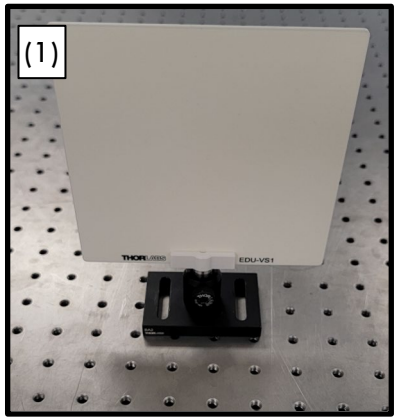
Connect the S120C to the PM400



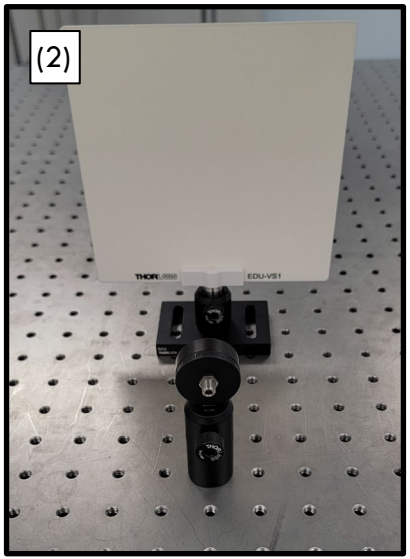
Change the PM400 settings

Lab Work - Part 3: Coupling Conditions & Mode Mixing

- Coupling Conditions & Mode Mixing: **Setup**
  - Mount the EDU-VS1viewing screen on a post and post holder off to the side [1]
  - Mount a LMR1, SM1 threaded mounted with a SM1SMA adapter in a post and post holder in front of viewing screen [2]



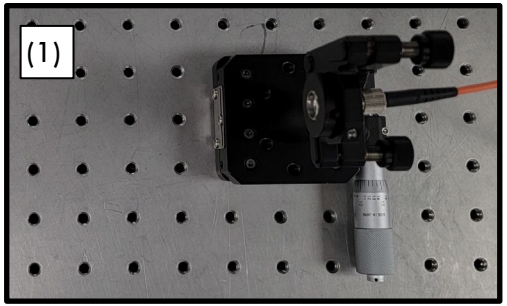
Mount the viewing screen



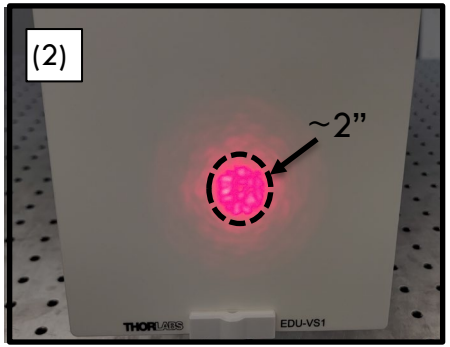
Mount the LMR1and SM1SMA

Lab Work - Part 3: Coupling Conditions & Mode Mixing

- Coupling Conditions & Mode Mixing: **Test Procedure**
  - Keeping the collimator translation stage in its center neutral position, enable the S1FC635 laser source and set the output power to ~1mW (1)
  - As done in the previous lab parts, use the kinematic adjusters of the collimating and coupling collimators to adjust and optimize the alignment to maximize the power coupled into the 400 µm core fiber
  - Once optimized, disable the S1FC635 source
  - Disconnect the 400 µm fiber from the power sensor and connect it to SM1SMA adapter in the LMR1
  - Enable the S1FC635 source and make any adjustments required to view the output profile effectively
  - Set the distance between the viewing screen fiber so the beam diameter is ~2" in diameter or slightly larger (2)



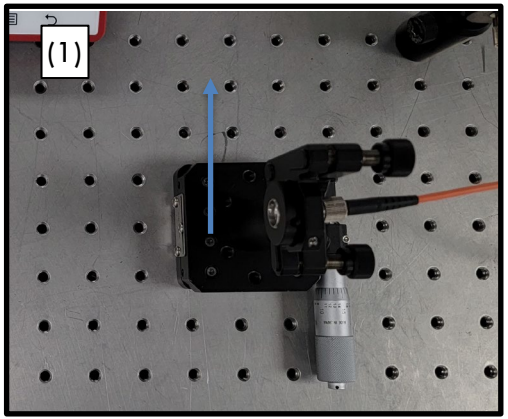
Stage in centered position on row of holes



Create a large spot on the viewing screen

Lab Work - Part 3: Coupling Conditions & Mode Mixing

- Coupling Conditions & Mode Mixing: **Test Procedure**
  - Observe the beam profile:
    - Does it seem well mode mixed or have a more Gaussian distribution?
    - If you agitate the fiber, how does the profile change?
  - Start adjusting the translation stage of the recieving collimator to offset the beam into the coupling collimator and make observations about the beam: (1)
    - What changes do you observe in the beam profile intensity distribution?
    - Does the beam diameter stay constant while adjusting?
    - Is there a critical point where the beam diameter starts becoming bigger?



Translate the receiving collimator

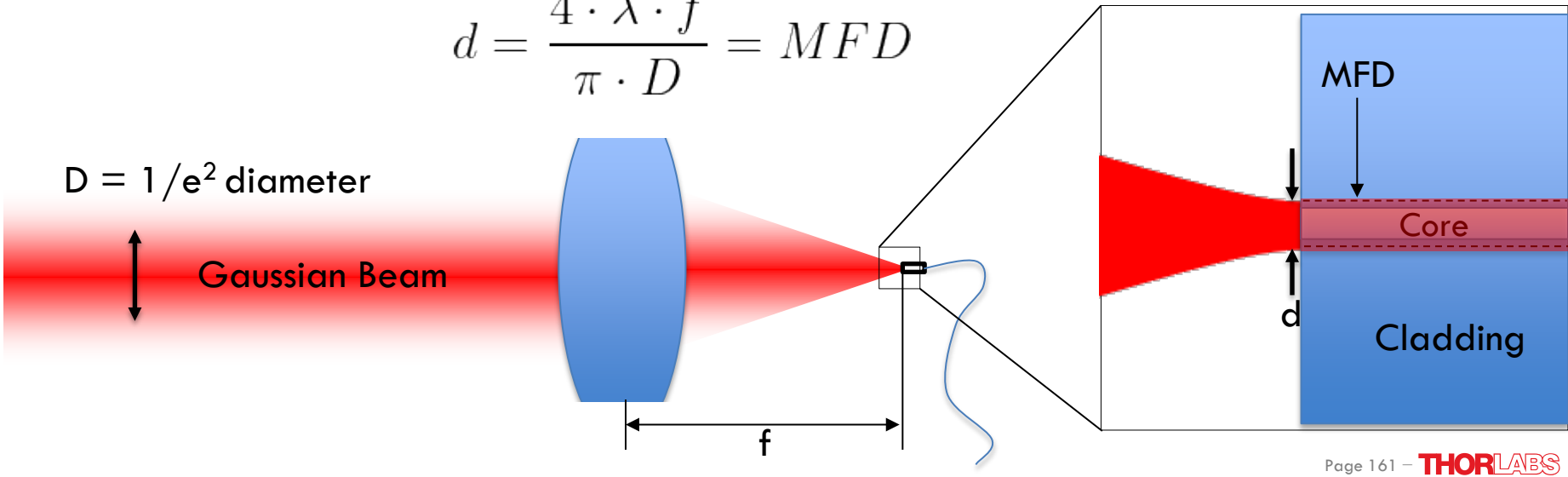
Lab Work - Part 3: Coupling Conditions & Mode Mixing

- Coupling Conditions & Mode Mixing: **Analysis**
  - Why does translating the incident beam on the coupling collimator change the mode distribution in the fiber?
  - When translating the beam, was there a discernable point where light started coupling into cladding modes?
  - Based on the 0.22 NA of the fiber, how big should the anticipated beam diameter be at the viewing screen?
  - Based on the collimated beam diameter spec of the collimator used (0.8 mm), the focal length used to couple into the fiber (f = 35.41 mm), and 0.22 NA fiber used, is there certain translation distance you'd expect before light should start coupling into cladding modes?

Lab Work - Part 4 : SM to SM Freespace Coupling - Background

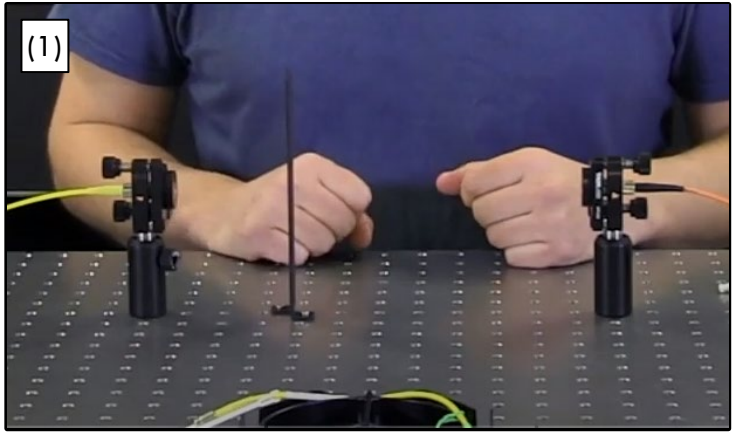
- Single mode coupling is less trivial than coupling into multimode fiber
- Unlike multimode fiber, light simply being incident on the fiber does not guarantee coupling efficiency since the fibers can only propagate the LP01 Gaussian mode
- Coupling into single mode fiber with high efficiency requires the source being a single mode Gaussian and focusing to the correct spot size to match the mode field diameter (MFD) of the receiving fiber

$$d = \frac{4 \cdot \lambda \cdot f}{\pi \cdot D} = MFD$$



Lab Work - Part 4: SM to SM Freespace Coupling

- Freespace Fiber Coupling Single Mode into Single Mode Fiber: **Setup**
  - Mount the F230FC-B collimator into the AD11F adapter and KM100T kinematic mount onto a post and post holder [1]
  - Mount the other F230FC-B collimator into the AD11F adapter and KM100T kinematic mount onto a post and post holder in the same row of mounting holes as the first at least 6” inches apart [1]
  - Connect the single mode patch cable from the S1FC635 to the source F230FC-A collimator [1]



Mount the collimators and connect the source fiber

Lab Work - Part 4: SM to SM Freespace Coupling

- Freespace Fiber Coupling Single Mode into Single Mode Fiber: **Setup**
  - Connect the S120C power sensor to the PM400 Power Meter Console and turn on the PM400 (1)
  - Hit the wavelength setting (λ) and set the laser's wavelength (635 nm). (2)
  - Change the PM400's Bandwidth Setting to HIGH and power range to auto range, if not already set (2)
  - With the laser off, Zero the PM400 to offset the dark current and ambient coupled light (2)



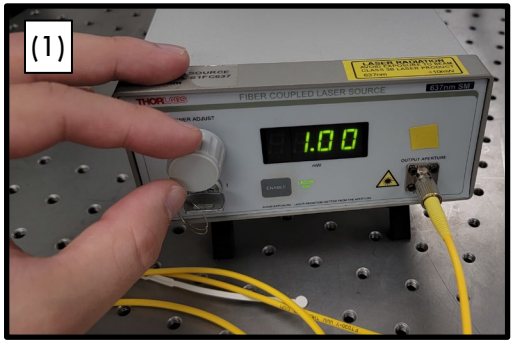
Connect the S120C to the PM400



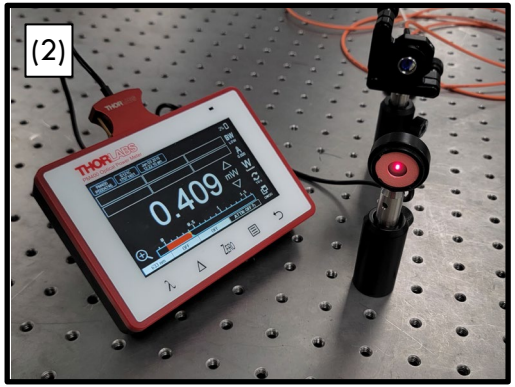
Change the PM400 settings

Lab Work - Part 4: SM to SM Freespace Coupling

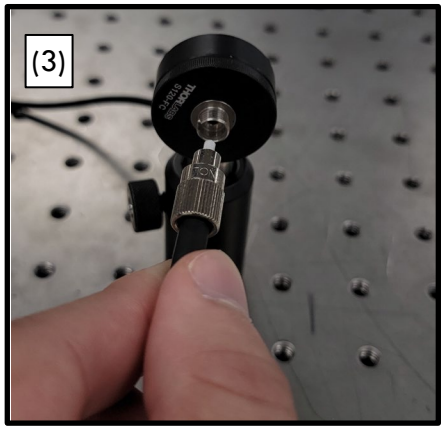
- Freespace Fiber Coupling Single Mode into Single Mode Fiber: **Setup**
  - Turn on the S1FC635 benchtop laser source and turn on to 1 mW setting [1]
  - Place the S120C power sensor in the beam path to measure the output power
  - Record the power measured by the PM400 as the baseline incident power
  - Remove the S120C from the beam path, thread on the S120-FC adapter to the S120C, and connect the 50 µm, 0.22 NA multimode fiber to the receiving collimator and to the S120-FC [3]
  - Re-Zero the PM400 to account for reduced ambient light



Enable the S1FC635 to ~1 mW



Measured the collimated beams' power



Connect the multimode patch cable to the S120C

Lab Work - Part 4: SM to SM Freespace Coupling

- Freespace Fiber Coupling Single Mode into Multimode Fiber: **Test Procedure**
  - Record the maximized power achieved from alignment
  - Disable the laser and replace the 50 µm, 0.22 NA patch cable with the single mode patch cable
  - Since an alignment was already performed with the multimode fiber, this bring the alignment for the single mode fiber to a much closer starting point than from scratch
  - Enable the laser and repeat the same alignment procedure to maximize coupling with the single mode fiber and record the final power

Lab Work - Part 4: SM to SM Freespace Coupling

- Freespace Fiber Coupling Single Mode into Single Mode Fiber: **Analysis**
  - **Below are useful specifications of the hardware used for this lab:**
    - **F230FC-B**
      - **Focal Length: 4.43**
      - **Collimated Beam Diameter ($1/e^2$): 0.8 mm**
    - **SM600**
      - **Mode Field Diameter: ~4.5 µm**
  - What efficiency was achieved with the 50 µm, 0.22 NA fiber?
  - What efficiency was achieved with the single mode fiber?
  - Calculate the diffraction limited spot size of the focused beam
  - Calculate the theoretical efficiency of this spot size to fiber MFD
  - If we're using the same fiber (MFD) for collimating and coupling and the focal length to both collimate and couple, the calculated theoretical efficiency expected?